Libertinus and diacritics

To do

Audit mark anchors to fit roman and italic anchor models; patch as necessary.

Add alt cap dot above for italic and semibold italic.

Revise fontmetrics table per anchor model with good candidates.

Anchor model

Libertinus has an anchor model where the base anchors are set to locations where the marks would be placed relative to the base. See the blue dots in the left panel of Figure 1; from top to bottom, the above-anchor is the blue dot above the base glyph, the overlay-anchor is blue dot inside the base glyph outline, and the below-anchor is the blue dot below the base glyph. The mark anchor is generally near the optical center of the mark glyph outline. See the red square in the middle panel of Figure 1; the mark's below-anchor is at the optical center of the mark. The base and mark are combined in the right panel of Figure 1 by superimposing their below-anchors.

The clearance between a base and a mark is therefore encoded in the base anchor, not in the mark. The Y coordinate of an above- or below-anchor is the vertical aspect of the base glyph plus a clearance; see the section on vertical aspects below. The X coordinate is at the horizontal center of the base glyph. For a mostly symmetric letterform in the roman styles, a candidate for this X coordinate is the midpoint of the bounding box of the glyph.

The anchor of a mark is at the horizontal center of the mark, if the mark is symmetric. An asymmetric mark, such as the acute or grave, has its anchor a few units to one side of the center of its outline. Vertically, the anchor of an above-mark is about 140 units above the bottom of the mark outline. That is near the top of a tall mark, such as the circumflex, and above the top of a short mark, such as the macron or dot. Because the above-anchor of a base at the x-height is 216 units above the x-height, the bottom of a tall mark is about 75 units above the base, and a short mark sits a little higher.

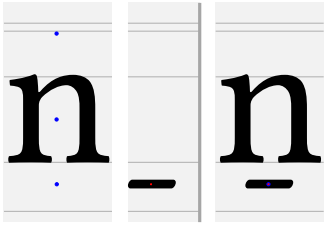


Figure 1: Libertinus n and macron with anchors.

Other fonts use a different geometric model for anchors. In Noto, the base's below-anchor is at the baseline, the overlay-anchor is inside the glyph, and the above-anchor is at the x-height. See the blue dots in the left panel of Figure 2; from the bottom to the top, the mark's below-anchor is the blue dot on the baseline, the overlay-anchor is blue dot inside the base glyph outline, and the above-anchor is the blue dot at the x-height. In Noto, the mark's below-anchor is at the origin $(0, 0)$, and the mark glyph outline is drawn below the origin. In the middle panel of Figure 2, the mark's below-anchor is the red square at the origin. The base and mark are combined in the right panel of Figure 2 by superimposing their below-anchors. So in Noto the clearance is encoded in the mark glyph, and in Libertinus it is encoded in the base anchor.

Italic anchor model

In italic, the Libertinus anchor model is the same as in the roman styles, but it is applied along the italic axis. The italic lane of a base glyph is its advance-width box, slanted by the italic angle θ (12°). The midline of the lane is the line $X = W/2 + Y \tan\theta$, where W is the advance width of the glyph. See the gray band and the dashed blue line in the left panel of Figure 3.

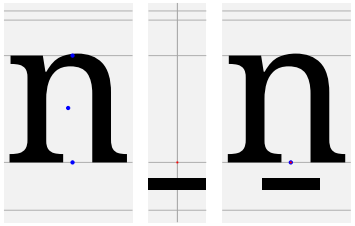


Figure 2: Noto n and macron with anchors.

The Y coordinate of an above- or below-anchor is set exactly as in the roman styles: the vertical aspect of the glyph plus the clearance, measured vertically. So marks sit at the same heights in roman and italic. The X coordinate is the point of the midline at that height: $ax = W/2 + ay \tan\theta$ and $bx = W/2 + by \tan\theta$. For italic n, $W = 518$, so $ax = 259 + 645 \times 0.2126 = 396$ and $bx = 259 - 110 \times 0.2126 = 236$; the font has $ax = 394$ and $bx = 235$. The above-anchor is therefore right of the below-anchor by $(ay - by) \tan\theta$, which is about 160 units for a letter at the x-height.

The height in this formula is the height of the anchor itself, that is, the vertical aspect plus the clearance, not the height of the vertical aspect. So the correct vertical aspect of the glyph must be determined before the X coordinate can be calculated. In italic, the above-anchor is at $W/2 + 137$ for a letter at the x-height ($ay = 645$), at $W/2 + 181$ for a capital ($ay = 850$), and at $W/2 + 189$ for an ascender ($ay = 890$). The below-anchor is at $W/2 - 23$ for a letter on the baseline ($by = -110$) and at $W/2 - 68$ for a descender ($by = -319$). An error of 10 units in the Y coordinate moves the X coordinate by about 2 units.

The anchor of a mark lies on the mark's own slanted axis, through the center of its outline if the mark is symmetric. See the dashed red lines in the middle panel of Figure 3. That is, the mark anchor is at $X = C + Y \tan\theta$, where Y is the height of the anchor, as in the roman styles, and C is where the axis of the mark meets the baseline: the horizontal center of the mark outline after the slant is removed. An asymmetric mark keeps the same offset from its axis as in the roman styles. Superimposing the anchors lays the axis of the mark on the midline of the base, as in the right panel. The mark is displaced from the base along the italic axis, not vertically above or below it.

The midline is a geometric center, not an optical center, but it is a good candidate for the anchor X coordinate of a mostly symmetric letterform. Asymmetric letterforms, and letters that overhang their advance width such as f and j, need curated anchors.

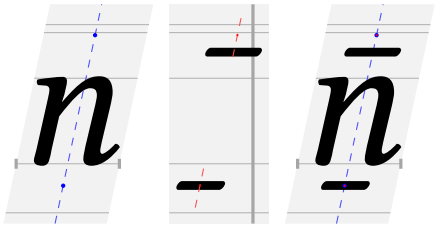


Figure 3: Libertinus italic n, macron and macron below, with anchors. The gray band is the italic lane and the dashed blue line is its midline.

Semibold italic anchor model

Semibold italic has its own anchor model, which is not the italic anchor model, and the patch keeps it. The designer's anchors are the anchors of regular, set on slanted outlines (the italic angle θ is 11.5°): 42 of the 44 above- and below-mark anchors, and 319 of the 384 above- and below-anchors of bases, have the coordinates of regular exactly. So neither anchor carries the term $Y \tan\theta$. A mark anchor is left of the mark's axis by a shift K, and a base anchor is left of the midline of its lane by about the same shift, so the two errors cancel when the anchors are superimposed. See Figure 4, where the hollow symbols are the positions given by the italic anchor model.

In this model, the mark anchor is at $X = C + Y \tan\theta - K$ and the base anchor is at $X = W/2 + Y \tan\theta - K$, with $\tan\theta = 0.2035$ and the same Y coordinates as in the other styles. The shift is $K_a = 150$ for above-marks (the median; most are between 132 and 168) and $K_b = -12$ for below-marks (most are between -23 and -4). The italic anchor model is the case $K = 0$. The candidates for a base are $ax = W/2 + ay \tan\theta - 150$ and $bx = W/2 + by \tan\theta + 12$. So the above-anchor is at $W/2 - 19$ for a letter at the x-height ($ay = 645$), at $W/2 + 23$ for a capital ($ay = 850$), and at $W/2 + 31$ for an ascender ($ay = 890$). The below-anchor is at $W/2 - 10$ for a letter on the baseline ($by = -110$) and at $W/2 - 53$ for a descender ($by = -319$). For semibold italic n, $W = 606$, so $ax = 284$ and $bx = 293$; the font has $ax = 264$ and $bx = 265$, the values of regular n, and the macron lands 29 units left of the midline.

The cancellation is only approximate, because K is not one number. For a mark, $K = Y \tan\theta$ plus a constant, where Y is the height of the mark anchor; the designer's base anchors are near $W/2 - 9$ at every height, so the error of a mark on a base grows with the difference in height of the two anchors, by 2 units for every 10. The median distance of a designer's base anchor from the candidate is about 20 units for a lowercase letter and 30 units for a capital. The patch does not move this style to the italic anchor model. It keeps the designer's anchors where they conform to the semibold italic anchor model, and only corrects the anchors that do not, so that they conform to it: a base anchor is moved to the candidate above, and a mark anchor is moved to $X = C + Y \tan\theta - 150$ for an above-mark or $X = C + Y \tan\theta + 12$ for a below-mark. An anchor set by the italic anchor model would be wrong here: it would be about 150 units too far right for an above-mark.

The following do not conform to the semibold italic anchor model and probably need to be patched, some of them manually.

- Capitals. The designer's above-anchors are about 25 units left of the candidate, since they are 205 units higher than the anchors of lowercase letters but no further right. The worst are B, D, I, N, O, R, W (40 to 60 units left) and L (159 units left; its anchor is over the stem, as in regular).
- The capital forms of the grave, acute, circumflex and breve (`grave.cap`, `acute.cap`, `circumflex.cap`, `breve.cap`), which the font substitutes over capitals. Their mark anchors were set with the slant (K between -13 and 29), but the above-anchors of capitals were not, so these four marks land 160 to 190 units left of the middle of a capital. The other above-marks have no capital form in use and land 20 to 50 units left. The capital forms need mark anchors at $X = C + Y \tan\theta - 150$; `caron.cap` and `dieresis.cap` are not substituted.
- Above-anchors that were set with the slant, as in italic, so that an above-mark lands 130 to 190 units too far right: a, s, h and J.
- Other base anchors more than 40 units from the candidate. Above: c, g, Æ, Ɖ, ɦ, Ɠ, Ƥ, ɳ, ɗ, ɦ, ʄ, Ħ, ɵ, ɧ. Below: A, C, D, J, a, d, f, j, q, r, t, Æ, ĸ, ɦ, Ɠ, Ƥ, ɳ, ɾ, ɑ, ɖ, ɗ, ɚ, ɥ, ɯ, ɻ, Ħ, ɵ. Several of these are copies of the anchors of another glyph. The anchors of ɥ and ɻ are transposed, as in the other styles.
- Marks whose shift is far from the median, so that no base anchor can be right for them and for the other marks: dieresis U+0308 ($K = 111$, so it lands about 40 units left of the other above-marks), comma above U+0313 (125), inverted breve U+0311 (132), hook above U+0309 and double grave U+030F (168, about 18 units right), koronis U+0343 (371), right tack below U+0319 (-30) and tilde below U+0330 ($+10$). The double acute U+030B (156) conforms.
- Tilde U+0303, one of the two mark anchors that is not a copy of regular; its shift is 139.
- The Hebrew ligatures U+05F0, U+05F1, U+05F2, which have above- and below-anchors in this lookup table in semibold italic only.
- Letters that overhang their advance width, such as f and j, which need curated anchors in every italic style.

The anchors of precomposed glyphs were not compared with these candidates.

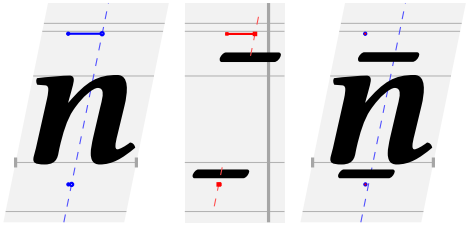


Figure 4: Libertinus semibold italic n, macron and macron below, with the designer's anchors (filled) and the positions given by the italic anchor model (hollow). The above-anchor of n is 170 units left of the midline and the anchor of the macron is 141 units left of its axis.

Font metrics per glyph

See above- and below-anchor coordinates for A–Z, a–z, and IPA letters for Libertinus serif fonts (regular, italic, semibold, semibold italic) in four tables (Tables 1, 2, 3, 4, 5, 6). Recall that Language Science Press uses semibold Libertinus fonts, e.g. `LibertinusSerif-Semibold.otf`, for all boldface style.

Table row headings

hex: Unicode character codepoint in 4-hex.

reg, it, sb, si: regular, italic, semibold, semibold italic. In that row, the character is printed alone, with combining dot above, and with combining macron below, to show any anchor positioning.

ax, ay: X and Y coordinates of the above-anchor.

bx, by: X and Y coordinates of the below-anchor.

The next few rows use the bounding box of the glyph, because the midpoint of the width of the bounding box is a naive candidate for ax and bx, and the bounding box coordinates are easy to determine.

xm is a horizontal midpoint relative to the bounding box of the glyph. For roman font styles (regular and semibold), xm is the x-midpoint of the bounding box, $(X_{\text{min}} + X_{\text{max}})/2$, where the bounding box is characterized by $(X_{\text{min}}, Y_{\text{min}})$ and $(X_{\text{max}}, Y_{\text{max}})$. For italic font styles (italic and semibold italic), there are two midpoints, axm and bxm, at the top and bottom of the italic lane where the glyph lives.

The δ row – axδ and bxδ – is the offset of the anchor x-value from the relevant x-midpoint value.

These x-midpoints are geometric centers, not optical centers, but they are reasonable candidate for the anchor x-coordinate.

Vertical aspects, anchor Y coordinates and reference, and clearance

A font has at least five vertical aspects where glyph outlines live inside: descender height, baseline, x-height, ascender height, capital height. The values or ranges of these Y coordinates are in the first five rows of Table 9. These vertical aspects are the gray horizontal lines of Figure 1. These vertical aspects are displayed in a FontForge glyph outline window as black horizontal guidelines with red or blue shaded bands for ranges. I denote these as yra, yrc, yrx, $yr0$, yrd because they are Y values, and r indicates a reference value. a, c, x, 0, d indicate the five vertical aspects. These values (and the ranges that set the bands) are in the SFD, in a line that starts with `BlueValues`, e.g. in Libertinus serif regular,

```
BlueValues 47 [-12 0 429 442 460 472 600 610 645 658 688 698]
OtherBlues 11 [-238 -227]
```

means $yra = 698, [688, 698]$; $yrc = 645, [645, 658]$; $yrx = 429, [429, 442]$; $yr0 = 0, [-12, 0]$; and $yrd = -232, [-238, -227]$.

Table 1: Font metrics for A–Z

hex	0041	0042	0043	0044	0045	0046	0047	0048	0049	004A	004B	004C	004D	004E	004F	0050	0051	0052	0053	0054	0055	0056	0057	0058	0059	005A
cat	Lu	Lu	Lu	Lu	Lu	Lu	Lu	Lu	Lu	Lu	Lu	Lu	Lu	Lu	Lu	Lu	Lu	Lu	Lu	Lu	Lu	Lu	Lu	Lu	Lu	Lu
reg	AȦẠ	BḂḄ	CĊÇ	DḊḌ	EĖẸ	FḞḞ	GĠĢ	HḢḤ	IİỊ	JJ̇J̣	KḲḴ	LĿḶ	MṀṂ	NṄṆ	OȮỌ	PṖṔ	QQ̇Ǫ	RṘṚ	SṠṢ	TṪṰ	UU̇Ụ	VV̇Ṿ	WẆẈ	XẊẌ	YẎỴ	ZŻẔ
aa	c	c	c	c	c	c	c	c	c	c	c	c	c	c	c	c	c	c	c	c	c	c	c	c	c	c
ba	b	b	b	b	b	b	b	b	b	d+60	b	b	b	b	b	b	d+23	b	b	b	b	b	b	b	b	b
ax	354	267	340	306	281	284	365	361	150	182	336	151	431	346	335	277	348	269	232	307	346	336	450	332	295	320
ay	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850
bx	308	268	363	309	287	209	367	362	147	109	342	262	393	348	350	240		258	234	300	328	318	479	307	295	287
by	-110	-110	-110	-110	-110	-110	-110	-110	-110	-319	-110	-110	-110	-110	-110	-110		-110	-110	-110	-110	-110	-110	-110	-110	-110
xm	346	280	324	336	271	244	351	362	148	120	323	260	418	353	351	259	359	301	235	297	331	325	474	323	287	309
axδ	8	-13	16	-30	10	39	13	-1	2	62	13	-109	13	-7	-16	17	-11	-32	-3	10	15	10	-24	9	7	10
bxδ	-38	-12	39	-27	16	-35	15	0	-1	-11	19	1	-25	-5	-1	-19		-43	-1	3	-3	-7	5	-16	7	-22
it	*AȦẠ*	*BḂḄ*	*CĊÇ*	*DḊḌ*	*EĖẸ*	*FḞḞ*	*GĠĢ*	*HḢḤ*	*IİỊ*	*JJ̇J̣*	*KḲḴ*	*LĿḶ*	*MṀṂ*	*NṄṆ*	*OȮỌ*	*PṖṔ*	*QQ̇Ǫ*	*RṘṚ*	*SṠṢ*	*TṪṰ*	*UU̇Ụ*	*VV̇Ṿ*	*WẆẈ*	*XẊẌ*	*YẎỴ*	*ZŻẔ*
aa	c	c	c	c	c	c	c	c	c	a-21	c	c	c	c	c	c	c	c	c	c	c	c	c	c	c	c
ba	b	b	b	b	b	b	b	b	b	d+32	b	b	b	b	b	b	d+23	b	b	b	b	b	b	b	b	b
ax	505	447	500	396	413	406	513	481	316	392	466	389	557	476	515	426	498	419	402	457	486	481	610	482	435	450
ay	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850
bx	308	238	333	279	269	131	345	320	127	109	292	252	363	328	310	140		248	204	277	308	263	439	287	270	267
by	-108	-108	-110	-107	-107	-111	-110	-107	-109	-319	-110	-110	-110	-109	-113	-110		-111	-111	-109	-109	-110	-107	-107	-109	-107
axm	445	401	491	467	393	394	506	510	315	393	482	380	557	512	508	406	529	407	383	494	553	531	699	479	505	467
axδ	59	45	8	-71	19	11	6	-29	0	-1	-16	8	0	-36	6	19	-31	11	18	-37	-67	-50	-89	2	-70	-17
bxm	241	197	287	263	189	190	302	306	111	189	278	176	353	308	304	202	325	203	179	290	349	327	495	275	301	263
bxδ	66	40	45	15	79	-59	42	13	15	-80	13	75	9	19	5	-62		44	24	-13	-41	-64	-56	11	-31	3
sb	**AȦẠ**	**BḂḄ**	**CĊÇ**	**DḊḌ**	**EĖẸ**	**FḞḞ**	**GĠĢ**	**HḢḤ**	**IİỊ**	**JJ̇J̣**	**KḲḴ**	**LĿḶ**	**MṀṂ**	**NṄṆ**	**OȮỌ**	**PṖṔ**	**QQ̇Ǫ**	**RṘṚ**	**SṠṢ**	**TṪṰ**	**UU̇Ụ**	**VV̇Ṿ**	**WẆẈ**	**XẊẌ**	**YẎỴ**	**ZŻẔ**
aa	c	c	c	c	c	c	c	c	c	c	c	c	c	c	c	c	c	c	c	a-25	c	c	c	c	c	c
ba	b	b	b	b	b	b	b	b	b	d+50	b	b	b	b	b	b	d	b	b	b	b	b	b	b	b	b
ax										203					363		363	304	252	298		331	493	364	316	321
ay										806					802		802	808	802	802		805	805	804	805	803
bx								362		109								292	234	301		303	470	318	316	308
by								-110		-319								-111	-111	-107		-108	-107	-107	-109	-107
xm	349	299	332	351	283	268	368	362	160	131	340	267	427	354	365	277	365	320	237	299	339	318	489	354	308	309
axδ										72					-2		-2	-16	14	-1		12	4	9	7	12
bxδ								0		-22								-28	-3	2		-15	-19	-36	7	-1
si	***AȦẠ***	***BḂḄ***	***CĊÇ***	***DḊḌ***	***EĖẸ***	***FḞḞ***	***GĠĢ***	***HḢḤ***	***IİỊ***	***JJ̇J̣***	***KḲḴ***	***LĿḶ***	***MṀṂ***	***NṄṆ***	***OȮỌ***	***PṖṔ***	***QQ̇Ǫ***	***RṘṚ***	***SṠṢ***	***TṪṰ***	***UU̇Ụ***	***VV̇Ṿ***	***WẆẈ***	***XẊẌ***	***YẎỴ***	***ZŻẔ***
aa	c	c	c	c	c	c	c	c	c	a-19	c	c	c	c	c	c	c	c	c	a-25	c	c	c	c	c	c
ba	b	b	b	b	b	b	b	b	b	d+68	b	b	b	b	b	b	d	b	b	b	b	b	b	b	b	b
ax	354	267	340	306	273	276	365	361	150	432	336	149	431	346	335	316	348	269	232	307	346	336	450	332	295	320
ay	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850
bx	308	268	363	279	279	201	367	365	147	109	342	260	393	348	350	240		258	234	300	328	318	479	307	295	287
by	-110	-110	-110	-110	-110	-110	-110	-110	-110	-319	-110	-110	-110	-110	-110	-110		-110	-110	-110	-110	-110	-110	-110	-110	-110
axm	470	405	485	474	412	411	512	552	351	408	528	419	557	560	542	414	563	416	370	486	578	525	689	494	514	456
axδ	-116	-138	-145	-168	-139	-135	-147	-191	-201	23	-192	-270	-126	-214	-207	-98	-215	-147	-138	-179	-232	-189	-239	-162	-219	-136
bxm	275	209	289	278	216	216	316	357	156	213	333	223	362	365	346	218	367	221	175	290	383	330	494	299	319	261
bxδ	32	58	73	0	62	-15	50	7	-9	-104	8	36	30	-17	3	21		36	58	9	-55	-12	-15	7	-24	25

Table 2: Font metrics for a–z

hex	0061	0062	0063	0064	0065	0066	0067	0068	0069	006A	006B	006C	006D	006E	006F	0070	0071	0072	0073	0074	0075	0076	0077	0078	0079	007A
cat	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll
reg	aȧạ	bḃḅ	cċç	dḋḍ	eėẹ	fḟ	gġg	hḣḥ	iiį	jjj	kḳķ	lŀļ	mṁṃ	nṅṇ	oȯọ	pṗp	qq̇q	rṙṛ	sṡṣ	tṫṭ	uu̇ụ	vv̇ṿ	wẘẉ	xẋx	yẏy	zżẓ
aa	x	a	x	a	x	a	x+33	a	a-51	a-51	a	a	x	x	x	x	x	x	x	a-116	x	x	x	x	x	x
ba	b	b	b	b	b	b	d	b	b	d+9	b	b	b	b	b	d	d	b	b	b	b	b	b	b	d	b
ax	209		242	218	243		231	299	137		306	135	392	264	238	271	238	191	206		255	256	345	256	262	220
ay	645		645	645	645		645	645	795		784	885	645	646	645	646	645	646	645		645	646	646	646	645	646
bx	196	234	236	221	234	132	229	272	136	109	290	135	408	265	243	179	348	135	196	163	239	241	352	224	242	207
by	-110	-110	-110	-110	-110	-110	-319	-110	-110	-319	-214	-110	-110	-110	-110	-319	-319	-110	-110	-110	-110	-110	-110	-110	-319	-110
xm	245	231	217	270	222	208	256	270	142	75	261	134	401	275	252	242	268	190	199	163	271	249	373	245	260	220
axδ	-36		24	-52	21		-25	28	-5		45	1	-9	-11	-14	28	-30	1	7		-16	7	-28	11	2	0
bxδ	-49	2	18	-49	12	-76	-27	1	-6	33	29	1	6	-10	-9	-63	79	-55	-3	0	-32	-8	-21	-21	-18	-13
it	*aȧạ*	*bḃḅ*	*cċç*	*dḋḍ*	*eėẹ*	*fḟ*	*gġg*	*hḣḥ*	*iiį*	*jjj*	*kḳķ*	*lŀļ*	*mṁṃ*	*nṅṇ*	*oȯọ*	*pṗp*	*qq̇q*	*rṙṛ*	*sṡṣ*	*tṫṭ*	*uu̇ụ*	*vv̇ṿ*	*wẘẉ*	*xẋx*	*yẏy*	*zżẓ*
aa	x	a	x	a	x	a	x	a	a-80	a-80	a	a	x	x	x	x+62	x+17	x	x	x+132	x	x	x	x	x	x+15
ba	b	b	b	b	b	d	d	b	b	d+36	b	b	b	b	b	d	d	b	b	b	b	b	b	b	d	b
ax	341		343	218	332		360	409			405	292	522	394	359	411	369	312	327		386		348	366	367	360
ay	647		645	645	645		645	645			646	890	645	646	645	646	645	646	645		646		646	646	644	646
bx	174	214	163	221	158	222	197	229	136	0	223	115	382	235	186	248	169	113	169	133	226		355	224	368	187
by	-108	-108	-113	-107	-108	-111	-319	-111	-107	-319	-107	-107	-105	-111	-107	-110	-110	-110	-111	-107	-111		-110	-110	-107	-111
axm	376	411	344	430	345	362	377	424	274	204	414	318	540	396	363	355	409	350	309	311	402	392	500	377	414	351
axδ	-35		-1	-212	-13		-17	-15			-9	-26	-18	-2	-4	55	-40	-38	17		-16		-152	-11	-47	8
bxm	216	198	184	217	184	150	216	211	113	43	201	105	380	235	202	194	248	190	149	150	242	231	339	216	253	191
bxδ	-42	15	-21	3	-26	71	-19	17	22	-43	21	9	1	0	-16	53	-79	-77	19	-17	-16		15	7	114	-4
sb	**aȧạ**	**bḃḅ**	**cċç**	**dḋḍ**	**eėẹ**	**fḟ**	**gġg**	**hḣḥ**	**iiį**	**jjj**	**kḳķ**	**lŀļ**	**mṁṃ**	**nṅṇ**	**oȯọ**	**pṗp**	**qq̇q**	**rṙṛ**	**sṡṣ**	**tṫṭ**	**uu̇ụ**	**vv̇ṿ**	**wẘẉ**	**xẋx**	**yẏy**	**zżẓ**
aa	x	a	x	a	x	a	x+33	a	a-48	a-48	a	a	x	x	x	x	x	x	x	a-103	x	x	x	x	x	x
ba	b	b-13	b	b	b	b	d	b	b	d	b	b	b	b	b	d	d	b	b	b	b	b	b	b	d	b
ax							231		137			142	396	264							264					
ay							645		795			885	645	646							646					
bx							229	277	136	109		135	396	265						163	248					
by							-319	-110	-107	-319		-110	-105	-111						-107	-111					
xm	255	235	224	273	232	225	263	275	155	78	261	148	408	289	259	254	273	212	201	168	275	262	384	260	264	221
axδ							-32		-18			-6	-12	-25							-11					
bxδ							-34	2	-19	31		-13	-12	-24						-5	-27					
si	***aȧạ***	***bḃḅ***	***cċç***	***dḋḍ***	***eėẹ***	***fḟ***	***gġg***	***hḣḥ***	***iiį***	***jjj***	***kḳķ***	***lŀļ***	***mṁṃ***	***nṅṇ***	***oȯọ***	***pṗp***	***qq̇q***	***rṙṛ***	***sṡṣ***	***tṫṭ***	***uu̇ụ***	***vv̇ṿ***	***wẘẉ***	***xẋx***	***yẏy***	***zżẓ***
aa	x	a	x	a	x	a	x	a	a-67	a-67	a	a	x	x	x	x+36	x+24	x	x	a-125	x	x	x	x	x	x+23
ba	b	b	b	b	b	d	d	b	b	d+25	b	b	b	b	b	d	d	b	b	b	b	b	b	b	d	b
ax	379		242	218	243		291	469	137		306	174	392	264	238	271	238	191	326		255	256	345		262	220
ay	645		645	645	645		645	645	795		784	890	645	646	645	646	645	646	645		645	646	646		645	646
bx	196	234	236	221	234	132	199	272	136	67	290	135	400	265	243	179	348	135	196	133	239	241	352		242	207
by	-110	-110	-110	-110	-110	-110	-319	-110	-110	-319	-214	-110	-110	-110	-110	-319	-319	-110	-110	-107	-110	-110	-110		-319	-110
axm	404	414	364	447	364	353	380	449	289	220	450	337	560	432	381	364	421	383	332	333	411	398	505	385	392	353
axδ	-25		-122	-229	-121		-89	19	-152		-144	-163	-168	-168	-143	-93	-183	-192	-6		-156	-142	-160		-130	-133
bxm	250	211	210	243	210	149	226	246	135	66	247	134	406	278	227	211	267	229	179	179	257	244	352	231	238	200
bxδ	-54	22	25	-22	23	-17	-27	25	0	0	42	0	-6	-13	15	-32	80	-94	16	-46	-18	-3	0		3	6

Table 3: Font metrics for Latin and IPA 1/4

hex	00C6	00D0	00D8	00DE	00DF	00E6	00F0	00F8	00FE	0110	0111	0126	0127	0131	0141	0142	0152	0153	017F	0181	0182	0186	0187	0189	018A	018B	018E	018F	0190	0191	0192	0193
cat	Lu	Lu	Lu	Lu	Ll	Ll	Ll	Ll	Ll	Lu	Ll	Lu	Ll	Ll	Lu	Ll	Lu	Ll	Ll	Lu	Lu	Lu	Lu	Lu	Lu	Lu	Lu	Lu	Lu	Lu	Ll	Lu
reg	ÆǼÆ̨	ĐĐ́Đ̨	ØǾØ̨	ÞÞ́Þ̨	ßßß̨	æǽæ̨	ðð́ð̨	øǿø̨	þþ́þ̨	ĐĐ́Đ̨	đđ́đ̨	ĦĦ́Ħ̨	ħħ́ħ̨	ıíı̨	ŁŁ́Ł̨	łł́ł̨	ŒŒ́Œ̨	œœ́œ̨	ſſ́ſ̨	ƁƁ́Ɓ̨	ƂƂ́Ƃ̨	ƆƆ́Ɔ̨	ƇƇ́Ƈ̨	ƉƉ́Ɖ̨	ƊƊ́Ɗ̨	ƋƋ́Ƌ̨	ƎƎ́Ǝ̨	ƏƏ́Ə̨	ƐƐ́Ɛ̨	ƑƑ́Ƒ̨	ƒƒ́ƒ̨	ƓƓ́Ɠ̨	ƔƔ́Ɣ̨
aa	c	c	c	c	a	x	a	x+25	a	c	a	c	a	x	c	a	c	x	a	c	c	c	a+45	c	c	c	c	c	c	c	a	a+45
ba	b	b	b-13	b	b	b	b	b-28	d	b	b	b	b	b	b	b	b	b	b	b	b	b	b	b	b	b	b	b	b	d+60	d	b
ax	480	341	351	281		350			278					132			460	405		327					306							
ay	850	850	850	850		645			646					645			850	645		805					805							
bx	549		350			327		243						139						328					369							
by	-107		-110			-109		-110						-108						-108					-107							
xm	426	333	351	249	259	343	240	250	236	336	270	362	270	142	256	133	436	387	190	303	273	322	369	333	368	292	280	326	256	183	201	371
axδ	53	8	0	31		6			41					-10			24	17		24					-62							
bxδ	122		-1			-16		-7						-3						25					0							
it	*ÆǼÆ̨*	*ĐĐ́Đ̨*	*ØǾØ̨*	*ÞÞ́Þ̨*	*ßßß̨*	*æǽæ̨*	*ðð́ð̨*	*øǿø̨*	*þþ́þ̨*	*ĐĐ́Đ̨*	*đđ́đ̨*	*ĦĦ́Ħ̨*	*ħħ́ħ̨*	*ıíı̨*	*ŁŁ́Ł̨*	*łł́ł̨*	*ŒŒ́Œ̨*	*œœ́œ̨*	*ſſ́ſ̨*	*ƁƁ́Ɓ̨*	*ƂƂ́Ƃ̨*	*ƆƆ́Ɔ̨*	*ƇƇ́Ƈ̨*	*ĐĐ́Đ̨*	*ƊƊ́Ɗ̨*	*ƋƋ́Ƌ̨*	*ƎƎ́Ǝ̨*	*ƏƏ́Ə̨*	*ƐƐ́Ɛ̨*	*ƑƑ́Ƒ̨*	*ƒƒ́ƒ̨*	*ƓƓ́Ɠ̨*	*ƔƔ́Ɣ̨*
aa	c	c	c+25	c	a	x	a+6	x+48	a	c	a	c	a	x	c	a	c	x	a	c	c	c+14	a+45	c	c	c	c	c	c	c	a	a+45
ba	b	b	b-43	b	d	b	b	b-48	d-7	b	b	b	b	b	b	b	b	b	d	b	b	b	b	b	b	b	b	b	b	d+60	d+60	b
ax	615					480			278					267			575	480		327					306							
ay	850					645			646					645			850	645		805					805							
bx	549		310			327		186						139						328					369							
by	-107		-113			-109		-107						-108						-108					-107							
axm	545	467	525	386	384	486	414	374	387	467	449	510	424	288	380	315	593	490	357	459	426	498	595	487	545	468	489	523	419	358	401	573
axδ	69					-6			-109					-21			-18	-10		-132					-239							
bxm	341	263	320	182	172	325	201	214	175	263	236	306	211	128	176	102	389	330	144	255	222	294	382	283	341	264	285	319	214	153	188	360
bxδ	207		-10			1		-28						10						72					27							
sb	**ÆǼÆ̨**	**ĐĐ́Đ̨**	**ØǾØ̨**	**ÞÞ́Þ̨**	**ßßß̨**	**æǽæ̨**	**ðð́ð̨**	**øǿø̨**	**þþ́þ̨**	**ĐĐ́Đ̨**	**đđ́đ̨**	**ĦĦ́Ħ̨**	**ħħ́ħ̨**	**ıíı̨**	**ŁŁ́Ł̨**	**łł́ł̨**	**ŒŒ́Œ̨**	**œœ́œ̨**	**ſſ́ſ̨**	**ƁƁ́Ɓ̨**	**ƂƂ́Ƃ̨**	**ƆƆ́Ɔ̨**	**ƇƇ́Ƈ̨**	**ĐĐ́Đ̨**	**ƊƊ́Ɗ̨**	**ƋƋ́Ƌ̨**	**ƎƎ́Ǝ̨**	**ƏƏ́Ə̨**	**ƐƐ́Ɛ̨**	**ƑƑ́Ƒ̨**	**ƒƒ́ƒ̨**	**ƓƓ́Ɠ̨**	**ƔƔ́Ɣ̨**
aa	c	c	c+25	c	a	x	a+19	x+41	a	c	a	c	a	x	c	a	c	x	a	c	c	c	a+45	c	c	c	c	c	c	c	a	a+45
ba	b	b	b-45	b	b	b	b	b-34	d	b	b	b	b	b	b	b	b	b	b	b	b	b	b	b	b	b	b	b	b	d+36	d+62	b
ax	510					355								151			450	430				275										
ay	850					645								645			850	645				833										
bx														135																		
by														-108																		
xm	446	349	365	287	284	362	256	259	260	349	273	362	275	155	258	142	432	401	217	316	267	313	368	349	381	292	302	353	269	236	221	373
axδ	64					-7								-4			18	28				-38										
bxδ														-20																		
si	*ÆǼÆ̨*	*ĐĐ́Đ̨*	*ØǾØ̨*	*ÞÞ́Þ̨*	*ßßß̨*	*æǽæ̨*	*ðð́ð̨*	*øǿø̨*	*þþ́þ̨*	*ĐĐ́Đ̨*	*đđ́đ̨*	*ĦĦ́Ħ̨*	*ħħ́ħ̨*	*ıíı̨*	*ŁŁ́Ł̨*	*łł́ł̨*	*ŒŒ́Œ̨*	*œœ́œ̨*	*ſſ́ſ̨*	*ƁƁ́Ɓ̨*	*ƂƂ́Ƃ̨*	*ƆƆ́Ɔ̨*	*ƇƇ́Ƈ̨*	*ĐĐ́Đ̨*	*ƊƊ́Ɗ̨*	*ƋƋ́Ƌ̨*	*ƎƎ́Ǝ̨*	*ƏƏ́Ə̨*	*ƐƐ́Ɛ̨*	*ƑƑ́Ƒ̨*	*ƒƒ́ƒ̨*	*ƓƓ́Ɠ̨*	*ƔƔ́Ɣ̨*
aa	c	c	c+25	c	a	x	a+19	x+22	a	c	a	c	a	x	c	a	c	x	a+13	c	c	c	a+45	c	c	c	c	c	c	c	a	a+45
ba	b	b	b-40	b	d	b	b	b-30	d	b	b	b	b	b	b	b	b	b	d-9	b	b	b	b	b	b	b	b	b	b	d+36	d+62	b
ax	500	341	351	281		350			278					132			480	415		327					306							
ay	850	850	850	850		645			646					645			850	645		805					805							
bx	549		350			327		243						139						328					369							
by	-107		-110			-109		-110						-108						-108					-107							
axm	568	474	526	414	390	500	426	398	396	474	472	564	449	311	431	344	620	544	367	483	404	488	583	509	562	471	489	513	426	401	408	579
axδ	-68	-133	-175	-133		-150			-118					-179			-140	-129		-156					-256							
bxm	372	278	331	218	186	346	223	244	193	278	269	369	246	157	235	141	425	390	163	288	208	292	379	313	367	275	293	318	231	206	204	376
bxδ	176		18			-19		-1						-18						39					1							

Table 4: Font metrics for Latin and IPA 2/4

hex	0194	0196	0197	0198	019C	019D	019F	01B7	0237	0250	0251	0252	0253	0254	0255	0256	0257	0258	0259	025A	025B	025C	025D	025E	025F	0260	0261	0262	0263	0264	0265	0266
cat	Lu	Lu	Lu	Lu	Lu	Lu	Lu	Lu	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll
reg	ƔƳƴ	lll	ꟾ	ꞞꞢꞠ	ꟿꟽꟽ	ꞐꞑꞒ	ꝊꝎꝐ	ȝȝȝ	jjȷ	ɐɐɐ	ɑɑɑ	ɒɒɒ	ɓɓɓ	ɔɔɔ	ɕɕɕ	ɖɖɖ	ɗɗɗ	ɘɘɘ	ɚɚɚ	ɚɚɚ	ɛɛɛ	ɜɜɜ	ʩʩʩ	ɞɞɞ	ɟɟɟ	ɠɠɠ	ɡɡɡ	ɢɢɢ	ɣɣɣ	ɤɤɤ	ɥɥɥ	ɦɦɦ
aa	c	a	c	c	c	c	c	c+22	x	x	x	x	a	x	x	a	a	x	x	x+45	x	x	a-90	x	x	a	x	x	x	x	x	a
ba	d	b	b	b	b	d+45	b	d+21	d+9	b	b	b	b	b	b-106	d+44	b	b	b	b	b	b	b	b	d+26	d	d	b	d+30	b	d	b
ax	331				431				140	228	228	264	480	216	209	218	218	204	195	229	209	194	209	239	169	209	239	265	259	248	268	299
ay	805				801				645	645	645	645	630	645	645	645	645	645	645	645	645	645	645	645	645	645	645	645	645	645	645	565
bx					450		350	272		229	212	259	212	177	212	212	212	212	186	212	212	184	212	256	196	236	226	248	251	216	219	240
by					-108		-110	-319		-114	-114	-114	-114	-114	-114	-114	-114	-114	-114	-114	-114	-114	-114	-114	-110	-110	-320	-110	-277	-114	-129	-114
xm	308	151	148	339	433	324	351	303	74	216	272	247	265	215	222	314	322	220	232	306	201	195	234	241	124	320	228	259	258	221	271	262
axδ	22				-2				66	11	-44	17	215	0	-13	-96	-104	-16	-37	-77	7	-1	-25	-2	44	-111	11	6	1	27	-3	36
bxδ					16		-1	-31		12	-60	12	-53	-38	-10	-102	-110	-8	-46	-94	10	-11	-22	14	71	-84	-2	-11	-7	-5	-52	-22
it	*ƳƳƴ*	*lll*	*ꟾ*	*ꞞꞢꞠ*	*ꟿꟽꟽ*	*ꞐꞑꞒ*	*ꝊꝎꝐ*	*ȝȝȝ*	*jjȷ*	*ɐɐɐ*	*ɑɑɑ*	*ɒɒɒ*	*ɓɓɓ*	*ɔɔɔ*	*ɕɕɕ*	*ɖɖɖ*	*ɗɗɗ*	*ɘɘɘ*	*ɚɚɚ*	*ɚɚɚ*	*ɛɛɛ*	*ɜɜɜ*	*ʩʩʩ*	*ɞɞɞ*	*ɟɟɟ*	*ɠɠɠ*	*ɡɡɡ*	*ɢɢɢ*	*ɣɣɣ*	*ɤɤɤ*	*ɥɥɥ*	*ɦɦɦ*
aa	c	a	c	c	c	c	c	c+22	x	x+16	x	x	a+6	x	x	a+9	a	x	x	x+44	x	x	x+132	x+17	x	a+6	x+17	x	x	x	x	a
ba	d+20	b	b	b	b	d+45	b	d+21	d+36	b	b	b	b	b	b-100	d+44	b	b	b	b	b	b	b	b-25	d	d+31	d+31	b	d+45	b+2	d	b
ax	331				431				140					322		218	218	204			322							265			258	299
ay	805				801				645					654		645	645	645			654							645			645	565
bx					450					229	212	243	212	172	212	212	212	212	212	212	212	184	212	256				248		177	219	240
by					-108					-114	-114	-114	-114	-114	-114	-114	-114	-114	-114	-114	-114	-114	-114	-114				-114		-114	-129	-114
axm	527	318	315	486	649	500	517	473	208	389	388	394	401	327	354	459	512	351	359	459	344	319	392	384	273	537	390	382	456	388	429	425
axδ	-196				-218				-68					-5		-241	-294	-147			-22							-117			-171	-126
bxm	322	105	111	282	445	296	313	269	47	229	227	233	188	166	194	246	299	191	199	299	183	158	232	224	113	325	230	221	296	227	268	212
bxδ					4					0	-15	9	23	5	17	-34	-87	20	12	-87	28	25	-20	31				26		-50	-49	27
sb	**ƳƳƴ**	**lll**	**ꟾ**	**ꞞꞢꞠ**	**ꟿꟽꟽ**	**ꞐꞑꞒ**	**ꝊꝎꝐ**	**ȝȝȝ**	**jjȷ**	**ɐɐɐ**	**ɑɑɑ**	**ɒɒɒ**	**ɓɓɓ**	**ɔɔɔ**	**ɕɕɕ**	**ɖɖɖ**	**ɗɗɗ**	**ɘɘɘ**	**ɚɚɚ**	**ɚɚɚ**	**ɛɛɛ**	**ɜɜɜ**	**ʩʩʩ**	**ɞɞɞ**	**ɟɟɟ**	**ɠɠɠ**	**ɡɡɡ**	**ɢɢɢ**	**ɣɣɣ**	**ɤɤɤ**	**ɥɥɥ**	**ɦɦɦ**
aa	c	a	c	c	c	c	c	a-24	x	x	x	x	a	x	x	a	a	x	x	x+50	x	x	a-128	x	x	a+6	x	x+23	x	x	x	a
ba	d+22	b	b	b	b	d+47	b	d+24	d	b-14	b	b	b	b-14	b-101	d+46	b	b	b	b	b	b	b	b	d	d+33	d	b	d+32	b	d	b
ax									163					199							229											
ay									645					645							645											
bx								272		228				184					212													
by								-319		-114				-114					-114													
xm	293	169	158	337	444	336	365	315	78	207	276	251	268	199	232	322	323	230	228	319	212	215	276	249	144	336	235	265	272	202	277	274
axδ									85					0							16											
bxδ								-43		21				-15					-16													
si	***ƳƳƴ***	***lll***	***ꟾ***	***ꞞꞢꞠ***	***ꟿꟽꟽ***	***ꞐꞑꞒ***	***ꝊꝎꝐ***	***ȝȝȝ***	***jjȷ***	***ɐɐɐ***	***ɑɑɑ***	***ɒɒɒ***	***ɓɓɓ***	***ɔɔɔ***	***ɕɕɕ***	***ɖɖɖ***	***ɗɗɗ***	***ɘɘɘ***	***ɚɚɚ***	***ɚɚɚ***	***ɛɛɛ***	***ɜɜɜ***	***ʩʩʩ***	***ɞɞɞ***	***ɟɟɟ***	***ɠɠɠ***	***ɡɡɡ***	***ɢɢɢ***	***ɣɣɣ***	***ɤɤɤ***	***ɥɥɥ***	***ɦɦɦ***
aa	c	a	c	c	c	c	c	a-25	x	x	x	x	a	x	x	a	a	x	x	x+50	x	x	a-128	x	x	a	x	x+23	x	x	x	a
ba	d+22	b	b	b	b	d+47	b	d+24	d+25	b	b	b	b	b	b-101	d+46	b	b	b	b	b	b	b	b	d	d+33	d	b	d+32	b	d	b
ax	331				431				140		228	264		160		218	218	204			209	194						265			258	299
ay	805				801				645		645	645		645		645	645	645			645	645						645			645	565
bx					450			236		229	212	259	212	177	212	212	212	212		212	212	184	212	256				248		177	219	240
by					-108			-319		-114	-114	-114	-114	-114	-114	-114	-114	-114		-114	-114	-114	-114	-114				-110		-114	-129	-114
axm	533	337	334	494	655	506	543	467	243	377	397	404	408	325	388	468	527	353	365	466	352	337	408	395	274	544	397	397	476	386	479	387
axδ	-202				-224				-103		-169	-140		-165		-250	-309	-149			-143	-143						-132			-221	-88
bxm	338	134	139	298	460	311	347	271	89	224	243	250	204	171	235	264	324	199	212	312	198	183	254	241	120	341	244	243	322	233	325	184
bxδ					-10			-35		4	-31	8	7	5	-23	-52	-112	12		-100	13	0	-42	14				4		-56	-106	55

Table 5: Font metrics for Latin and IPA 3/4

hex	0267	0268	0269	026A	026B	026C	026D	026E	026F	0270	0271	0272	0273	0274	0275	0277	0278	0279	027A	027B	027C	027D	027E	027F	0280	0281	0282	0283	0284	0285	0286	0287
cat	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll
reg	ɧɧɧ	ɨɨɨ	ᵢ̈ᵢ̈ᵢ̈	ɪ̈ɪ̈ɪ̈	ɪ̈ɪ̈ɪ̈	ɪ̈ɪ̈ɪ̈	ɪ̈ɪ̈ɪ̈	ɪ̈ɪ̈ɪ̈	ɪ̈ɪ̈ɪ̈	ɯ̈ɯ̈ɯ̈	ɯ̈ɯ̈ɯ̈	ɯ̈ɯ̈ɯ̈	ɯ̈ɯ̈ɯ̈	ɴ̈ɴ̈ɴ̈	ɵ̈ɵ̈ɵ̈	ɷ̈ɷ̈ɷ̈	ɸ̈ɸ̈ɸ̈	ɾ̈ɾ̈ɾ̈	ɾ̈ɾ̈ɾ̈	ɾ̈ɾ̈ɾ̈	ɭ̈ɭ̈ɭ̈	ɭ̈ɭ̈ɭ̈	ɻ̈ɻ̈ɻ̈	ɻ̈ɻ̈ɻ̈	ʁ̈ʁ̈ʁ̈	ʁ̈ʁ̈ʁ̈	ʂ̈ʂ̈ʂ̈	ʃ̈ʃ̈ʃ̈	ʃ̈ʃ̈ʃ̈	ɳ̈ɳ̈ɳ̈	ʃ̈ʃ̈ʃ̈	ʃ̈ʃ̈ʃ̈	ʈ̈ʈ̈ʈ̈
aa	a	a-51	x	x	a	a	a	a	x	x	x	x	x	x	x	x	a	x	a-51	x	x	x	x	x	x	x	x	a	a	x+121	a	x
ba	d+11	b	b	b	b	b	d+44	d	b-13	d	d+11	d+11	d+11	b	b	b	d	b-13	b-13	d+58	d	d	b	b	b	b	d+56	d+18	d+18	d+18	d+7	d+79
ax	299	155	123	133	209	209	209	209	407	209	383	264	263	209	209	209	209	209	209		222	209	209	209	209	209	209	209	209	209	209	209
ay	565	795	645	645	645	645	645	645	645	645	645	646	645	645	645	645	645	645	645		703	645	645	645	645	645	645	645	645	645	645	645
bx	240	134	153	139	146	218	196	196	410	294	383	265	262	247	196	341	196	190	203	196	127	196	196	196	196	196	196	196	196	196	196	196
by	-114	-114	-114	-114	-114	-120	-110	-110	-114	-114	-105	-111	-112	-114	-110	-114	-110	-114	-114	-110	-305	-110	-110	-110	-110	-110	-110	-110	-110	-110	-110	-110
xm	227	149	170	136	144	182	182	250	390	379	365	234	295	278	252	357	285	177	179	225	186	186	174	170	238	238	199	140	140	170	149	150
axδ	71	5	-47	-3	65	27	27	-41	16	-170	17	29	-32	-69	-43	-148	-76	31	29		35	22	34	38	-29	-29	10	68	68	38	59	59
bxδ	12	-15	-17	3	2	36	14	-54	19	-85	17	30	-33	-31	-56	-16	-89	12	23	-29	-59	9	21	25	-42	-42	-3	55	55	25	46	46
it	*ɧɧɧ*	*ɨɨɨ*	*ᵢ̈ᵢ̈ᵢ̈*	*ɪ̈ɪ̈ɪ̈*	*ɪ̈ɪ̈ɪ̈*	*ɪ̈ɪ̈ɪ̈*	*ɪ̈ɪ̈ɪ̈*	*ɪ̈ɪ̈ɪ̈*	*ɯ̈ɯ̈ɯ̈*	*ɯ̈ɯ̈ɯ̈*	*ɯ̈ɯ̈ɯ̈*	*ɯ̈ɯ̈ɯ̈*	*ɯ̈ɯ̈ɯ̈*	*ɴ̈ɴ̈ɴ̈*	*ɵ̈ɵ̈ɵ̈*	*ɷ̈ɷ̈ɷ̈*	*ɸ̈ɸ̈ɸ̈*	*ɾ̈ɾ̈ɾ̈*	*ɾ̈ɾ̈ɾ̈*	*ɾ̈ɾ̈ɾ̈*	*ɭ̈ɭ̈ɭ̈*	*ɭ̈ɭ̈ɭ̈*	*ɻ̈ɻ̈ɻ̈*	*ɻ̈ɻ̈ɻ̈*	*ʁ̈ʁ̈ʁ̈*	*ʁ̈ʁ̈ʁ̈*	*ʂ̈ʂ̈ʂ̈*	*ʃ̈ʃ̈ʃ̈*	*ʃ̈ʃ̈ʃ̈*	*ɳ̈ɳ̈ɳ̈*	*ʃ̈ʃ̈ʃ̈*	*ʃ̈ʃ̈ʃ̈*	*ʈ̈ʈ̈ʈ̈*
aa	a	a-80	x	x	a	a	a	a	x	x	x+21	x+19	x	x	x+14	x	a	x	a-71	x	x+19	x+17	x	x	x	x	x+15	a+10	a+10	a+10	a+10	x
ba	d	b	b	b	b	b	d+30	d+8	b-15	d	d	d	d	b	b	b-18	d+32	b	b-19	d+58	d+34	d+38	b	b	b	b	d+52	d+18	d+18	d+18	d+34	d+78
ax			123	128					407		383	264	263								222											
ay			645	645					645		645	646	645								703											
bx		134	153	139	146	218			410	294	383	265	262	247		341		190	203		127											
by		-114	-114	-114	-114	-120			-114	-114	-105	-111	-112	-114		-114		-114	-114		-305											
axm	437	276	269	277	326	356	329	423	565	589	502	324	409	420	380	490	478	189	340	343	327	375	308	306	337	351	326	333	333	334	341	277
axδ			-146	-149					-158		-119	-60	-146								-105											
bxm	224	115	109	116	113	143	116	210	404	428	341	163	249	259	220	330	265	29	179	182	167	215	148	145	177	191	165	121	121	122	128	116
bxδ		18	43	22	32	74			5	-134	41	101	12	-12		10		160	23		-40											
sb	**ɧɧɧ**	**ɨɨɨ**	**ᵢ̈ᵢ̈ᵢ̈**	**ɪ̈ɪ̈ɪ̈**	**ɪ̈ɪ̈ɪ̈**	**ɪ̈ɪ̈ɪ̈**	**ɪ̈ɪ̈ɪ̈**	**ɪ̈ɪ̈ɪ̈**	**ɯ̈ɯ̈ɯ̈**	**ɯ̈ɯ̈ɯ̈**	**ɯ̈ɯ̈ɯ̈**	**ɯ̈ɯ̈ɯ̈**	**ɯ̈ɯ̈ɯ̈**	**ɴ̈ɴ̈ɴ̈**	**ɵ̈ɵ̈ɵ̈**	**ɷ̈ɷ̈ɷ̈**	**ɸ̈ɸ̈ɸ̈**	**ɾ̈ɾ̈ɾ̈**	**ɾ̈ɾ̈ɾ̈**	**ɾ̈ɾ̈ɾ̈**	**ɭ̈ɭ̈ɭ̈**	**ɭ̈ɭ̈ɭ̈**	**ɻ̈ɻ̈ɻ̈**	**ɻ̈ɻ̈ɻ̈**	**ʁ̈ʁ̈ʁ̈**	**ʁ̈ʁ̈ʁ̈**	**ʂ̈ʂ̈ʂ̈**	**ʃ̈ʃ̈ʃ̈**	**ʃ̈ʃ̈ʃ̈**	**ɳ̈ɳ̈ɳ̈**	**ʃ̈ʃ̈ʃ̈**	**ʃ̈ʃ̈ʃ̈**	**ʈ̈ʈ̈ʈ̈**
aa	a	a-67	x	x	a	a	a	a	x-2	x	x	x	x	x	x	x	a	x-2	a-69	x	x	x	x	x	x	x	x	a	a	a	a	x
ba	d	b	b	b	b	b	d	d	b-17	d	d	d	d	b	b	b	d+34	b-17	b	d+60	d+36	d	b	b	b	b	d+53	d+21	d+21	d+21	d	d+67
ax				146					377																							
ay				645					645																							
bx		134		156	146	218			380						229			190														
by		-114		-114	-114	-120			-114						-114			-114														
xm	243	152	164	152	156	190	193	273	354	388	363	245	306	290	259	369	297	144	205	237	205	198	186	182	235	220	201	152	152	152	161	157
axδ				-6					23																							
bxδ		-18		4	-10	27			26						-30			45														
si	***ɧɧɧ***	***ɨɨɨ***	***ᵢ̈ᵢ̈ᵢ̈***	***ɪ̈ɪ̈ɪ̈***	***ɪ̈ɪ̈ɪ̈***	***ɪ̈ɪ̈ɪ̈***	***ɪ̈ɪ̈ɪ̈***	***ɪ̈ɪ̈ɪ̈***	***ɯ̈ɯ̈ɯ̈***	***ɯ̈ɯ̈ɯ̈***	***ɯ̈ɯ̈ɯ̈***	***ɯ̈ɯ̈ɯ̈***	***ɯ̈ɯ̈ɯ̈***	***ɴ̈ɴ̈ɴ̈***	***ɵ̈ɵ̈ɵ̈***	***ɷ̈ɷ̈ɷ̈***	***ɸ̈ɸ̈ɸ̈***	***ɾ̈ɾ̈ɾ̈***	***ɾ̈ɾ̈ɾ̈***	***ɾ̈ɾ̈ɾ̈***	***ɭ̈ɭ̈ɭ̈***	***ɭ̈ɭ̈ɭ̈***	***ɻ̈ɻ̈ɻ̈***	***ɻ̈ɻ̈ɻ̈***	***ʁ̈ʁ̈ʁ̈***	***ʁ̈ʁ̈ʁ̈***	***ʂ̈ʂ̈ʂ̈***	***ʃ̈ʃ̈ʃ̈***	***ʃ̈ʃ̈ʃ̈***	***ɳ̈ɳ̈ɳ̈***	***ʃ̈ʃ̈ʃ̈***	***ʃ̈ʃ̈ʃ̈***	***ʈ̈ʈ̈ʈ̈***
aa	a	a-67	x	x	a	a	a	a	x-3	x	x	x	x	x	x	x	a	x-3	a-69	x	x	x	x	x	x-1	x	x	a	a	a	a	x
ba	d	b	b	b	b	b	d	d	b	d	d	d	d	b	b	b	d+34	b	b	d+60	d+36	d	b	b	b	b	d+52	d+21	d+21	d+21	d	d+67
ax	299	137	123	128					407		383	264	263								222											
ay	565	795	645	645					645		645	646	645								703											
bx	240	136	153	139	146	218			410	294	383	265	262	247		341		190	203		127											
by	-114	-110	-114	-114	-114	-120			-114	-114	-105	-111	-112	-114		-114		-114	-114		-305											
axm	413	276	302	285	337	398	369	431	598	594	510	336	430	428	408	499	477	355	349	351	336	382	316	314	387	360	348	340	340	340	347	265
axδ	-114	-139	-179	-157					-191		-127	-72	-167								-114											
bxm	209	122	148	131	133	195	165	227	402	440	356	182	276	275	254	345	274	160	195	197	182	228	162	160	192	206	194	136	136	136	144	111
bxδ	30	13	4	7	12	22			7	-146	26	82	-14	-28		-4		29	7		-55											

Table 6: Font metrics for Latin and IPA 4/4

hex	0288	0289	028A	028B	028C	028D	028E	028F	0290	0291	0292	0293	0294	0295	0296	0297	0299	029B	029C	029D	029F
cat	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Lu	Lu	Lu	Ll	Ll	Ll	Ll	Ll	Ll
reg	ʈʈ	ʉʉ	ʊʊ	ʋʋ	ʌʌ	ʍʍ	ʎʎ	ʏʏ	ʐʐ	ʑʑ	ʒʒ	ʓʓ	ʔʔ	ʕʕ	ʖʖ	ɕɕ	ʙʙ	ɠɠ	ʜʜ	ʝʝ	ʟʟ
aa	a-114	x	x	x	x	x	a-37	x	x	x+19	x	x	c	c	c	x	x	a-121	x	a-51	x
ba	d+11	b	b	b	b	b	b	b	d+47	d+110	d	d	b	b	b	d+11	b	b	b	d+55	b
ax	209	209	209	259	209	209	209	209	209	209	218	209	209	209	209	209	209	209	209	209	209
ay	645	645	645	645	645	645	645	645	645	645	646	645	645	645	645	645	645	645	645	645	645
bx	196	196	258	251	196	196	196	239	196	196	185	196	202	230	196	196	196	196	305	196	196
by	-110	-110	-114	-114	-110	-110	-110	-107	-110	-110	-319	-110	-107	-107	-110	-110	-110	-110	-110	-110	-110
xm	157	271	253	235	244	367	248	236	262	217	216	220	211	221	208	221	232	294	293	126	207
axδ	51	-62	-44	24	-35	-158	-39	-27	-53	-8	1	-11	-2	-12	1	-12	-23	-85	-84	83	2
bxδ	38	-75	4	16	-48	-171	-52	3	-66	-21	-31	-24	-9	9	-12	-25	-36	-98	12	70	-11
it	*ʈʈ*	*ʉʉ*	*ʊʊ*	*ʋʋ*	*ʌʌ*	*ʍʍ*	*ʎʎ*	*ʏʏ*	*ʐʐ*	*ʑʑ*	*ʒʒ*	*ʓʓ*	*ʔʔ*	*ʕʕ*	*ʖʖ*	*ɕɕ*	*ʙʙ*	*ɠɠ*	*ʜʜ*	*ʝʝ*	*ʟʟ*
aa	a-114	x	x	x	x	x	a-52	x	x+19	x+19	x+15	x	a-26	a-15	c	x	x	a-94	x	a-80	x
ba	d+6	b	b-13	b-16	b	b	b	b	d+41	d+110	d	d-80	b	b	b-23	d	b	b	b	d+63	b
ax				259							328										
ay				645							646										
bx			258	251				239			130		202	230					273		
by			-114	-114				-107			-319		-107	-107					-110		
axm	323	410	404	414	338	462	417	423	360	353	350	370	395	401	356	405	336	435	421	234	313
axδ				-155							-22										
bxm	162	249	244	254	178	301	213	263	200	193	190	210	191	197	152	245	176	274	260	73	153
bxδ			13	-3				-24			-60		10	32					12		
sb	**ʈʈ**	**ʉʉ**	**ʊʊ**	**ʋʋ**	**ʌʌ**	**ʍʍ**	**ʎʎ**	**ʏʏ**	**ʐʐ**	**ʑʑ**	**ʒʒ**	**ʓʓ**	**ʔʔ**	**ʕʕ**	**ʖʖ**	**ɕɕ**	**ʙʙ**	**ɠɠ**	**ʜʜ**	**ʝʝ**	**ʟʟ**
aa	a-103	x	x	x	x	x	a-36	x	x	x+23	x+20	x+15	c	c	c	x	x	a-121	x	a-64	x
ba	d	b	b	b	b	b	b	b	d+41	d+107	d	d-96	b	b	b	d	b	b	b	d+65	b
ax			266								234										
ay			645								645										
bx											193								351		
by											-319								-110		
xm	182	274	267	255	231	355	243	244	254	229	227	236	206	206	206	273	245	290	316	132	219
axδ			-1								7										
bxδ											-34								35		
si	***ʈʈ***	***ʉʉ***	***ʊʊ***	***ʋʋ***	***ʌʌ***	***ʍʍ***	***ʎʎ***	***ʏʏ***	***ʐʐ***	***ʑʑ***	***ʒʒ***	***ʓʓ***	***ʔʔ***	***ʕʕ***	***ʖʖ***	***ɕɕ***	***ʙʙ***	***ɠɠ***	***ʜʜ***	***ʝʝ***	***ʟʟ***
aa	a-103	x	x-1	x	x	x	a-56	x	x+23	x+23	x+24	x+15	c	c	c	x	x	a-94	x+28	a-78	x
ba	d	b	b	b	b	b	b	b	d+38	d+107	d	d-96	b	b	b	d+15	b	b	b	d+65	b
ax				259							218										
ay				645							646										
bx			258	251				239			141		202	230					327		
by			-114	-114				-107			-319		-107	-107					-110		
axm	330	437	455	423	327	453	369	428	386	362	345	378	403	412	363	411	354	461	488	244	322
axδ				-164							-127										
bxm	176	283	259	269	173	299	215	274	232	208	191	224	208	216	168	257	200	307	334	90	168
bxδ			-1	-18				-35			-50		-6	13					-7		

Table 7: Font metrics for above marks

hex	0300	0301	0302	0303	0304	0306	0307	0308	0309	030A	030B	030C	030D	030F	0310	0311	0312	033D	033E	033F	030E	0346	034A	034B	034C	0350	0351	0352	0357	035B
cat	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn
reg	◌̀	◌́	◌̂	◌̃	◌̄	◌̆	◌̇	◌̈	◌̉	◌̊	◌̋	◌̌	◌̍	◌̏	◌̐	◌̑	◌̒	◌̽	◌̾	◌̿	◌̎	◌͆	◌͊	◌͋	◌͌	◌͍	◌͐	◌͑	◌͒	◌͗	◌͛
ax	-142	-191	-139	-149	-209	-161	-180	-210	-160	-184	-219	-205	-155	-140	-251	-205	-110	-265	-265	-235	-245	-251	-215	-205	-235	-201	-171	-251	-171	-164
ay	683	682	697	697	644	681	667	645	765	708	714	676	688	758	675	645	605	605	605	611	719	621	645	645	645	645	645	645	645	645
xm	-159	-159	-140	-149	-211	-161	-180	-209	-161	-182	-169	-203	-154	-167	-251	-208	-101	-260	-247	-240	-244	-255	-217	-217	-238	-183	-155	-251	-141	-174
axδ	17	-31	1	0	2	0	0	0	1	-2	-49	-1	-1	27	0	3	-8	-4	-18	5	0	4	2	12	3	-18	-15	0	-30	10
it	◌̀	◌́	◌̂	◌̃	◌̄	◌̆	◌̇	◌̈	◌̉	◌̊	◌̋	◌̌	◌̍	◌̏	◌̐	◌̑	◌̒	◌̽	◌̾	◌̿	◌̎	◌͆	◌͊	◌͋	◌͌	◌͍	◌͐	◌͑	◌͒	◌͗	◌͛
ax	-3	-52	3	-17	-78	-22	-40	-79	-4	-40	-74	-67	-15	14	-114	-74	13	-142	-142	-111	-99	-125	-84	-74	-104	-70	-40	-120	-40	-33
ay	683	682	697	697	645	681	689	645	765	708	714	676	688	758	675	645	605	605	605	611	719	621	645	645	645	645	645	645	645	645
xm	-32	-33	-24	-22	-97	-26	-55	-94	-13	-48	-41	-68	-26	-34	-116	-104	16	-150	-134	-129	-116	-145	-98	-96	-119	-70	-31	-131	-24	-51
axδ	29	-18	27	5	19	4	15	15	9	8	-33	1	11	48	2	30	-3	8	-8	18	17	20	14	22	15	0	-9	11	-15	18
sb	◌̀	◌́	◌̂	◌̃	◌̄	◌̆	◌̇	◌̈	◌̉	◌̊	◌̋	◌̌	◌̍	◌̏	◌̐	◌̑	◌̒	□	□	□	◌̎	□	□	□	□	□	◌͆	□	◌͗	◌͛
ax	-127	-180	-127	-133	-205	-149					-197	-193	-137	-127	-239	-205					-232									-180
ay	713	713	700	679	625	684					733	676	719	758	675	645					719									639
xm	-144	-144	-127	-136	-199	-149	-168	-185	-148	-169	-157	-192	-137	-152	-235	-220	-101				-232						105		111	-184
axδ	17	-36	0	3	-5	0					-39	0	0	25	-4	15					0									4
si	◌̀	◌́	◌̂	◌̃	◌̄	◌̆	◌̇	◌̈	◌̉	◌̊	◌̋	◌̌	◌̍	◌̏	◌̐	◌̑	◌̒	□	□	□	◌̎	□	□	□	□	□	□	□	□	□
ax	-142	-191	-139	-134	-209	-161	-180	-210	-160	-184	-219	-205	-155	-140	-251	-205	-110				-245									
ay	683	682	697	700	645	681	667	645	765	708	714	676	688	758	675	645	605				719									
xm	-18	-18	-12	-10	-91	-14	-44	-112	-1	-37	-30	-59	-9	-22	-100	-102	26				-104									
axδ	-124	-172	-127	-123	-117	-146	-135	-98	-159	-147	-189	-146	-146	-118	-150	-102	-136				-140									

Table 8: Font metrics for below marks

hex	0316	0317	0318	0319	031C	0320	0323	0324	0325	0326	0329	032A	032B	032C	032D	032E	032F	031D	031E	031F	0330	0331	0333	0339	033A	033B	033C	0347	0348	0349	034D	034E
cat	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn
reg	◌̀	◌́	◌̄	◌̱	◌̓	◌̄	◌̣	◌̤	◌̥	◌̦	◌̠	◌̺	◌̼	◌̬	◌̂	◌̯	◌̂	◌̗	◌̞	◌̟	◌̰	◌̱	◌̿	◌̾	◌͂	◌̽	◌͇	◌̪	◌̺	◌̿	◌̋	◌̀	◌͈	◌́
bx	-219	-210	-224	-285	-216	-253	-216	-218	-218	-255	-257	-253	-251	-204	-195	-251	-250	-249	-247	-249	-213	-244	-244	-216	-253	-246	-251	-244	-197	-192	-222	-227
by	-133	-132	-148	-147	-146	-148	-113	-116	-116	-82	-122	-113	-108	-116	-133	-110	-116	-143	-106	-145	-70	-110	-110	-146	-113	-113	-108	-110	-110	-110	-110	-110
xm	-248	-224	-249	-249	-262	-251	-216	-217	-217	-245	-256	-255	-251	-203	-195	-252	-251	-249	-249	-251	-205	-240	-240	-178	-255	-252	-251	-240	-195	-189	-228	-228
bxδ	29	14	25	-36	46	-1	0	-1	-1	-10	-1	2	0	0	0	1	1	0	2	2	-8	-4	-4	-38	2	6	0	-4	-2	-3	6	1
it	◌̀	◌́	◌̄	◌̱	◌̓	◌̄	◌̣	◌̤	◌̥	◌̦	◌̠	◌̺	◌̼	◌̬	◌̂	◌̯	◌̂	◌̗	◌̞	◌̟	◌̰	◌̱	◌̿	◌̾	◌͂	◌̽	◌͇	◌̪	◌̺	◌̿	◌̋	◌̀	◌͈	◌́
bx	-246	-237	-254	-315	-246	-283	-239	-242	-242	-272	-282	-276	-273	-228	-222	-275	-274	-278	-269	-278	-227	-266	-266	-246	-276	-269	-273	-266	-219	-214	-244	-249
by	-133	-132	-148	-147	-146	-148	-113	-116	-116	-82	-122	-113	-108	-116	-133	-110	-116	-143	-106	-145	-70	-110	-110	-146	-113	-113	-108	-110	-110	-110	-110	-110
xm	-281	-257	-270	-287	-283	-281	-239	-240	-240	-277	-286	-286	-270	-221	-238	-261	-294	-291	-267	-281	-220	-262	-269	-216	-284	-283	-292	-269	-223	-206	-255	-259
bxδ	35	20	16	-28	37	-2	0	-2	-1	5	4	10	-2	-6	16	-14	20	13	-2	3	-7	-4	3	-29	8	14	19	3	4	-8	11	10
sb	◌̀	◌́	◌̄	◌̱	◌̓	◌̄	◌̣	◌̤	◌̥	◌̦	◌̠	◌̺	◌̼	◌̬	◌̂	◌̯	◌̂	◌̗	◌̞	◌̟	◌̰	◌̱	◌̿	◌̾	◌͂	◌̽	◌͇	◌̪	◌̺	◌̿	◌̋	◌̀	◌͈	◌́
bx	-219	-210	-224	-285	-216	-253	-216	-218	-218	-255	-257	-253	-251	-208	-195	-239	-250	-249	-247	-249	-213	-244										
by	-133	-132	-148	-147	-146	-149	-113	-116	-116	-73	-124	-113	-108	-116	-133	-110	-116	-143	-106	-146	-70	-58										
xm	-247	-224	-248	-249	-262	-251	-216	-217	-216	-244	-256	-255	-251	-203	-195	-240	-263	-248	-249	-251	-192	-240										
bxδ	28	14	24	-35	46	-2	0	0	-1	-10	-1	2	0	-4	0	1	13	0	2	2	-20	-4										
si	◌̀	◌́	◌̄	◌̱	◌̓	◌̄	◌̣	◌̤	◌̥	◌̦	◌̠	◌̺	◌̼	◌̬	◌̂	◌̯	◌̂	◌̗	◌̞	◌̟	◌̰	◌̱	◌̿	◌̾	◌͂	◌̽	◌͇	◌̪	◌̺	◌̿	◌̋	◌̀	◌͈	◌́
bx	-219	-210	-224	-285	-216	-253	-216	-218	-218	-255	-257	-253	-251	-204	-195	-251	-250	-249	-247	-249	-213	-244										
by	-133	-132	-148	-147	-146	-148	-113	-116	-116	-82	-122	-113	-108	-116	-133	-110	-116	-143	-106	-145	-70	-110										
xm	-269	-245	-258	-287	-271	-269	-227	-228	-228	-261	-274	-274	-258	-211	-226	-248	-293	-281	-252	-269	-195	-239										
bxδ	50	35	34	2	55	16	11	10	10	6	17	21	7	7	31	-2	43	32	5	20	-17	-5										

In the anchor model for Libertinus, the Y coordinate of an above-anchor (ay) or a below-anchor (by) is at a stable value depending on where the glyph outline lives relative to a vertical aspect. See the rows for ay and by in Tables 1 and 2 and find the common values according to the glyph's vertical aspects. There is no set name in font design for these above- or below-anchor Y coordinate reference values by ascender aspect, so I will denote them $ayra$, $ayrc$, $ayrx$, $byr0$, $byrd$, These are the middle five rows of Table 9. For example, all glyphs that sit on the baseline have below-anchor Y coordinate $by = -110$, and thus $byr0 = -110$. All capitals A–Z have $ay = 850$, and thus $ayrc = 850$. Any glyph that descends to the descender value $yrd = -232$ or descender range $[-238, -227]$ should have $by = byrd = -319$.

The clearance is defined as the difference between the anchor reference value and the vertical aspect Y value. I will denote these clearances as $ay\delta a$, $ay\delta c$, $ay\delta x$, $by\delta 0$, $by\delta d$. So, $ay\delta a = ayra - yra$ etc. For example, capital letters A–Z have a capital height $yrc = 645$, and all capital letters A–Z have an above-anchor Y coordinate at $ayrc = 850$. So the above-anchor clearance for capitals is $ay\delta c = ayrc - yec = 850 - 645 = 205$. Because of the weird anchor model for Libertinus, this clearance value is not the exact vertical distance between a base and a mark, but this clearance value is still useful.

A glyph may not be a full ascender or a full descender, and its bounding box may not live closely within two vertical aspects. In such a case, the anchor Y coordinate is set by looking for the nearest, or most suitable, vertical aspect, and considering the clearance of that vertical aspect relative to the bounding box of the glyph.

Here is the heuristic algorithm for calculating the Y coordinate of the above- or below-anchor (ay or by) of a glyph that is not a full ascender or descender.

Consider a glyph that is not a full ascender. Find the nearest, or most suitable, vertical aspect – ascender a, capital c, or x-height x. Then $ay = Y_{\text{max}} + ay\delta[acx] - \epsilon$, where Y_{max} is the maximum Y coordinate of bounding box of the glyph, and ϵ is the correction for the vertical aspect range, usually for an overshoot.

Consider a glyph that is not a full descender. Find the nearest, or most suitable, vertical aspect – baseline 0 or desender d. Then $by = Y_{\text{min}} + by\delta[0d] + \epsilon$, where Y_{min} in the minimum Y coordinate of the bounding box of the glyph, and ϵ is the correction for the vertical aspect range, usually for an overshoot.

Table 9: Vertical aspects, anchor Y references, and clearances of Libertinus

	v. aspect	regular	italic	semibold	semibold italic
yra	ascender	$698, [688, 698]$	$698, [688, 698]$	$698, [690, 698]$	$698, [696, 705]$
yrc	capital	$645, [645, 658]$	$645, [645, 658]$	$645, [645, 662]$	$645, [645, 662]$
yrx	x-height	$429, [429, 442]$	$429, [429, 442]$	$434, [433, 447]$	$434, [434, 447]$
$yr0$	baseline	$0, [-12, 0]$	$0, [-12, 1]$	$0, [-12, 1]$,	$0, [-20, 0]$
yrd	descender	$-232, [-238, -227]$	$-232, [-238, -227]$	$-232, [-238, -212]$	$-232, [-239, -219]$
$ayra$	ascender	885	890	885	890
$ayrc$	capital	850	850	805	850
$ayrx$	x-height	645	645	645	645
$byr0$	baseline	-110	-110	-110	-110
$byrd$	descender	-319	-319	-319	-319
$ay\delta a$	ascender	187	192	187	192
$ay\delta c$	capital	205	205	160	205
$ay\delta x$	x-height	216	216	211	211
$by\delta 0$	baseline	110	110	110	110
$by\delta d$	descender	87	87	87	87

For the vertical aspect, the first single value is the optical value of the vertical aspect. In practice, this is the y-coordinate of the central control point of a (vertical) stem that meets that aspect. In the table, the range for the vertical aspects are the BlueValues set by the designer in the SFD and seen as guidelines in a FontForge glyph outline window view.

I need to keep track of serif and bowl corrections relative to the (single) vertical aspect value. In Libertinus serif regular, a glyph that meets an aspect with a stem needs a 2 unit correction. For example, the letter x meets the baseline and x-height aspects, with stems, but $y_{\text{min}} = 0 - 2 = -2$ and $y_{\text{max}} = 429 + 2 = 431$. For bowls, the corrections tend to be different for each aspect. For Libertinus serif regular, a glyph that is a bowl full-descender (usually) has a 6 unit overshoot, i.e. $y_{\text{min}} = -232 - 6 = -238$. A bowl on the baseline (usually) has a 10 unit

overshoot, i.e. $y_{\text{min}} = 0 - 10 = -10$. A bowl at the x-height aspect (usually) has a 10 unit overshoot, i.e. $y_{\text{max}} = 429 + 10 = 439$. A bowl at the capital aspect (usually) has a 13 unit overshoot, i.e. $y_{\text{max}} = 645 + 13 = 658$. For full ascenders, there is no correction in practice, regardless of the shape of the glyph's ascender, because $y_{\text{max}} = 698$ for all such glyphs.

That is, these stem and bowl corrections are 2 and 6, 10, 10, 13.

For the other font styles (italic, semibold, semibold italic), these corrections are remarkably consistent for all aspects, even though e.g. italic small letters do not really have flat terminal stems. But I think that the correction principle is valid for glyphs that are not within the threshold of an expected vertical aspect. For a non-full descender, if y_{min} is closest to the descender aspect, and the glyph looks like a bowl at this lowest point, then the correction is 6; otherwise I assume that the glyph must look like a flat stem and the correction is 2. But for a non-full descender that is closer to the baseline, the bowl correction is 10. Full descender bowls tend to be flatter because the lower threhold of the descender really is a hard lower bound.

Copied anchors

Regular, above

(281, 850) ĖĚÊḖP̌P̂

(336, 850) ḰǨK̂V̌V̂

(346, 850) ṄŇN̂ṺǓÛ

(209, 645) ȧăâǎçĉċčéĕêḗȝȝ̂ǧǧĝḧḧḧḧḫḫẖẖẗŭűûǔůṅňn̂ṅȯŏôǒȱȱ̂ȱṗṗp̂ṙṙr̂ṛṛȓȑȓȑȓȑȓȑṣṣŝṣſſſ

Italic, above

(481, 850) ḢḤḪḨV̇V̌V̂
(218, 645) ḋḑḏḍḓḍ̌ḓ̂
(480, 645) ȧǡâȩǎêȇ
(322, 654) ȯǒôȅȇ̂

Italic, below

(310, -113) ọǫǫǫǫǫ
(186, -107) ọǫǫǫǫǫ
(212, -114) ạąɑḁḇḅḇ̧ḉḉ̧ḍḍḍḍḍḑ̣ḑ̣ẹęɘɘ̧ɘ̧ɘ̧ɘ̧ɚɚ̧ɛɛɛ̧ɜ̧ɜ̧ɜ̧

Semibold, above

(363, 802) ȮȎÔǪǑǪ̂
(264, 646) ṅňn̂ůǔû

Semibold, below

(109, -319) J̰J̰j̰j̰

Semibolditalic, above

No clusters detected.

Semibolditalic, below

No clusters detected.
ƀ ƅ ƀ ƅ̀ ƅ ƀ ƅ̀ Ɓ Ɓ Ɓ Ɓ̄ Ɓ̄ Ɓ Ɓ̄ Ɗ Ɗ Ɗ Ɗ̄ Ɗ̄ Ɗ Ɗ̄ Ɯ Ɯ Ɯ Ɯ̄ Ɯ̄ Ɯ Ɯ̄ j j j̄ j̄ j̄ j̄ ɘ̀ ɘ̀ ɘ̄ ɘ̄ ɘ̄ ɘ̄ ɡ̇ ɡ̇ ɡ̇ Ḡ Ḡ Ḡ i i̇ ī ī̇ ī ī ī ɯ̀ ɯ̀ ɯ̀ ɯ̄ ɯ̄ ɯ̄ ɯ̄ ɱ ɱ̀ ɱ̀ ɱ̄ ɱ̄ ɱ̄
ṇ ṇ ṇ̀ ṇ̄ ɳ ɳ̄ ɳ̄ ɾ̇ ɾ̇ ɾ̇ ɾ̄ ɾ̄ ɾ̄
Ị Ƒ̣ Ị Ƒ̣ Ị Ƒ̣ ị ƒ̣ ị ƒ̣ Æ Æ̣ Æ Æ̣ Æ Æ̣ Æ Æ̣ æ æ̣ æ æ̣ æ æ̣ ɪ ɪ̣ ɪ ɪ̣ ɪ ɪ̣ Ɓ Ɓ̣ Ɓ Ɓ̣ Ɓ Ɓ̣ Ɗ Ɗ̣ Ɗ Ɗ̣ Ɗ Ɗ̣ Ɯ Ɯ̣ Ɯ Ɯ̣ Ɯ Ɯ̣ ẹ ẹ̀ ẹ̄ ẹ̄ ẹ̄ ẹ̄ ɑ ɑ̣ ɑ ɑ̣ ɑ ɑ̣
ɓ ɓ̀ ɓ ɓ̄ ɓ ɓ̄ ç ç̀ ç ç̄ ç ç̄ ɖ ɖ̀ ɖ ɖ̄ ɖ ɖ̄ ɗ ɗ̀ ɗ ɗ̄ ɗ ɗ̄ ɘ ɘ̀ ɘ ɘ̄ ɘ ɘ̄ ɚ ɚ̀ ɚ ɚ̄ ɚ ɚ̄ ɛ ɛ̀ ɛ ɛ̄ ɛ ɛ̄ ʒ ʒ̀ ʒ ʒ̄ ʒ ʒ̄ ɔ ɔ̀ ɔ ɔ̄ ɔ ɔ̄ ʂ ʂ̀ ʂ ʂ̄ ʂ ʂ̄
ɵ ɵ̀ ɵ ɵ̄ ɵ ɵ̄ ɥ ɥ̀ ɥ ɥ̄ ɥ ɥ̄ ɦ ɦ̀ ɦ ɦ̄ ɦ ɦ̄ ɧ ɧ̀ ɧ ɧ̄ ɧ ɧ̄ i̇ i̇ i̇ ī̇ ī̇ ī̇ ɨ ɨ̀ ɨ ɨ̄ ɨ ɨ̄ ɪ ɪ̀ ɪ ɪ̄ ɪ ɪ̄ ɯ ɯ̀ ɯ ɯ̄ ɯ ɯ̄ ɯ
ɱ ɱ̀ ɱ ɱ̄ ɱ ɱ̄ ɱ ɱ̄ ɳ ɳ̀ ɳ ɳ̄ ɳ ɳ̄ ɽ ɽ̀ ɽ ɽ̄ ɽ ɽ̄ ɽ ɽ̄ ɹ ɹ̀ ɹ ɹ̄ ɹ ɹ̄ ɹ ɹ̄ ʊ ʊ̀ ʊ ʊ̄ ʊ ʊ̄ ʊ ʊ̄ ʊ ʊ̄
ụ ụ̀ ụ ụ̄ ụ ụ̄ ỵ ỵ̀ ỵ ỵ̄ ỵ ỵ̄ ỵ ỵ̄ ʔ ʔ̣ ʔ ʔ̣ ʔ ʔ̣ ʕ ʕ̣ ʕ ʕ̣ ʕ ʕ̣ ʃ ʃ̣ ʃ ʃ̣ ʃ ʃ̣

Copied above-anchors

ȧ ā ȼ c̄ ɛ̄ ē ʒ̄ ʒ̄ ɠ ɠ ɬ̄ ɬ̄ ɬ̄ ɭ ɭ ɮ ɮ ŋ̄ ɯ ū ɳ Ń Ñ ȯ ō ѡ ѡ ɸ ɸ ɪ̄ ɪ ɪ ɻ ɻ ɾ ɾ ɾ̄ ʁ ʁ̄ ʂ ʂ̄ ʃ ʃ ʃ̄ ʃ̄ ɭ ɭ ʃ ʃ ɨ̄ ɨ̄ ʈ ʈ ʉ ʉ̄ ʊ̄ ʊ̄ ʌ ʌ̄ ʍ ʍ̄ ʎ ʎ̄ ʏ ʏ̄ ʑ ʑ̄ ʑ̄ ʒ ʒ̄ ʔ ʔ ʕ ʕ ɔ ɔ̄ ʙ ʙ̄ ɠ ɠ̄ ʜ ʜ̄ ʝ ʝ̄ ɭ ɭ̄

Copied below-anchors

ạ ą ș ʂ ɟ ɟ̄ ɭ ɭ̄ ɮ ɮ̄ ɵ ɵ ɸ ɸ ɻ ɻ̄ ɾ ɾ̄ ɹ ɹ̄ ʁ ʁ̄ ʁ ʁ̄ ʂ ʂ̄ ʃ ʃ̄ ʃ ʃ̄ ɭ ɭ̄ ʃ ʃ̄ ɨ ɨ̄ ʈ ʈ̄ ʉ ʉ̄ ʌ ʌ̄ ʍ ʍ̄ ʎ ʎ̄ ʏ ʏ̄ ʐ ʐ̄ ʐ ʐ̄ ʒ ʒ̄ ʔ ʔ̄ ɔ ɔ̄ ʙ ʙ̄ ɠ ɠ̄ ɟ ɟ̄ ɭ ɭ̄

ɑ ɑ ɓ ɓ ɕ ɕ ɖ ɖ ɗ ɗ ɘ ɘ ɚ ɚ ɛ ɛ ɟ ɟ

Base coverage

Libertinus serif regular has good above- and below-anchor base coverage for A–Z, a–z, and IPA letters, in the sense that most anchors are defined. But I will show later that some anchors are not set to good values. The base coverage gets worse for italic and semibold. Observe that many IPA letters reuse $ax = 209$ and $bx = 196$. These are good anchors for the letter a, but these were just copied to some IPA letters without regard to the letterform and are therefore bad anchors for those IPA letters.

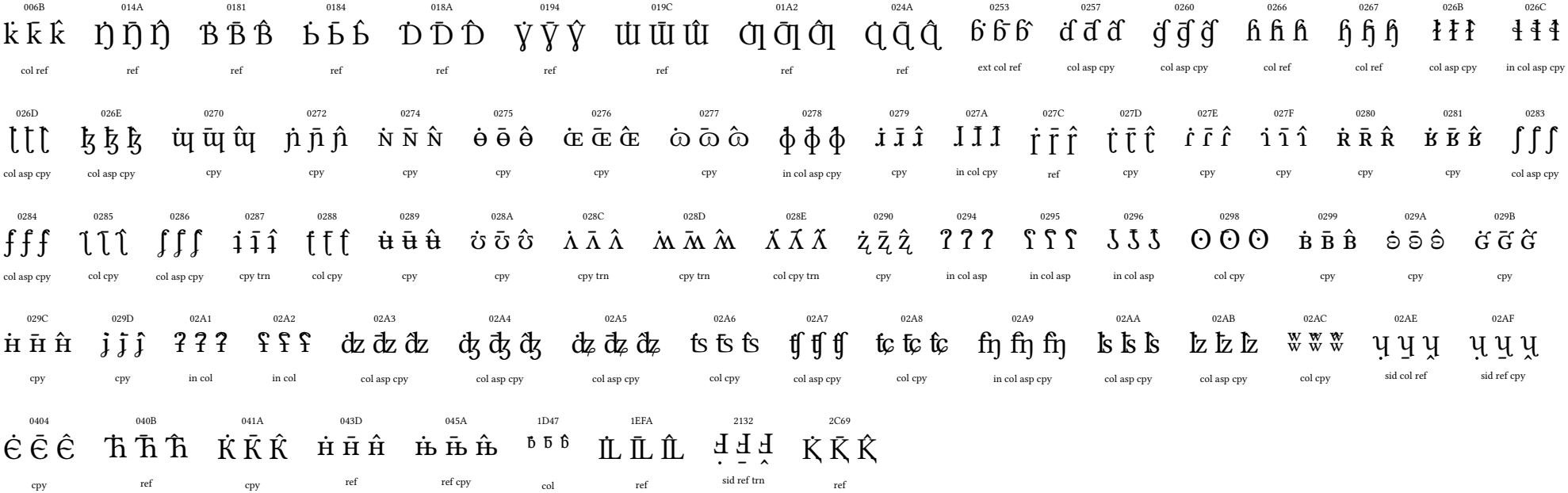
Anchor audit

These are the designer-set above- and below-anchors that `run_anchor_audit.py` classifies as bad or suspect, in the original fonts. Each cell shows the codepoint, then the base with a few marks (dot, macron and circumflex above; dot and macron below), then tags for the tests that flagged the anchor. The combining grapheme joiner separates base and mark, so every mark is positioned by its anchor. The reasons in full are in `data/fontanchors_audit/`.

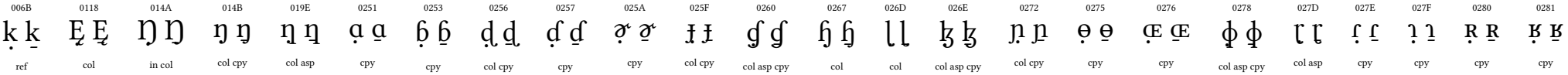
Tags: **sid** anchor on the wrong side of the glyph; **in** anchor inside the outline; **ext** anchor x outside the glyph; **ctr** far from the centre line; **ref** y far from the reference values; **asp** beside an ascender or descender; **clr** floats far from the glyph; **col** a mark overlaps or crowds the glyph; **cpy** copied from another glyph; **trn** turned letter not following its source; **sty** copied from another style; **pre** disagrees with precomposed glyphs; **pos** position differs between styles.

Regular

Bad, above-anchor (77)



Bad, below-anchor (62)



0282 0283 0284 0285 0286 0287 0288 0289 028C 028D 028E 0290 0291 0293 0297 0298 0299 029A 029B 029D 02A0 02A2 02A3

ʂ ʂ̄ | ʃ ʃ̄ | ʄ ʄ̄ | ɭ ɭ̄ | ʃ̩ ʃ̩̄ | ɨ ɨ̄ | ʈ ʈ̄ | ʱ ʱ̄ | ʏ ʏ̄ | ʷ ʷ̄ | ʸ ʸ̄ | ʐ ʐ̄ | ʑ ʑ̄ | ʒ ʒ̄ | ƈ ƈ̄ | ʘ ʘ̄ | ʙ ʙ̄ | ɘ ɘ̄ | ɠ ɠ̄ | ɟ ɟ̄ | ɖ ɖ̄ | ʡ ʡ̄ | ʣ ʣ̄

in col | col cpy | col cpy | in col cpy | col cpy | in col cpy | col cpy | cpy | cpy trn | cpy trn | cpy trn | col cpy | col cpy | col asp cpy | col cpy | cpy | cpy | cpy | cpy | col cpy | col asp cpy | cpy trn | cpy

02A4 02A5 02A6 02A7 02A8 02A9 02AA 02AB 02AC 02AE 02AF 0404 041A 045A 2132

ʤ ʤ̄ | ʣ ʣ̄ | ʦ ʦ̄ | ʧ ʧ̄ | ʨ ʨ̄ | ɲ ɲ̄ | ʂ ʂ̄ | ʐ ʐ̄ | ʷ̤ ʷ̤̄ | ᷇ ᷇̄ | ᷈ ᷈̄ | Ҷ Ҷ̄ | Ҟ Ҟ̄ | ҥ ҥ̄ | Ⅎ Ⅎ̄

asp cpy | cpy | cpy | col cpy | col cpy | cpy | cpy | cpy | cpy | sid col ref | sid col ref | cpy | cpy | cpy | sid col ref trn

Suspect, above-anchor (21)

0064 0068 0071 00C6 00DE 00E7 00FE 018C 0195 0237 0250 0252 0256 025D 028B 028F 02A0

d̋ đ̋ ɖ̋ | ẖ ħ̋ ẖ | q̋ q̄ q̂ | Æ̋ Ǣ Æ̂ | þ̋ þ̄ þ̂ | ç̋ ç̄ ĉ | þ̋ þ̄ þ̋ | ƌ̋ ƌ̄ ƌ̋ | ɦ̋ ħ̄ ɦ̋ | j̋ j̄ ĵ | ɐ̋ ɐ̄ ɐ̂ | ɒ̋ ɒ̄ ɒ̂ | ɖ̋ ɖ̄ ɖ̋ | ʒ̋ ʒ̄ ʒ̋ | ʊ̋ ʊ̄ ʊ̂ | ʏ̋ ʏ̄ ʏ̂ | ɋ̋ ɋ̄ ɋ̋

col asp pre | col asp | cpy | pre | cpy | ref | col asp | col asp | col asp | pos | trn | cpy | col asp | col cpy | cpy | cpy | col asp cpy

1ECD 2C68 2C6A 2C6B

ọ̋ ō̋ ô̋ | ẖ̋ ħ̋ ẖ̋ | ⱪ̋ ⱪ̄ ⱪ̋ | Ⱬ̋ Ⱬ̄ Ⱬ̋

cpy | col asp | col asp | ref

Suspect, below-anchor (33)

0052 005A 006A 0070 0079 0105 0119 0138 0172 0181 01A4 0235 0236 0238 024A 024B 0252 0255 025D 0265 026C 0270 0271

Ḳ̠ Ḵ̠ | Ẓ̠ Ƶ̠ | j̠ j̱ | p̠ p̱ | y̠ y̱ | ą̠ ą̱ | ę̠ ę̱ | ķ̠ ķ̱ | Ų̠ Ų̱ | Ḅ̠ Ḇ̠ | Ṛ̠ Ṟ̠ | ṇ̠ ṉ̠ | ṭ̠ ṯ̠ | ȸ̠ ȸ̱ | Ɋ̠ Ɋ̱ | ɋ̠ ɋ̱ | ɒ̠ ɒ̱ | ɕ̠ ɕ̱ | ʒ̠ ʒ̱ | ɻ̠ ɻ̱ | ɬ̠ ɬ̱ | ɯ̠ ɯ̱ | ŋ̠ ŋ̱

pre | cpy | cpy | pos | pre | col | ctr col | cpy | col | cpy | cpy | col cpy | col cpy | cpy | col | col | trn | col | cpy | col ref | ref | asp | col

0273 027B 027C 029E 03B2 2184 2C69 2C6C 2C6E 2C74

ɳ̠ ɳ̱ | ɾ̠ ɾ̱ | ɾ̠ ɾ̱ | ɮ̠ ɮ̱ | β̠ β̱ | ɔ̠ ɔ̱ | Ⱪ̠ Ⱪ̱ | ⱬ̠ ⱬ̱ | Ɱ̠ Ɱ̱ | ⱶ̠ ⱶ̱

col cpy | col cpy | ref pos | asp | col asp | cpy | cpy | col | col | cpy

Italic

Bad, above-anchor (47)

0064 00FE 014A 014B 0181 0184 018A 018C 0194 0195 019C 019E 01A2 01B4 0235 0239

d̈ d̄ d̂ | þ̈ þ̄ þ̂ | ŋ̈ ŋ̄ ŋ̂ | ɲ̈ ɲ̄ ɲ̂ | ɓ̈ ɓ̄ ɓ̂ | ƅ̈ ƅ̄ ƅ̂ | ɗ̈ ɗ̄ ɗ̂ | ƌ̈ ƌ̄ ƌ̂ | ɣ̈ ɣ̄ ɣ̂ | ɦ̈ ɦ̄ ɦ̂ | ɯ̈ ɯ̄ ɯ̂ | ɳ̈ ɳ̄ ɳ̂ | ƣ̈ ƣ̄ ƣ̂ | ƥ̈ ƥ̄ ƥ̂ | ɲ̈ n̄̈ ɲ̂ | ɸ̈ ɸ̄ ɸ̂

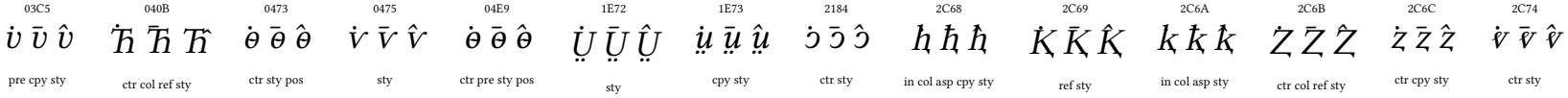
ctr asp pre sty | in col asp sty | ref sty | sty | ref sty | ctr ref sty | ref sty | ctr col asp | ref sty | col asp sty | ref sty | cpy | ctr ref | sty | ctr sty | sty

024D 0256 0257 0258 0262 0266 0269 026A 026F 0271 0272 0273 027C 028B 02A0 02AE 02AF

ɟ̈ ɟ̄ ɟ̂ | ɖ̈ ɖ̄ ɖ̂ | ɗ̈ ɗ̄ ɗ̂ | ɘ̈ ɘ̄ ɘ̂ | ɢ̈ ɢ̄ ɢ̂ | ɦ̈ ɦ̄ ɦ̂ | ɪ̈ ɪ̄ ɪ̂ | ɪ̈ ɪ̄ ɪ̂ | ɯ̈ ɯ̄ ɯ̂ | ɱ̈ ɱ̄ ɱ̂ | ɲ̈ ɲ̄ ɲ̂ | ɳ̈ ɳ̄ ɳ̂ | ɾ̈ ɾ̄ ɾ̂ | ʊ̈ ʊ̄ ʊ̂ | ɋ̈ ɋ̄ ɋ̂ | ɥ̈ ɥ̄ ɥ̲ | ɥ̈ ɥ̄ ɥ̲

col cpy | ctr cpy sty | ctr asp cpy sty | ctr sty | sty | col ref sty | ext sty | ext pos | trn sty pos | sty | cpy sty | sty | ctr ref sty | sty | ctr sty | sid col ref sty | sid ref

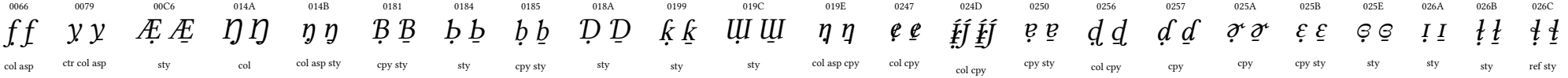
03C5 040B 0473 0475 04E9 1E72 1E73 2184 2C68 2C69 2C6A 2C6B 2C6C 2C74



pre cpy sty | ctr col ref sty | ctr sty pos | sty | ctr pre sty pos | sty | cpy sty | ctr sty | in col asp cpy sty | ref sty | in col asp sty | ctr col ref sty | ctr cpy sty | ctr sty

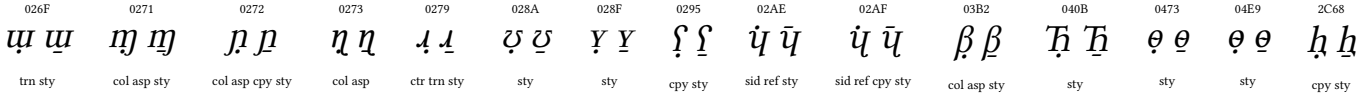
Bad, below-anchor (38)

0066 0079 00C6 014A 014B 0181 0184 0185 018A 0199 019C 019E 0247 024D 0250 0256 0257 025A 025B 025E 026A 026B 026C



col asp | ctr col asp | sty | col | col asp sty | cpy sty | sty | cpy sty | sty | sty | sty | col asp cpy | col cpy | col cpy | cpy sty | col cpy | cpy | cpy | cpy sty | sty | sty | sty | ref sty

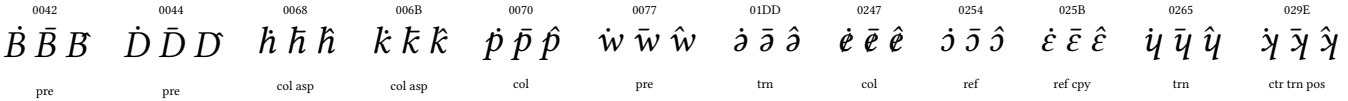
026F 0271 0272 0273 0279 028A 028F 0295 02AE 02AF 03B2 040B 0473 04E9 2C68



trn sty | col asp sty | col asp cpy sty | col asp | ctr trn sty | sty | sty | cpy sty | sid ref sty | sid ref cpy sty | col asp sty | sty | sty | sty | cpy sty

Suspect, above-anchor (12)

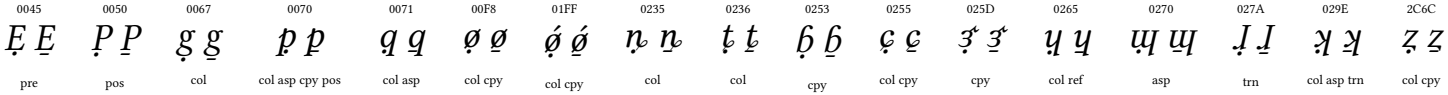
0042 0044 0068 006B 0070 0077 01DD 0247 0254 025B 0265 029E



pre | pre | col asp | col asp | col | pre | trn | col | ref | ref cpy | trn | ctr trn pos

Suspect, below-anchor (17)

0045 0050 0067 0070 0071 00F8 01FF 0235 0236 0253 0255 025D 0265 0270 027A 029E 2C6C

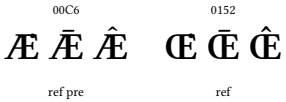


pre | pos | col | col asp cpy pos | col asp | col cpy | col cpy | col | col | cpy | col cpy | cpy | col ref | asp | trn | col asp trn | col cpy

Semibold

Bad, above-anchor (2)

00C6 0152



ref pre | ref

Bad, below-anchor (1)

0279



sty

Suspect, above-anchor (4)

0186	0237	1ECC	1EE4
Ɔ̇ Ɔ̄ Ɔ̂	j̇ j̄ ĵ	Ọ̇ Ọ̄ Ộ	Ụ̇ Ụ̄ Ự̂
ref	ctr pos	ref	ref

Suspect, below-anchor (5)

0067	006A	0131	026C	2139
g̣ g̱	j̣ j̱	ı̣ ı̱	ɬ̣ ɬ̱	ị̇ i̱
col	cpy	cpy	ref	cpy

Semibold italic

Bad, above-anchor (122)

0041	0042	0043	0044	0047	0048	0049	004B	004C	004D	004E	004F	0051	0052	0053
Ȧ Ā Â	*Ḃ B̄ Ḃ*	*Ċ C̄ Ĉ*	*Ḋ D̄ Ḋ*	*Ġ Ḡ Ĝ*	*Ḣ H̄ Ḣ*	*İ Ī Î*	*Ḳ K̄ Ḳ*	*L̇ L̄ L̇*	*Ṁ M̄ Ṁ*	*Ṅ N̄ Ṅ*	*Ȯ Ō Ȯ*	*Q̇ Q̄ Q̇*	*Ṙ R̄ Ṙ*	*Ṡ S̄ Ṡ*
pre sty pos	ctr pre sty pos	pre sty pos	pre sty pos	pre sty pos	pre sty pos	ext pre sty pos	pre sty pos	ext pos	pre sty pos	pre sty pos	ctr pre sty pos	sty pos	ctr pre sty pos	ctr pre sty pos

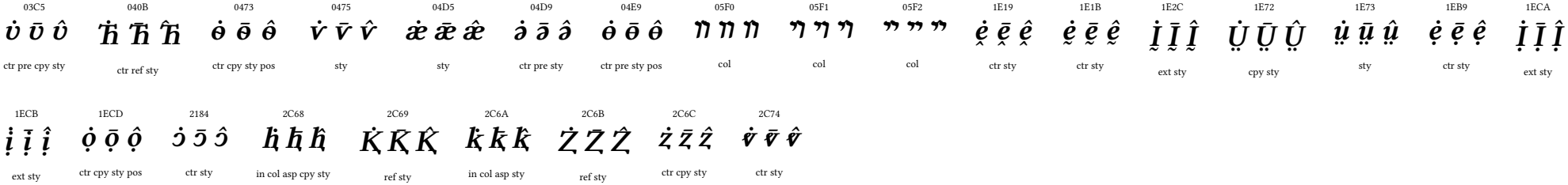
0054	0055	0056	0057	0058	0059	005A	0063	0064	0065	0069	006B	006C	006D	006E	006F
Ṫ T̄ Ṫ	*U̇ Ū Û*	*V̇ V̄ V̂*	*Ẇ W̄ Ŵ*	*Ẋ X̄ X̂*	*Ẏ Ȳ Ŷ*	*Ż Z̄ Ẑ*	*ċ c̄ ĉ*	*d̄ d̄ d̄*	*ė ē ê*	*i̇ ī î*	*k̇ k̄ k̂*	*l̇ l̄ l̂*	*ṁ m̄ m̂*	*ṅ n̄ n̂*	*ȯ ō ô*
pre sty pos	pre cpy sty pos	pre cpy sty pos	pre sty pos	pre sty pos	pre sty pos	pre sty pos	ctr pre sty pos	ctr col asp pre sty	ctr pre sty pos	ext sty pos	col ref pre sty	ext pos	pre sty pos	ctr pre sty pos	ctr pre sty pos

0070	0071	0072	0075	0076	0077	0079	007A	00C7	00D0	00D8	00DE	00E6	00E7	00FE	0104
ṗ p̄ p̂	*q̇ q̄ q̂*	*ṙ r̄ r̂*	*u̇ ū û*	*v̇ v̄ v̂*	*ẇ w̄ ŵ*	*ẏ ȳ ŷ*	*ż z̄ ẑ*	*Ç Ç̄ Ç*	*Ḋ Đ Ḋ*	*Ø Ø̄ Ø̂*	*Þ P̄ Þ*	*æ̇ ǣ æ̂*	*ç̇ ç̄ ĉ*	*þ̇ þ̄ þ̂*	*Ą Ą̄ Ą̂*
col pre sty pos	ctr cpy sty pos	ctr pre sty pos	sty pos	pre sty	pre sty	pre sty pos	ctr pre sty pos	sty	sty	sty	sty	pre sty pos	ctr ref sty	in col asp sty	cpy sty

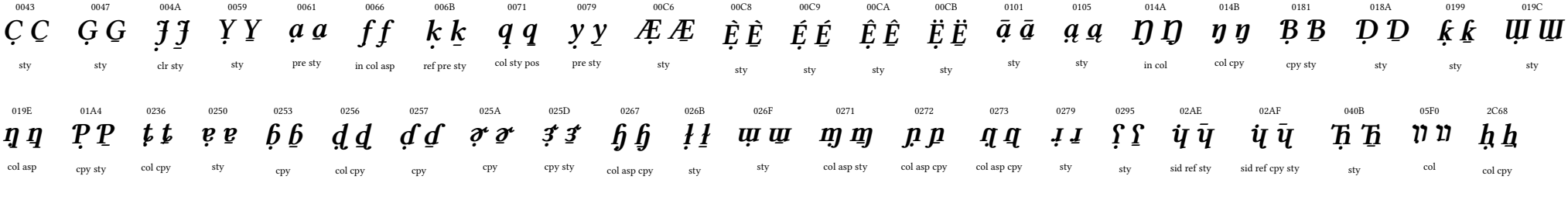
0105	0119	012E	012F	0131	0138	014A	014B	0181	0184	018A	018C	0194	0195	019C	019E
ą̇ ą̄ ą̂	*ę̇ ē ê*	*į̇ į̄ į̂*	*į̇ į̄ į̂*	*i̇ ī î*	*ḳ k̄ k̂*	*ŋ̇ Ŋ̄ Ŋ̂*	*ṅ ŋ̄ n̂*	*Ḃ B̄ B̂*	*ƅ Ƃ ƃ*	*Ḋ Đ Ḋ*	*đ̄ đ̄ đ̄*	*ẏ ȳ ŷ*	*ħ̇ ħ̄ ħ̂*	*ɯ̇ ɯ̄ ɯ̂*	*ṅ ŋ̄ n̂*
ctr sty	ctr sty	ext sty	ext	ext sty pos	ctr sty pos	ref	sty	ref sty	col ref sty	ref sty	in ctr col sty	ref sty	in col asp sty	ref sty	cpy sty

01A2	01A3	01B4	0228	0229	0235	0237	0239	0247	0251	0252	0256	0257	0258	025B	025C
Ơ̇ Ơ̄ Ơ̂	*ơ̇ ơ̄ ơ̂*	*ư̇ ư̄ ư̂*	*Ḝ Ē Ȇ*	*ḝ ē ê*	*ṋ n̄ n̂*	*j̇ j̄ ĵ*	*ṗ p̄ p̂*	*ė ē ê*	*ȧ ā â*	*ɒ̇ ɒ̄ ɒ̂*	*ɖ̇ ɖ̄ ɖ̂*	*ɗ̇ ɗ̄ ɗ̂*	*ɘ̇ ɘ̄ ɘ̂*	*ɛ̇ ɛ̄ ɛ̂*	*ɜ̇ ɜ̄ ɜ̂*
ref cpy sty	sty	col sty	ctr sty	ctr sty	ctr sty	ctr sty pos	sty	ctr cpy	ctr sty	cpy trn sty	ctr col asp cpy sty	ctr col asp cpy sty	ctr sty	ctr cpy sty pos	ctr trn sty

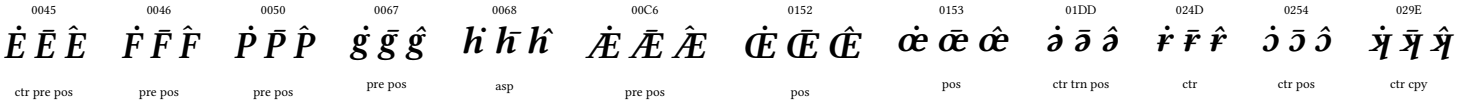
0262	0265	0266	0267	0268	0269	026A	026F	0271	0272	0273	027C	028B	0292	02A0	02AE	02AF
ġ ḡ ĝ	*q̇ q̄ q̂*	*ɦ̇ ɦ̄ ɦ̂*	*ɧ̇ ɧ̄ ɧ̂*	*ɨ̇ ɨ̄ ɨ̂*	*i̇ ī î*	*i̇ ī î*	*ɯ̇ ɯ̄ ɯ̂*	*ɱ̇ ɱ̄ ɱ̂*	*ɲ̇ ɲ̄ ɲ̂*	*ɳ̇ ɳ̄ ɳ̂*	*ɾ̇ ɾ̄ ɾ̂*	*ʋ̇ ʋ̄ ʋ̂*	*ʒ̇ ʒ̄ ʒ̂*	*ʠ̇ ʠ̄ ʠ̂*	*ɥ̇ ɥ̲ ɥ̂*	*ɥ̇ ɥ̲ ɥ̂*
sty	ctr sty pos	in col ref sty	in col ref	ext	ext sty	ext pos	sty pos	sty	cpy sty	sty	ctr ref sty	sty	ctr pre cpy sty pos	ctr col asp cpy sty	sid col ref	sid col ref cpy



Bad, below-anchor (44)



Suspect, above-anchor (12)



Suspect, below-anchor (27)



Anchor survey, with patches

I will systematically survey above- and below anchors in the four font font styles, commenting on any features and patching any problems. Refer to Tables 1 and 2 for anchor values. Missing anchors are readily apparaent as blank entries. Large anchor-midpoint offset values ($axδ$, $bxδ$) indicate possible problems, but these can often be explained by glyph symmetry.

A–Z, regular, above- and below-anchor

The above-anchors of A–Z regular are good.

The above-anchor of D is just right of the major stem, not right enough to be centered over the bowl. For D, $ax = 306$. Consider the precomposed characters Ď (U+010E) and Ḋ (U+1E0A). The designer drew their glyphs with the mark centered at $X = 312$ and 296 respectively. So, this is a deliberate design decision to keep the mark closer to the stem, not centered over the bowl.

Similarly, the above anchor of F is just right of the major stem, not right enough to be centered over cross strokes. For F, $ax = 284$. For Ḟ (U+1E1E), the dot is centered at $X = 254$. So, the designer is deliberately choosing to keep the mark cloder to the stem.

The above-anchor of L is centered over the major stem, which is a valid design choice.

The below-anchors of A–Z regular are good. But J's below anchor is too low, and Q is missing a below-anchor. Even though J and Q arguably do not need below-anchors for linguistic notation, I will set this anchor by the heuristic algorithm.

Patch J regular, below-anchor

J's below-anchor is too low. For J, $by = -319$, but J is not a full descender; its bbox $Y_{\text{min}} = -172$, which is well above $yrd = -232$. Yes, the designer set $by = byrd = -319$ for J in all fout font styles, but this is arguably too low. According to the heuristic algorithm, the below-anchor anchor should be at $by = Y_{\text{min}} - byδd + ϵ = -172 - 87 + 5 = -254$, where $ϵ = 5$ accounts for the overshoot of J's tail.

Patch Q regular, below-anchor

Q has bbox $Y_{\text{min}} = -209$. Again, according to the heuristic algorithm, the below-anchor anchor should be at $by = Y_{\text{min}} - byδd + ϵ = -209 - 87 + 5 = -291$, where $ϵ = 5$ accounts for the overshoot of Q's tail. For bx, observe that O and Q have the same counter outline, literally. So, the below-anchor for Q should be at $bx = 350$, the same as that for O.

a–z, regular, above- and below-anchor

b, f, j, and t are missing above-anchors. Some letters (n, p, r, v, w, x, z) should have ay regularized to 645, not 646. For d, h, and k, the above-anchor is centered above the body, not the ascender; this is good design choice, but care must be taken to avoid collision between an above-mark and an ascender. For d and h, ay is at the x-height anchor reference, and therefore wide marks collide with the ascender. For k, ay is a bit higher, but not at the full ascender anchor references, and therefore the widest marks collide with the ascender.

The below-anchor of k is too low. Otherwise, the below-anchors of a–z are good.

Patch k regular, below-anchor

For k, $by - 214$. But it should be at $byr0 = -110$, like all other letters that sit on the baseline.

Here is this positioning of k in a patched version of the font (with o for comparison only).

Original: k̲ k̲o ok̲

Patched: k̲ k̲o ok̲

Patch b, d, f, h regular, above-anchor

For b, d, f, and h, set $ay = ayra = 885$.

For ax for b, d, f, and h, there are at least two strategies. First, set ax at the bowl apex. So, for f, set $ax = 279$ at the apex of the bowl. But for b, d, and h, you could set ax centered horizontally over the bowl, in which case you also need to decide which two vertical aspects of the bowl should be used for the centering.

Patch j regular, above-anchor

j needs an above anchor for the rare case of an above-mark that is placed over the dot. i does have an above-anchor at $(137, 795)$, and the outlines of the top of

Patch n, p, r, v, w, x, z regular, above-anchor

In my patched version of the font, I regularize ay to $ayrx = 645$ for all letters that sit at the x-height, including n, p, r, v, w, x, z.

Above-anchor, a–z regular

b, f, j, and t are missing the above-anchor. d and h have an above-anchor above the meanline but not above the ascender. l has an above-anchor above the ascender. k has an above-anchor, but it is just below the ascender, so any marks other than the do crash into the ascender. These are all legitimate design choices. There are no above-marks, other than those with precomposed characters, the would ever be combined with b, f, or t.

Patch j dot removal for three above-marks

The dot of i and j should be removed before adding any above-mark. But the dot is not removed for three above-marks: x, turned tilde, double macron. So the GSUB table should be patched to fir the dotless j substitution in this context.

To do.

Capital mark alternates

Libertinus has custom glyphs for some marks (acute, grave, cicrumflex, breve, a.o.) that are designed better for capital letters. These alternates are flatter. GSUB lookups 23 and 24 implement the substitution. This substitution is not fired on some capital consonants (T, V, W, X) because these marks are not expected on these bases.

Libertinus has custom glyphs for some marks that are designed better for superscripts. These alternates are smaller. Regrettably?, no GSUB table implements their substitution.

A cirmumflex Â Ầ Ấ Ầ Ấ̉

O circumflex Ô Ồ Ố Ồ Ố̉

AE ligature Æ Æ̀ Æ Æ̀ Ǽ

S̊ T̊ V̊ S̊ T̊ V̊ S̊ T̊ V̊ S̊ T̊ V̊

S̊ T̊ V̊ S̊ T̊ V̊ S̊ T̊ V̊ S̊ T̊ V̊

Regular, above-anchor, capitals, X coordinate

Figure 5 (left panel) shows pairs of bases aligned at their above-anchor. If a capital letter has a bad X coordinate for its above-anchor, then it will look always off, right or left, in its black column relative to the gray letters that it overlays. Note that the above-anchor X coordinate is not in the geometric (horizontal) center of the glyph, by design. The above-anchor X coordinate is always off center by a few UPM units, sometimes left and sometimes right. But it very hard to discern a few UPM units. See the right panel of Figure 5 for an illustration of increasingly bad (spoofed) X coordinates of N's above-anchor.

Note that the above-anchor of L is above the major (vertical) stem of the L. For E, F, P and R, the above-anchor is optically centered above the glyph, not above the major stem.

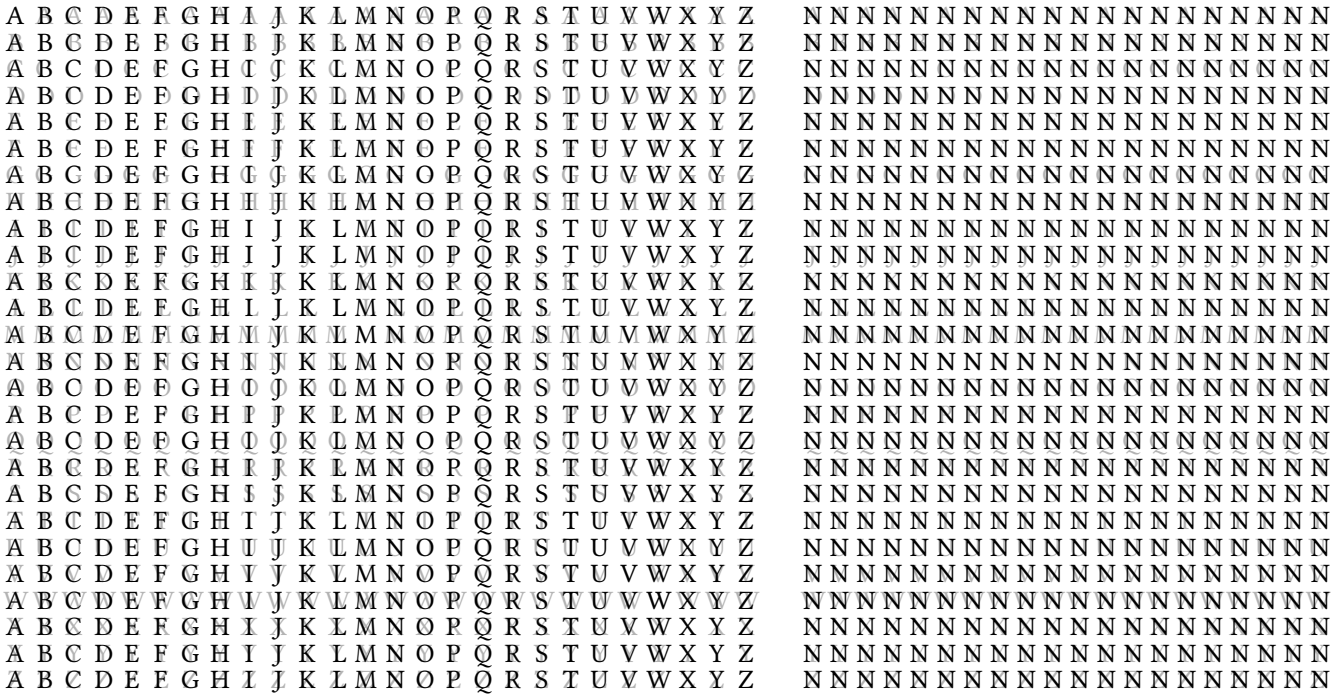


Figure 5: Above-anchor X coordinate alignment. In the left panel, the 26×26 grid shows letters A–Z aligned at the their above-anchor. The right panel shows how hard it is to discern bad X coordinates. The right panel shows N aligned with letters at their above-anchor, starting at the first column with Ns above-anchor as currently set by the font, and then spoofing increasing bad values for the X coordinate of N's above-anchor, by 5 UPM per column. At the rightmost column, it becomes evident that the (spoofed) X coordinate of N's above-anchor is bad, because in that column the black Ns are too far right of the gray letters below them.

Comma-style marks

Libertinus has a some (precomposed) glyphs for Unicode characters (with code points) where the caron or cedilla has a comma-style shape, instead of a wedge or hook as might be naively expected. But all of these are appropriate for Czech/Slovak orthography (caron or *háček*) or Latvian orthography (cedilla, but really the comma-above mark or *komatiņš*, by a known mistake in Unicode).

Czech/Slovak *háček*

In Czech/Slovak orthography, the d, l, L, n and t may take a reduced caron that may look like a comma, not a wedge. In Libertinus, only the d, L, l, and t have the reduced comma-style caron in their precomposed form, and the n has wedge caron precomposed form.

`\char"010F` ď *ď* **ď** ***ď***

`\char"013D` Ľ *Ľ* **Ľ** ***Ľ***

`\char"013E` ľ *ľ* **ľ** ***ľ***

`\char"0148` ň *ň* **ň** ***ň***

`\char"0165` ť *ť* **ť** ***ť***

Libertinus even has a custom glyph for a small capital L with a caron, and it is comma-style.

`\textsc{l\char"030C}` ʟ̬

`\textsc{\char"013E}` ʟ̬

Cedilla and Latvian *komatiņš*

Here are all precomposed characters with a cedilla.

`\char"00C7` Ç *Ç* **Ç** ***Ç***	`\char"00E7` ç *ç* **ç** ***ç***
`\char"1E08` Ḉ *Ḉ* **Ḉ** ***Ḉ***	`\char"1E08` ḉ *ḉ* **ḉ** ***ḉ***
`\char"1E10` Ḑ *Ḑ* **Ḑ** ***Ḑ***	`\char"1E11` ḑ *ḑ* **ḑ** ***ḑ***
`\char"0228` Ȩ *Ȩ* **Ȩ** ***Ȩ***	`\char"0229` ȩ *ȩ* **ȩ** ***ȩ***
`\char"1E1C` Ḝ *Ḝ* **Ḝ** ***Ḝ***	`\char"1E1D` ḝ *ḝ* **ḝ** ***ḝ***
`\char"0122` Ģ *Ģ* **Ģ** ***Ģ***	`\char"0123` ģ *ģ* **ģ** ***ģ***
`\char"1E28` Ḩ *Ḩ* **Ḩ** ***Ḩ***	`\char"1E29` ḩ *ḩ* **ḩ** ***ḩ***
`\char"0136` Ķ *Ķ* **Ķ** ***Ķ***	`\char"0137` ķ *ķ* **ķ** ***ķ***
`\char"013B` Ļ *Ļ* **Ļ** ***Ļ***	`\char"013C` ļ *ļ* **ļ** ***ļ***
`\char"0145` Ņ *Ņ* **Ņ** ***Ņ***	`\char"0146` ņ *ņ* **ņ** ***ņ***
`\char"0156` Ŗ *Ŗ* **Ŗ** ***Ŗ***	`\char"0157` ŗ *ŗ* **ŗ** ***ŗ***
`\char"015E` Ş *Ş* **Ş** ***Ş***	`\char"015F` ş *ş* **ş** ***ş***
`\char"0162` Ţ *Ţ* **Ţ** ***Ţ***	`\char"0163` ţ *ţ* **ţ** ***ţ***

These characters – Ģ ģ Ķ ķ Ļ ļ Ņ ņ Ŗ ŗ – use the Latvian *komatiņš* which is proper for Latvian orthography, but the result of a historical mistake in Unicode naming.

The Unicode characters of D and d with cedilla exist for scholarly and historical reasons, not Latvian orthography. Fonts design these Unicode characters differently. Libertinus designs them inconsistently: comma-style in regular and italic, and hook-style in (semi)bold and (semi)bold italic. I think the cedilla on a D or d should be hook-like, but I do not really know.

The other characters – c, e, h, s, t – with a cedilla should definitely be hook-like, and they are so in Libertinus.

Somewhat mysteriously, if the LaTeX source tries to combine any of these letters – c, d, e, g, h, k, l, n, r, s, t – with the combining cedilla, U+0327, then the precomposed character is produced.

`c\char"0327` ç ç **ç** ***ç***
`d\char"0327` ḑ *ḑ* **ḑ** ***ḑ***
`e\char"0327` ȩ *ȩ* **ȩ** ***ȩ***
`g\char"0327` ģ *ģ* **ģ** ***ģ***
`h\char"0327` ḫ *ḫ* **ḫ** ***ḫ***
`k\char"0327` ķ *ķ* **ķ** ***ķ***
`l\char"0327` ļ *ļ* **ļ** ***ļ***
`n\char"0327` ņ *ņ* **ņ** ***ņ***
`r\char"0327` ŗ *ŗ* **ŗ** ***ŗ***
`s\char"0327` ş *ş* **ş** ***ş***
`t\char"0327` ţ *ţ* **ţ** ***ţ***
This behavior does not make sense for the Latvian *komatiņš* bases – g, k, l, n, r – because any person trying to produce Latvian g with *komatiņš* would not try `g\char"0327` or `\c{g}`.
This behavior is tolerable for the other bases – c, d, e, h, s, t.

IJ

The ij and IJ ligatures are Unicode characters (with code points) and are supported by Libertinus.
IJ `\char"0132` U+0132 capital ligature IJ
ij `\char"0133` U+0133 small ligature ij
Libertinus has custom glyphs for these ligatures with an acute, and the small capital ligature with an acute. The GSUB table that activates these custom glyphs only works when the language is set to Dutch by a font loading command. For example,
`\newfontfamily\SerifRegularDutch[Language = Dutch]{LibertinusSerif-Regular.otf}`
ĲĲ́ `{\SerifRegularDutch \char"0132\char"0301}` Dutch capital ligature IJ with acute
ij́ `{\SerifRegularDutch \char"0133\char"0301}` Dutch small ligature ij with acute
ij́ `{\SerifRegularDutch \textsc{\char"0133\char"0301}}` Dutch small capital ligature IJ with acute
In an English LaTeX document, `ij` and `IJ` are kerned by the GPOS table to produce ij and IJ, so they look close together, but they are not substituted by the single glyph for the ligature.

Lowercase i with tilde below and grave above, in italic and semibold italic

First, start with i with tilde below (precomposed).
ḭ *ḭ* **ḭ** ***ḭ*** `\char"1E2D` i with tilde below (precomposed)
You cannot simply put a grave on it because it collides with the dot.
ḭ̀ *ḭ̀* **ḭ̀** ***ḭ̀*** `\char"1E2D\char"0300` i with tilde below (precomposed) with grave
Start over with i with grave (precomposed).
ì *ì* **ì** ***ì*** `\char"00EC` i with grave (precomposed)
Try putting a grave below.
ḭ̀ *ḭ̀* **ḭ̀** ***ḭ̀*** `\char"00EC\char"0330` i with grave (precomposed) with tilde below

This fails, because normalization breaks down the input “i with grave (precomposed) with tilde below” into “i with grave with tilde below”, then into canonical order “i with tilde below with grave”, the recomposes it into “i with tilde below (precomposed) with grave”, which reintroduces i’s dot.

Next, try to block normalization with the combining grapheme joiner (CGJ).

ḭ̀ *ḭ̀* **ḭ̀** ***ḭ̀*** `\char"00EC\cgj\char"0330` i with grave (precomposed) with combining grapheme joiner with tilde below

This looks bad because i with grave (precomposed) does not have a below-anchor and fallback shaping looks bad.

Finally, try starting with dotless i, which has OK above- and below-anchors.

ı̰̀ *ı̰̀* **ı̰̀** ***ı̰̀*** `\char"0131\char"0330\char"0300` dotless i with tilde below with grave

Normalization does not affect the dotless i, so it looks the same with the CGJ.

ı̰̀ *ı̰̀* **ı̰̀** ***ı̰̀*** `\char"0131\cgj\char"0330\char"0300` dotless i with CGJ with tilde below with grave

All together now, for comparison.

ḭìı̰̀ *ḭìı̰̀* **ḭìı̰̀** ***ḭìı̰̀*** `\char"1E2D \char"00EC \char"0131\char"0330\char"0300` i with tilde below (precomposed); i with grave (precomposed); dotless i with tilde below with grave

The same strategy works with other above marks.

ı̰̏ *ı̰̏* **ı̰̏** ***ı̰̏*** `\char"0131\char"0330\char"030F` dotless i with tilde below with double grave

ı̰́ *ı̰́* **ı̰́** ***ı̰́*** `\char"0131\char"0330\char"0301` dotless i with tilde below with acute

ı̰̋ *ı̰̋* **ı̰̋** ***ı̰̋*** `\char"0131\char"0330\char"030B` dotless i with tilde below with double acute

Lowercase a with double grave, followed by floating double grave, in italic

In italic, we want to produce a (precomposed) vowel with a double grave, followed by floating double grave. Ideally, the two double graves should be aligned. We cannot use the combining double grave as the floater, because it has nothing to combine to, and we cannot control its height. So we use a spacing character, specifically the double prime character (‶ `\char"2036`), which looks like a double grave, in italic.

ä‶ `\textit{ä\char"2036}}`

Then we lower it by a few points.

ä‶ `\textit{ä\raisebox{-0.05em}{\char"2036}}`

The distance to move the glyph can be calculated from its SFD splineset, or by trial and error.

Note that only a, e, and i have precomposed glyphs with double acute.

ä‶ `\textit{ä\raisebox{-0.05em}{\char"2036}}`

ë‶ `\textit{ë\raisebox{-0.05em}{\char"2036}}`

ë‶ `\textit{ë\raisebox{-0.045em}{\char"2036}}`

ɛ‶ `\textit{\char"025B\raisebox{-0.05em}{\char"2036}}`

Tilde below, double acute above, double grave above, on vowels, in italic and semibold italic

Consider the tilde below. There is the combining tilde below, U+0330, encoded in LaTeX as the codepoint `\char"0330`. Remember to put the base before the combining mark in LaTeX, e.g. `a\char"0330` a̰. Three vowels (e, i, u) have precomposed characters – lowercase and capital – encoded in LaTeX as these codepoints:

`\char"1E1A \char"1E1B \char"1E2C \char"1E2D \char"1E74 \char"1E75` Ḛ ḛ Ḭ ḭ Ṵ ṵ *Ḛ* *ḛ* *Ḭ* *ḭ* *Ṵ* *ṵ* **Ḛ** **ḛ** **Ḭ** **ḭ** **Ṵ** **ṵ** ***Ḛ*** ***ḛ*** ***Ḭ*** ***ḭ*** ***Ṵ*** ***ṵ***

E, e, I, i, U, or u followed by `\char"0330` will produce the precomposed character glyph. A, a, O, and o, as well as any other IPA vowel with a below-anchor, followed by `\char"0330` will be shaped by attaching the combining below tilde to that vowel at their below-anchor. Note the specific case of i. `i\char"0330 = \char"1E2D` ḭ = ḭ is a dotted i (a precomposed glyph).

Suppose you want a vowel with a tilde below, possibly also with a mark above, especially acute, grave, caron, double acute, or double grave. In the specific case of i, the above mark will crash into the dotted i unless you take care to encode the triplet starting with a dotless i, U+0131. For example, consider the combining acute U+0301.

`i\char"0330\char"0301 = \char"1E2D\char"0301` i̩ í̩ = ị ị́

`\char"0131\char"0330\char"0301` ı̩́ ị́

Recall that the canonical order for a triplet of a base and two combining marks (one below and one above) is base + combining below mark + combining above mark, not base + above + below.

Consider the wrong order:

`i\char"0301\char"0330` í̩ i\char"0301 í i\char"0330 í̩

`\char"00EC\char"0330` ì̩ `\char"00EC` ì ì̩

`\char"1E2D\char"0301` ḭ ḭ `\char"1E2D` ḭ ḭ

But note that the position of the combining tilde below is bad in italic and semibold italic. This is because the below-anchor for the combining tilde below mark is bad. For tilde below, the precomposed character glyphs with i, e, and u are sometimes good and sometimes bad in italic and semibold italic.

Precomposed characters are blue and therefore cannot be changed unless their glyph outline is changed. Black combinations of vowels and marks are shaped by anchors and therefore can be changed by setting good anchors.

Use the combining grapheme joiner (CGJ) character U+034F, `\char"034F`, or `\cgj` if loaded by some package, to force the shaper to use combining marks instead of precomposed glyphs.

A E I O U Ɛ Ɔ a e i o u ɛ ɔ *A E I O U Ɛ Ɔ a e i o u ɛ ɔ* **A E I O U Ɛ Ɔ a e i o u ɛ ɔ** ***A E I O U Ɛ Ɔ a e i o u ɛ ɔ***

1 Á É Í Ó Ú Ɛ Ɔ á é í ó ú ɛ ɔ *Á É Í Ó Ú Ɛ Ɔ á é í ó ú ɛ ɔ* Á̩ É̩ Í̩ Ó̩ Ú̩ Ɛ̩ Ɔ̩ á̩ é̩ í̩ ó̩ ú̩ ɛ̩ ɔ̩ *Á̩ É̩ Í̩ Ó̩ Ú̩ Ɛ̩ Ɔ̩ á̩ é̩ í̩ ó̩ ú̩ ɛ̩ ɔ̩*

2 À È Ì Ò Ù Ɛ Ɔ à è ì ò ù ɛ ɔ *À È Ì Ò Ù Ɛ Ɔ à è ì ò ù ɛ ɔ* À̩ È̩ Ì̩ Ò̩ Ù̩ Ɛ̩ Ɔ̩ à̩ è̩ ì̩ ò̩ ù̩ ɛ̩ ɔ̩ *À̩ È̩ Ì̩ Ò̩ Ù̩ Ɛ̩ Ɔ̩ à̩ è̩ ì̩ ò̩ ù̩ ɛ̩ ɔ̩*

3 Ă Ĕ Ĭ Ŏ Ŭ Ɛ̆ Ɔ̆ ă ĕ ĭ ŏ ŭ ɛ̆ ɔ̆ *Ă Ĕ Ĭ Ŏ Ŭ Ɛ̆ Ɔ̆ ă ĕ ĭ ŏ ŭ ɛ̆ ɔ̆* Ă̩ Ĕ̩ Ĭ̩ Ŏ̩ Ŭ̩ Ɛ̩̆ Ɔ̩̆ ă̩ ĕ̩ ĭ̩ ŏ̩ ŭ̩ ɛ̩̆ ɔ̩̆ *Ă̩ Ĕ̩ Ĭ̩ Ŏ̩ Ŭ̩ Ɛ̩̆ Ɔ̩̆ ă̩ ĕ̩ ĭ̩ ŏ̩ ŭ̩ ɛ̩̆ ɔ̩̆*

4 Ắ Ế Ḯ Ố Ű Ɛ́ Ɔ́ ắ ế ḯ ố ű ɛ́ ɔ́ *Ắ Ế Ḯ Ố Ű Ɛ́ Ɔ́ ắ ế ḯ ố ű ɛ́ ɔ́* Ắ̩ Ế̩ Ḯ̩ Ố̩ Ű̩ Ɛ̩́ Ɔ̩́ ắ̩ ế̩ ḯ̩ ố̩ ű̩ ɛ̩́ ɔ̩́ *Ắ̩ Ế̩ Ḯ̩ Ố̩ Ű̩ Ɛ̩́ Ɔ̩́ ắ̩ ế̩ ḯ̩ ố̩ ű̩ ɛ̩́ ɔ̩́*

5 Ằ Ề Ỉ Ồ Ừ Ɛ̀ Ɔ̀ ằ ề ỉ ồ ừ ɛ̀ ɔ̀ *Ằ Ề Ỉ Ồ Ừ Ɛ̀ Ɔ̀ ằ ề ỉ ồ ừ ɛ̀ ɔ̀* Ằ̩ Ề̩ Ỉ̩ Ồ̩ Ừ̩ Ɛ̩̀ Ɔ̩̀ ằ̩ ề̩ ỉ̩ ồ̩ ừ̩ ɛ̩̀ ɔ̩̀ *Ằ̩ Ề̩ Ỉ̩ Ồ̩ Ừ̩ Ɛ̩̀ Ɔ̩̀ ằ̩ ề̩ ỉ̩ ồ̩ ừ̩ ɛ̩̀ ɔ̩̀*

6 Ạ Ệ Ị Ọ Ự Ɛ̣ Ɔ̣ ạ ệ ị ọ ự ɛ̣ ɔ̣ *Ạ Ệ Ị Ọ Ự Ɛ̣ Ɔ̣ ạ ệ ị ọ ự ɛ̣ ɔ̣* Ạ Ệ Ị Ọ Ự Ɛ̣ Ɔ̣ ạ ệ ị ọ ự ɛ̣ ɔ̣ *Ạ Ệ Ị Ọ Ự Ɛ̣ Ɔ̣ ạ ệ ị ọ ự ɛ̣ ɔ̣*

7 Ấ Ế Ố Ứ Ɛ́ Ɔ́ á é í ó ú ɛ́ ɔ́ *Ấ Ế Ố Ứ Ɛ́ Ɔ́ á é í ó ú ɛ́ ɔ́* Ấ̩ Ế̩ Ố̩ Ứ̩ Ɛ̩́ Ɔ̩́ á̩ é̩ í̩ ó̩ ú̩ ɛ̩́ ɔ̩́ *Ấ̩ Ế̩ Ố̩ Ứ̩ Ɛ̩́ Ɔ̩́ á̩ é̩ í̩ ó̩ ú̩ ɛ̩́ ɔ̩́*

7c Ấ Ế Ố Ứ Ɛ́ Ɔ́ á é í ó ú ɛ́ ɔ́ *Ấ Ế Ố Ứ Ɛ́ Ɔ́ á é í ó ú ɛ́ ɔ́* Ấ̩ Ế̩ Ố̩ Ứ̩ Ɛ̩́ Ɔ̩́ á̩ é̩ í̩ ó̩ ú̩ ɛ̩́ ɔ̩́ *Ấ̩ Ế̩ Ố̩ Ứ̩ Ɛ̩́ Ɔ̩́ á̩ é̩ í̩ ó̩ ú̩ ɛ̩́ ɔ̩́*

8 Ầ Ề Ồ Ừ Ɛ̀ Ɔ̀ à è ì ò ù ɛ̀ ɔ̀ *Ầ Ề Ồ Ừ Ɛ̀ Ɔ̀ à è ì ò ù ɛ̀ ɔ̀* Ầ̩ Ề̩ Ồ̩ Ừ̩ Ɛ̩̀ Ɔ̩̀ à̩ è̩ ì̩ ò̩ ù̩ ɛ̩̀ ɔ̩̀ *Ầ̩ Ề̩ Ồ̩ Ừ̩ Ɛ̩̀ Ɔ̩̀ à̩ è̩ ì̩ ò̩ ù̩ ɛ̩̀ ɔ̩̀*

9 Ặ Ệ Ị Ọ Ự Ɛ̣̆ Ɔ̣̆ ặ ệ ị ọ ự ɛ̣̆ ɔ̣̆ *Ặ Ệ Ị Ọ Ự Ɛ̣̆ Ɔ̣̆ ặ ệ ị ọ ự ɛ̣̆ ɔ̣̆* Ặ Ệ Ị Ọ Ự Ɛ̣̆ Ɔ̣̆ ặ ệ ị ọ ự ɛ̣̆ ɔ̣̆ *Ặ Ệ Ị Ọ Ự Ɛ̣̆ Ɔ̣̆ ặ ệ ị ọ ự ɛ̣̆ ɔ̣̆*

10 Ắ Ệ Ị Ọ Ự Ɛ̣́ Ɔ̣́ ắ ệ ị ọ ự ɛ̣́ ɔ̣́ *Ắ Ệ Ị Ọ Ự Ɛ̣́ Ɔ̣́ ắ ệ ị ọ ự ɛ̣́ ɔ̣́* Ắ Ệ Ị Ọ Ự Ɛ̣́ Ɔ̣́ ắ ệ ị ọ ự ɛ̣́ ɔ̣́ *Ắ Ệ Ị Ọ Ự Ɛ̣́ Ɔ̣́ ắ ệ ị ọ ự ɛ̣́ ɔ̣́*

11 Ằ Ệ Ị Ọ Ự Ɛ̣̀ Ɔ̣̀ ằ ệ ị ọ ự ɛ̣̀ ɔ̣̀ *Ằ Ệ Ị Ọ Ự Ɛ̣̀ Ɔ̣̀ ằ ệ ị ọ ự ɛ̣̀ ɔ̣̀* Ằ Ệ Ị Ọ Ự Ɛ̣̀ Ɔ̣̀ ằ ệ ị ọ ự ɛ̣̀ ɔ̣̀ *Ằ Ệ Ị Ọ Ự Ɛ̣̀ Ɔ̣̀ ằ ệ ị ọ ự ɛ̣̀ ɔ̣̀*

Patched

A E I O U Ɛ Ɔ a e i o u ɛ ɔ *A E I O U Ɛ Ɔ a e i o u ɛ ɔ* **A E I O U Ɛ Ɔ a e i o u ɛ ɔ** ***A E I O U Ɛ Ɔ a e i o u ɛ ɔ***

Á É Í Ó Ú Ɛ́ Ɔ́ á é í ó ú ɛ́ ɔ́ *Á É Í Ó Ú Ɛ́ Ɔ́ á é í ó ú ɛ́ ɔ́* Á̩ É̩ Í̩ Ó̩ Ú̩ Ɛ̩́ Ɔ̩́ á̩ é̩ í̩ ó̩ ú̩ ɛ̩́ ɔ̩́ *Á̩ É̩ Í̩ Ó̩ Ú̩ Ɛ̩́ Ɔ̩́ á̩ é̩ í̩ ó̩ ú̩ ɛ̩́ ɔ̩́*

À È Ì Ò Ù Ɛ̀ Ɔ̀ à è ì ò ù ɛ̀ ɔ̀ *À È Ì Ò Ù Ɛ̀ Ɔ̀ à è ì ò ù ɛ̀ ɔ̀* À̩ È̩ Ì̩ Ò̩ Ù̩ Ɛ̩̀ Ɔ̩̀ à̩ è̩ ì̩ ò̩ ù̩ ɛ̩̀ ɔ̩̀ *À̩ È̩ Ì̩ Ò̩ Ù̩ Ɛ̩̀ Ɔ̩̀ à̩ è̩ ì̩ ò̩ ù̩ ɛ̩̀ ɔ̩̀*

Ă Ĕ Ĭ Ŏ Ŭ Ɛ̆ Ɔ̆ ă ĕ ĭ ŏ ŭ ɛ̆ ɔ̆ *Ă Ĕ Ĭ Ŏ Ŭ Ɛ̆ Ɔ̆ ă ĕ ĭ ŏ ŭ ɛ̆ ɔ̆* Ă̩ Ĕ̩ Ĭ̩ Ŏ̩ Ŭ̩ Ɛ̩̆ Ɔ̩̆ ă̩ ĕ̩ ĭ̩ ŏ̩ ŭ̩ ɛ̩̆ ɔ̩̆ *Ă̩ Ĕ̩ Ĭ̩ Ŏ̩ Ŭ̩ Ɛ̩̆ Ɔ̩̆ ă̩ ĕ̩ ĭ̩ ŏ̩ ŭ̩ ɛ̩̆ ɔ̩̆*

Ắ Ế Ḯ Ố Ű Ɛ́ Ɔ́ ắ ế ḯ ố ű ɛ́ ɔ́ *Ắ Ế Ḯ Ố Ű Ɛ́ Ɔ́ ắ ế ḯ ố ű ɛ́ ɔ́* Ắ̩ Ế̩ Ḯ̩ Ố̩ Ű̩ Ɛ̩́ Ɔ̩́ ắ̩ ế̩ ḯ̩ ố̩ ű̩ ɛ̩́ ɔ̩́ *Ắ̩ Ế̩ Ḯ̩ Ố̩ Ű̩ Ɛ̩́ Ɔ̩́ ắ̩ ế̩ ḯ̩ ố̩ ű̩ ɛ̩́ ɔ̩́*

Ằ Ề Ỉ Ồ Ừ Ɛ̀ Ɔ̀ ằ ề ỉ ồ ừ ɛ̀ ɔ̀ *Ằ Ề Ỉ Ồ Ừ Ɛ̀ Ɔ̀ ằ ề ỉ ồ ừ ɛ̀ ɔ̀* Ằ̩ Ề̩ Ỉ̩ Ồ̩ Ừ̩ Ɛ̩̀ Ɔ̩̀ ằ̩ ề̩ ỉ̩ ồ̩ ừ̩ ɛ̩̀ ɔ̩̀ *Ằ̩ Ề̩ Ỉ̩ Ồ̩ Ừ̩ Ɛ̩̀ Ɔ̩̀ ằ̩ ề̩ ỉ̩ ồ̩ ừ̩ ɛ̩̀ ɔ̩̀*

Ạ Ệ Ị Ọ Ự Ɛ̣ Ɔ̣ ạ ệ ị ọ ự ɛ̣ ɔ̣ *Ạ Ệ Ị Ọ Ự Ɛ̣ Ɔ̣ ạ ệ ị ọ ự ɛ̣ ɔ̣* Ạ Ệ Ị Ọ Ự Ɛ̣ Ɔ̣ ạ ệ ị ọ ự ɛ̣ ɔ̣ *Ạ Ệ Ị Ọ Ự Ɛ̣ Ɔ̣ ạ ệ ị ọ ự ɛ̣ ɔ̣*

cgj Ạ Ệ Ị Ọ Ự Ɛ̣ Ɔ̣ ạ ệ ị ọ ự ɛ̣ ɔ̣ *Ạ Ệ Ị Ọ Ự Ɛ̣ Ɔ̣ ạ ệ ị ọ ự ɛ̣ ɔ̣* Ạ Ệ Ị Ọ Ự Ɛ̣ Ɔ̣ ạ ệ ị ọ ự ɛ̣ ɔ̣ *Ạ Ệ Ị Ọ Ự Ɛ̣ Ɔ̣ ạ ệ ị ọ ự ɛ̣ ɔ̣*

Ą́ Ę́ Į́ Ǫ́ Ų́ Ȩ́ Ɔ́ ą́ ę́ į́ ǫ́ ų́ ȩ́ ɔ́ *Ą́ Ę́ Į́ Ǫ́ Ų́ Ȩ́ Ɔ́ ą́ ę́ į́ ǫ́ ų́ ȩ́ ɔ́* **Ą́ Ę́ Į́ Ǫ́ Ų́ Ȩ́ Ɔ́ ą́ ę́ į́ ǫ́ ų́ ȩ́ ɔ́** ***Ą́ Ę́ Į́ Ǫ́ Ų́ Ȩ́ Ɔ́ ą́ ę́ į́ ǫ́ ų́ ȩ́ ɔ́***
cgj Ą́ Ę́ Į́ Ǫ́ Ų́ Ȩ́ Ɔ́ ą́ ę́ į́ ǫ́ ų́ ȩ́ ɔ́ *Ą́ Ę́ Į́ Ǫ́ Ų́ Ȩ́ Ɔ́ ą́ ę́ į́ ǫ́ ų́ ȩ́ ɔ́* **Ą́ Ę́ Į́ Ǫ́ Ų́ Ȩ́ Ɔ́ ą́ ę́ į́ ǫ́ ų́ ȩ́ ɔ́** ***Ą́ Ę́ Į́ Ǫ́ Ų́ Ȩ́ Ɔ́ ą́ ę́ į́ ǫ́ ų́ ȩ́ ɔ́***
Ą́ Ę́ Į́ Ǫ́ Ų́ Ȩ́ Ɔ́ ą́ ę́ į́ ǫ́ ų́ ȩ́ ɔ́ *Ą́ Ę́ Į́ Ǫ́ Ų́ Ȩ́ Ɔ́ ą́ ę́ į́ ǫ́ ų́ ȩ́ ɔ́* **Ą́ Ę́ Į́ Ǫ́ Ų́ Ȩ́ Ɔ́ ą́ ę́ į́ ǫ́ ų́ ȩ́ ɔ́** ***Ą́ Ę́ Į́ Ǫ́ Ų́ Ȩ́ Ɔ́ ą́ ę́ į́ ǫ́ ų́ ȩ́ ɔ́***
Ą̀ Ę̀ Į̀ Ǫ̀ Ų̀ Ȩ̀ Ɔ̀ ą̀ ę̀ į̀ ǫ̀ ų̀ ȩ̀ ɔ̀ *Ą̀ Ę̀ Į̀ Ǫ̀ Ų̀ Ȩ̀ Ɔ̀ ą̀ ę̀ į̀ ǫ̀ ų̀ ȩ̀ ɔ̀* **Ą̀ Ę̀ Į̀ Ǫ̀ Ų̀ Ȩ̀ Ɔ̀ ą̀ ę̀ į̀ ǫ̀ ų̀ ȩ̀ ɔ̀** ***Ą̀ Ę̀ Į̀ Ǫ̀ Ų̀ Ȩ̀ Ɔ̀ ą̀ ę̀ į̀ ǫ̀ ų̀ ȩ̀ ɔ̀***
cgj Ą̀ Ę̀ Į̀ Ǫ̀ Ų̀ Ȩ̀ Ɔ̀ ą̀ ę̀ į̀ ǫ̀ ų̀ ȩ̀ ɔ̀ *Ą̀ Ę̀ Į̀ Ǫ̀ Ų̀ Ȩ̀ Ɔ̀ ą̀ ę̀ į̀ ǫ̀ ų̀ ȩ̀ ɔ̀* **Ą̀ Ę̀ Į̀ Ǫ̀ Ų̀ Ȩ̀ Ɔ̀ ą̀ ę̀ į̀ ǫ̀ ų̀ ȩ̀ ɔ̀** ***Ą̀ Ę̀ Į̀ Ǫ̀ Ų̀ Ȩ̀ Ɔ̀ ą̀ ę̀ į̀ ǫ̀ ų̀ ȩ̀ ɔ̀***
Ą̆ Ę̆ Į̆ Ǫ̆ Ų̆ Ḝ Ɔ̆ ą̆ ę̆ į̆ ǫ̆ ų̆ ḝ ɔ̆ *Ą̆ Ę̆ Į̆ Ǫ̆ Ų̆ Ḝ Ɔ̆ ą̆ ę̆ į̆ ǫ̆ ų̆ ḝ ɔ̆* **Ą̆ Ę̆ Į̆ Ǫ̆ Ų̆ Ḝ Ɔ̆ ą̆ ę̆ į̆ ǫ̆ ų̆ ḝ ɔ̆** ***Ą̆ Ę̆ Į̆ Ǫ̆ Ų̆ Ḝ Ɔ̆ ą̆ ę̆ į̆ ǫ̆ ų̆ ḝ ɔ̆***
cgj Ą̆ Ę̆ Į̆ Ǫ̆ Ų̆ Ḝ Ɔ̆ ą̆ ę̆ į̆ ǫ̆ ų̆ ḝ ɔ̆ *Ą̆ Ę̆ Į̆ Ǫ̆ Ų̆ Ḝ Ɔ̆ ą̆ ę̆ į̆ ǫ̆ ų̆ ḝ ɔ̆* **Ą̆ Ę̆ Į̆ Ǫ̆ Ų̆ Ḝ Ɔ̆ ą̆ ę̆ į̆ ǫ̆ ų̆ ḝ ɔ̆** ***Ą̆ Ę̆ Į̆ Ǫ̆ Ų̆ Ḝ Ɔ̆ ą̆ ę̆ į̆ ǫ̆ ų̆ ḝ ɔ̆***
Ą̋ Ę̋ Į̋ Ǫ̋ Ų̋ Ȩ̋ Ɔ̋ ą̋ ę̋ į̋ ǫ̋ ų̋ ȩ̋ ɔ̋ *Ą̋ Ę̋ Į̋ Ǫ̋ Ų̋ Ȩ̋ Ɔ̋ ą̋ ę̋ į̋ ǫ̋ ų̋ ȩ̋ ɔ̋* **Ą̋ Ę̋ Į̋ Ǫ̋ Ų̋ Ȩ̋ Ɔ̋ ą̋ ę̋ į̋ ǫ̋ ų̋ ȩ̋ ɔ̋** ***Ą̋ Ę̋ Į̋ Ǫ̋ Ų̋ Ȩ̋ Ɔ̋ ą̋ ę̋ į̋ ǫ̋ ų̋ ȩ̋ ɔ̋***
cgj Ą̋ Ę̋ Į̋ Ǫ̋ Ų̋ Ȩ̋ Ɔ̋ ą̋ ę̋ į̋ ǫ̋ ų̋ ȩ̋ ɔ̋ *Ą̋ Ę̋ Į̋ Ǫ̋ Ų̋ Ȩ̋ Ɔ̋ ą̋ ę̋ į̋ ǫ̋ ų̋ ȩ̋ ɔ̋* **Ą̋ Ę̋ Į̋ Ǫ̋ Ų̋ Ȩ̋ Ɔ̋ ą̋ ę̋ į̋ ǫ̋ ų̋ ȩ̋ ɔ̋** ***Ą̋ Ę̋ Į̋ Ǫ̋ Ų̋ Ȩ̋ Ɔ̋ ą̋ ę̋ į̋ ǫ̋ ų̋ ȩ̋ ɔ̋***
Ą̈ Ę̈ Į̈ Ǫ̈ Ų̈ Ȩ̈ Ɔ̈ ą̈ ę̈ į̈ ǫ̈ ų̈ ȩ̈ ɔ̈ *Ą̈ Ę̈ Į̈ Ǫ̈ Ų̈ Ȩ̈ Ɔ̈ ą̈ ę̈ į̈ ǫ̈ ų̈ ȩ̈ ɔ̈* **Ą̈ Ę̈ Į̈ Ǫ̈ Ų̈ Ȩ̈ Ɔ̈ ą̈ ę̈ į̈ ǫ̈ ų̈ ȩ̈ ɔ̈** ***Ą̈ Ę̈ Į̈ Ǫ̈ Ų̈ Ȩ̈ Ɔ̈ ą̈ ę̈ į̈ ǫ̈ ų̈ ȩ̈ ɔ̈***
cgj Ą̈ Ę̈ Į̈ Ǫ̈ Ų̈ Ȩ̈ Ɔ̈ ą̈ ę̈ į̈ ǫ̈ ų̈ ȩ̈ ɔ̈ *Ą̈ Ę̈ Į̈ Ǫ̈ Ų̈ Ȩ̈ Ɔ̈ ą̈ ę̈ į̈ ǫ̈ ų̈ ȩ̈ ɔ̈* **Ą̈ Ę̈ Į̈ Ǫ̈ Ų̈ Ȩ̈ Ɔ̈ ą̈ ę̈ į̈ ǫ̈ ų̈ ȩ̈ ɔ̈** ***Ą̈ Ę̈ Į̈ Ǫ̈ Ų̈ Ȩ̈ Ɔ̈ ą̈ ę̈ į̈ ǫ̈ ų̈ ȩ̈ ɔ̈***
Patched not grave
À È̀ Ì Ò Ù Ɛ̀ Ɔ̀ à è̀ ì ò ù ɛ̀ ɔ̀ *À È̀ Ì Ò Ù Ɛ̀ Ɔ̀ à è̀ ì ò ù ɛ̀ ɔ̀* **À È̀ Ì Ò Ù Ɛ̀ Ɔ̀ à è̀ ì ò ù ɛ̀ ɔ̀** ***À È̀ Ì Ò Ù Ɛ̀ Ɔ̀ à è̀ ì ò ù ɛ̀ ɔ̀***
Patched grave
À È̀ Ì Ò Ù Ɛ̀ Ɔ̀ à è̀ ì ò ù ɛ̀ ɔ̀ *À È̀ Ì Ò Ù Ɛ̀ Ɔ̀ à è̀ ì ò ù ɛ̀ ɔ̀* **À È̀ Ì Ò Ù Ɛ̀ Ɔ̀ à è̀ ì ò ù ɛ̀ ɔ̀** ***À È̀ Ì Ò Ù Ɛ̀ Ɔ̀ à è̀ ì ò ù ɛ̀ ɔ̀***

Mark over space

I added a new en-width x-height PUA character, U+E100, with anchors. To float an above mark, say double grave U+030F over a space, write `\char"E100\char"030F` But in practice, the floating mark carries the same tonal specification as the preceding vowel, and therefore must align exactly with the height of that vowel's combining mark. To ensure this mark alignment, insert a combining grapheme joiner, `\cgj`, in the prior vowel + mark combo, to force anchor positioning without substituting a precomposed glyph there. Note that in Libertinus, the two graves in the U+030F glyph are not translates of an identical outline; this is a deliberate design choice by the designer.

Original

`-ke\char"030F\ \char"030F` -kȅ̏ *-kȅ̏* **-kȅ̏** ***-kȅ̏***

Patched without CGJ

`-ke\char"030F \char"E100\char"030F` -kȅ̏ *-kȅ̏* **-kȅ̏** ***-kȅ̏***

Patched with CGJ

`-ke\cgj\char"030F \char"E100\char"030F` -kȅ̏ *-kȅ̏* **-kȅ̏** ***-kȅ̏***

f with dot below

tiɣə́ ṛfa

tiɣə́ ṛfa

tiɣə́ ṛfa

tiɣə́ ṛfa

ṛ = `\char"1E5B` is the precomposed character U+1E5B Latin small letter r with dot below.

f̣ = `f\char"0323` is f with U+0323 combining dot below.

Here are some letters with a dot below.

ạ ḅ ḍ ẹ ḥ ị ḳ ḷ ṃ ṇ ọ ṛ ṣ ṭ ụ ṿ ẉ ỵ ẓ ç f̧ g̦ j̨ p̧ q̧ x̧

ạ ḅ ḍ ẹ ḥ ị ḳ ḷ ṃ ṇ ọ ṛ ṣ ṭ ụ ṿ ẉ ỵ ẓ ç f̧ g̦ j̨ p̧ q̧ x̧

ạ ḅ ḍ ẹ ḥ ị ḳ ḷ ṃ ṇ ọ ṛ ṣ ṭ ụ ṿ ẉ ỵ ẓ ç f̧ g̦ j̨ p̧ q̧ x̧

ạ ḅ ḍ ẹ ḥ ị ḳ ḷ ṃ ṇ ọ ṛ ṣ ṭ ụ ṿ ẉ ỵ ẓ ç f̧ g̦ j̨ p̧ q̧ x̧

The left group of letters – a b d e h i k l m n o r s t u v w y z – have a precomposed character. For the right group of letters – c f g j p q x – the dot is placed according to the below-anchor (if the below-anchor is defined for the letter) or by HarfBuzz fallback shaping (placing the dot below the baseline, regardless of the letter's bounding box).

In all styles – regular, italic, semibold, and semibold italic – the precomposed glyph of y with dot below places the dot beside the descender, even though the y has a below-anchor below the descender.

Do the letters c, f, g, j, p, q, and x have good below-anchors? In regular, the below-anchor of all these combining bases is below the glyph outline (good). In italic, the below-anchor of f, p, and q is beside the descender (bad), the below-anchor of g and j is below the descender (good), and the below-anchor of c and x is below the baseline (good). In semibold, the below-anchor of g and j is below the descender (good), but the other bases – c, f, p, q – have no below-anchor, and HarfBuzz fallback positioning places the combining dot below the baseline, not below the bounding box; for c, f and x this is good, but for p and q this is bad, because of the descenders. In semibold italic, the below-anchor of c, g, j, p and q is below the glyph outline (good), the below-anchor of f is inside the descender (very bad), and x has no below-anchor but HarfBuzz fallback positioning places the combining dot below the baseline (good).

Which dots are used in precomposed characters and how big are the dots? In all styles (except semibold italic), the precomposed characters reference the dot of `uni0307`, combining dot above, U+0307, and take care to translate this above-mark to a position below the letter outline, except for m, which references `dotbelowcomb`, combining dot below, U+0323. In semibold italic, `dotaccent` is used instead of `uni0307`. These two dots are four-point Bézier circles (in regular and semibold, but ovals in italic and semibold italic), but of different diameter. In regular, `uni0307` has diameter 100, and `dotbelowcomb` has diameter 108. That is, the precomposed characters (except m) use a smaller dot, and the non-precomposed letters (and m) take a bigger dot. In semibold, the size difference is reversed: `uni0307` has larger diameter 124, and `dotbelowcomb` has small diameter 108. Actually, semibold `dotbelowcomb` is not a Bézier circle – the horizontal span is 109, and the vertical span is 108. In semibold italic, `dotaccent` is a skewed four-point Bézier oval – the horizontal and vertical spans are 120, offset by $(\pm 6, \pm 7.5)$ from the center — and `dotbelowcomb` is slightly larger, slightly flatter skewed six-point Bézier oval – horizontal span is 122 and the vertical span is 120. So, in semibold italic the two dots in precomposed and combining form are closest in size. Recall that f has a bad below-anchor, inside the descender, but g, j, p and q have a good below-anchor, below the glyph outline.

So, in order to solve the problem of the bad dot below f in italic, I need to decide the scope of the badness to correct. The simplest patch is move the below-anchor in italic and semibold italic below the descender. But this does not solve inconsistent below dot size across styles and letters. In italic and semibold italic, g and j have a (good) below-anchor at $Y = -319$, so that is also good Y coordinate for f's below-anchor.

I asked Copilot/GPT-5.1 to calculate a good X coordinate for f's below-anchor in italic and semibold italic, taking into account the U-pocket of f's descender and the below-anchor of `dotbelowcomb`. I decided that this U-pocket strategy was appropriate after testing other strategies that accounted for the slant of onf the major stroke. Here is a reasonbly faithful summary of the calculations and reasoning by Copilot/GPT-5.1.

> For italic, the descender's U-pocket is defined by the control points at $(14, -238)$ and $(33, -205)$, giving a U-center around $x \approx 24$. Using the (symmetric) italic `dotbelowcomb` anchor as the dot's optical center, the dot should sit under this U-pocket, not under the ball or the stem. A balanced, U-centered, slightly inner-biased choice is $X \approx 30$.
>
> For semibold italic, the U-pocket of the semibold italic descender is wider and more asymmetric, with a geometric U-center of $X = -33.5$. The semibold italic `dotbelowcomb` itself is also skewed, with its anchor already shifted relative to its mass. In practice, the visually correct position for the dot below f is under the open side of the U-pocket, clearly right of the ball and closer the right of the stem. The anchor candidate is $X \approx 62$.

Here are examples with the patched fonts, with f's below-anchor in italic at $(30, -319)$ and semibold italic at $(62, -319)$. This does not solve the problem of inconsistent dot size below letters, and I did not touch p, q, or y.

Original	Patched italics (f only)
tiɣə ɾfa	tiɣə ɾfa
tiɣə ɾfa	*tiɣə ɾfa*
tiɣə ɾfa	**tiɣə ɾfa**
tiɣə ɾfa	***tiɣə ɾfa***

Original (repeated from above)

ạ ḅ ḍ ẹ ḥ ị ḳ ḷ ṃ ṇ ọ ṛ ṣ ṭ ụ ṿ ẉ ỵ ẓ ç f ģ j ƥ q ẋ

ạ ḅ ḍ ẹ ḥ ị ḳ ḷ ṃ ṇ ọ ṛ ṣ ṭ ụ ṿ ẉ ỵ ẓ ç f ģ j ƥ q ẋ

ạ ḅ ḍ ẹ ḥ ị ḳ ḷ ṃ ṇ ọ ṛ ṣ ṭ ụ ṿ ẉ ỵ ẓ ç f ģ j ƥ q ẋ

ạ ḅ ḍ ẹ ḥ ị ḳ ḷ ṃ ṇ ọ ṛ ṣ ṭ ụ ṿ ẉ ỵ ẓ ç f ģ j ƥ q ẋ

Patched italics (f only)

ạ ḅ ḍ ẹ ḥ ị ḳ ḷ ṃ ṇ ọ ṛ ṣ ṭ ụ ṿ ẉ ỵ ẓ ç f ģ j ƥ q ẋ

ạ ḅ ḍ ẹ ḥ ị ḳ ḷ ṃ ṇ ọ ṛ ṣ ṭ ụ ṿ ẉ ỵ ẓ ç f ģ j ƥ q ẋ

Ashkun

`a\char"0304\char"0303` ã *ã* **ã** ***ã***

`\char"0101\char"0303` ã *ã* **ã** ***ã***

`u\char"0304\char"0303` ũ *ũ* **ũ** ***ũ***

`\char"016B\char"0303` ũ *ũ* **ũ** ***ũ***

Patched

`u\char"0304\char"0303` ũ *ũ* **ũ** ***ũ***

`u\cgj\char"0304\char"0303` ũ *ũ* **ũ** ***ũ***

`\char"016B\cgj\char"0303` ũ *ũ* **ũ** ***ũ***

`\char"016B\char"0303` ũ *ũ* **ũ** ***ũ***

Move box without patching font

`\char"016B\kern-4.132pt\raisebox{0.88pt}{\char"02DC}` ũ *ũ* **ũ** ***ũ***

Tacks

a̰ḛḭo̰ṵa̰ḛḭo̰ṵ

a̰ḛḭo̰ṵa̰ḛḭo̰ṵ

Upsilon and grave

Original, then patched, versions of upsilon and combining grave, after setting upsilon's above-anchor.

ὺ 𝜐́ **ὺ** **𝝊́**

ὺ 𝜐́ **ὺ** **𝝊́**

New upsilon above-anchor: regular $(253, 645)$; italic $(358, 645)$; semibold $(267, 645)$; semibold italic $(369, 645)$

Revised combining grave above-anchor: semibold italic $(0, 683)$. Moved from $(-142, 683)$, which is far left of the optical center of the mark, which has the effect of attaching the mark too far right.

Spoof to move modifier grave (which has the exact same outline as combining grave) to an anchored position, if upsilon had an above-anchor at $(358, 645)$ (keeping combining grave's above-anchor as is). Note upsilon is U+028A and modifier grave is U+02CB.

```
\textit{\char"028A\kern-4.66pt\raisebox{-0.38pt}{\char"02CB}}
```



Patch superscript consonant anchors

Libertinus serif regular

ʰ ʱ ʲ ʳ ʴ ʵ ʷ ʸ ˠ ˢ ˣ ˤ ᵇ ᵏ ᵐ ᵖ ᵗ ᵚ ᵛ ᵡ ᵝ ᵞ ᵟ

ʰ ʱ ʲ ʳ ʴ ʵ ʷ ʸ ˠ ˢ ˣ ˤ ᵇ ᵏ ᵐ ᵖ ᵗ ᵚ ᵛ ᵡ ᵝ ᵞ ᵟ

ʰ ʱ ʲ ʳ ʴ ʵ ʷ ʸ ˠ ˢ ˣ ˤ ᵇ ᵏ ᵐ ᵖ ᵗ ᵚ ᵛ ᵡ ᵝ ᵞ ᵟ

ḥ ḥ̱ ⱼ ṛ ṛ̱ ṣ ṣ̱ ẉ ẏ ẓ ẓ̱ ḅ ḳ ṃ ṗ ṭ ṭ̱ ṿ ᵡ ᵝ ᵞ ᵟ

ḥ ḥ̱ ⱼ ṛ ṛ̱ ṣ ṣ̱ ẃ ý ź ẕ́ ḅ ḳ ḿ ṕ ṭ ṭ̱ ṽ ᵡ ᵝ ᵞ ᵟ

ḥ ḥ̱ ⱼ̂ ṛ̂ ṛ̱̂ ṣ̂ ṣ̱̂ ẅ̂ ŷ ẑ ẕ̂ ḇ̂ ḳ̂ m̂ p̂ t̂ ṭ̱̂ v̂ ᵡ ᵝ ᵞ ᵟ

Libertinus serif regular, patched

ʰ᾽ ʱ᾽ ʲ᾽ ʳ᾽ ʴ᾽ ʵ᾽ ʷ᾽ ʸ᾽ ˠ᾽ ˢ᾽ ˣ᾽ ˤ᾽ ᵇ᾽ ᵏ᾽ ᵐ᾽ ᵖ᾽ ᵗ᾽ ᵚ᾽ ᵛ᾽ ᵡ᾽ ᵝ᾽ ᵞ᾽ ᵟ᾽

ʰ᾽ ʱ᾽ ʲ᾽ ʳ᾽ ʴ᾽ ʵ᾽ ʷ᾽ ʸ᾽ ˠ᾽ ˢ᾽ ˣ᾽ ˤ᾽ ᵇ᾽ ᵏ᾽ ᵐ᾽ ᵖ᾽ ᵗ᾽ ᵚ᾽ ᵛ᾽ ᵡ᾽ ᵝ᾽ ᵞ᾽ ᵟ᾽

ʰ᾽ ʱ᾽ ʲ᾽ ʳ᾽ ʴ᾽ ʵ᾽ ʷ᾽ ʸ᾽ ˠ᾽ ˢ᾽ ˣ᾽ ˤ᾽ ᵇ᾽ ᵏ᾽ ᵐ᾽ ᵖ᾽ ᵗ᾽ ᵚ᾽ ᵛ᾽ ᵡ᾽ ᵝ᾽ ᵞ᾽ ᵟ᾽

ḥ᾽ ḥ̱᾽ ⱼ᾽ ṛ᾽ ṛ̱᾽ ṣ᾽ ṣ̱᾽ ẉ᾽ ẏ᾽ ẓ᾽ ẓ̱᾽ ḅ᾽ ḳ᾽ ṃ᾽ ṗ᾽ ṭ᾽ ṭ̱᾽ ṿ᾽ ᵡ᾽ ᵝ᾽ ᵞ᾽ ᵟ᾽

ḥ᾽ ḥ̱᾽ ⱼ᾽ ṛ᾽ ṛ̱᾽ ṣ᾽ ṣ̱᾽ ẃ᾽ ý᾽ ź᾽ ẕ́᾽ ḅ᾽ ḳ᾽ ḿ᾽ ṕ᾽ ṭ᾽ ṭ̱᾽ ṽ᾽ ᵡ᾽ ᵝ᾽ ᵞ᾽ ᵟ᾽

ḥ᾽ ḥ̱᾽ ⱼ̂᾽ ṛ̂᾽ ṛ̱̂᾽ ṣ̂᾽ ṣ̱̂᾽ ẅ̂᾽ ŷ᾽ ẑ᾽ ẕ̂᾽ ḇ̂᾽ ḳ̂᾽ m̂᾽ p̂᾽ t̂᾽ ṭ̱̂᾽ v̂᾽ ᵡ᾽ ᵝ᾽ ᵞ᾽ ᵟ᾽

Gentium Plus

ʰ᾽ ʱ᾽ ʲ᾽ ʳ᾽ ʴ᾽ ʷ᾽ ʸ᾽ ˠ᾽ ˢ᾽ ˣ᾽ ˤ᾽ ᵇ᾽ ᵏ᾽ ᵐ᾽ ᵖ᾽ ᵗ᾽ ᵚ᾽ ᵛ᾽ ᵡ᾽ ᵝ᾽ ᵞ᾽ ᵟ᾽

ʰ᾽ ʱ᾽ ʲ᾽ ʳ᾽ ʴ᾽ ʷ᾽ ʸ᾽ ˠ᾽ ˢ᾽ ˣ᾽ ˤ᾽ ᵇ᾽ ᵏ᾽ ᵐ᾽ ᵖ᾽ ᵗ᾽ ᵚ᾽ ᵛ᾽ ᵡ᾽ ᵝ᾽ ᵞ᾽ ᵟ᾽

ʰ¯ ʱ¯ ʲ¯ ʳ¯ ʴ¯ ʷ¯ ʸ¯ ˠ¯ ˢ¯ ˣ¯ ˤ¯ ᵇ¯ ᵏ¯ ᵐ¯ ᵖ¯ ᵗ¯ ᵚ¯ ᵛ¯ ᵡ¯ ᵝ¯ ᵞ¯ ᵟ¯

ḥ̄ ḥ̱̄ ⱼ̄ ṝ ṝ̱ ṣ̄ ṣ̱̄ ẉ̄ ȳ ẓ̄ ẓ̱̄ ḇ̄ ḳ̄ m̄ p̄ ṭ̄ ṭ̱̄ v̄ ᵡ̄ ᵝ̄ ᵞ̄ ᵟ̄

ḥ́ ḥ̱́ ⱼ́ ṛ́ ṛ̱́ ṣ́ ṣ̱́ ẃ ý́ ź ẕ́́ ḅ́ ḳ́ ḿ ṕ ṭ́ ṭ̱́ ṽ́ ᵡ́ ᵝ́ ᵞ́ ᵟ́

ḥ̂ ḥ̱̂ ⱼ̂ ṛ̂ ṛ̱̂ ṣ̂ ṣ̱̂ ẅ̂ ŷ ẑ ẕ̂ ḇ̂ ḳ̂ m̂ p̂ ṭ̂ ṭ̱̂ v̂ ᵡ̂ ᵝ̂ ᵞ̂ ᵟ̂

A¯ B¯ C¯ D¯ E¯ F¯ G¯ H¯ I¯ J¯ K¯ L¯ M¯ N¯ O¯ P¯ Q¯ R¯ S¯ T¯ U¯ V¯ W¯ X¯ Y¯ Z¯ a¯ b¯ c¯ d¯ e¯ f¯ g¯ h¯ i¯ j¯ k¯ l¯ m¯ n¯ o¯ p¯ q¯ r¯ s¯ t¯ u¯ v¯ w¯ x¯ y¯ z¯ Æ¯ æ¯ Œ¯ œ¯ Ø¯ ø¯ Ð¯ ð¯ Þ¯ þ¯ ß¯ ſ¯ Ð¯ đ¯ Ħ¯ ħ¯ ı¯ j¯ Ł¯ ł¯ Ƀ¯ Ƃ¯
À̄ B̄ C̄ D̄ È̄ F̄ Ḡ H̄ Ì J̄ K̄ L̄ M̄ N̄ Ò P̄ Q̄ R̄ S̄ T̄ Ù V̄ W̄ X̄ Ỳ Z̄ à̄ b̄ c̄ d̄ è̄ f̄ ḡ h̄ ì j̄ k̄ l̄ m̄ n̄ ò p̄ q̄ r̄ s̄ t̄ ù v̄ w̄ x̄ ỳ z̄ Ǣ ǣ Œ̄ œ̄ Ø̄ ø̄ Ð̄ ð̄ Þ̄ þ̄ ß̄ ſ̄ Ð̄ đ̄ Ħ̄ ħ̄ ì j̄ Ł̄ ł̄ Ɓ̄ Ƃ̄

Dot removal

To do. Check GSUB to see if dot removal for i and j is complete.

Precomposed characters

To do. Check GSUB if an available precomposed character is encoded by a doublet of the base and a combining mark, or a triplet of the base and the two combining marks, in either mark order, then the precomposed character should be be produced by the font's GSUB table.

Marks

Here is a catalog of combining marks in Unicode, grouped by the seven GPOS anchor class indices in Libertinus (0–6).

Above marks – GPOS anchor 0

x̀ x́ x̂ x̃ x̄ x̆ ẋ ẍ x̊ x̋ x̌ x̍ x̎ x̏ x̐ x̑ x̒ x̓ x̔ x̕ x̖ x̗ x̘ x̙ x̚ x̛ x̜ x̝ x̞ x̟
x̀ x́ x̂ x̃ x̄ x̆ ẋ ẍ x̊ x̋ x̌ x̍ x̎ x̏ x̐ x̑ x̒ x̓ x̔ x̕ x̖ x̗ x̘ x̙ x̚ x̛ x̜ x̝ x̞ x̟
x̀ x́ x̂ x̃ x̄ x̆ ẋ ẍ x̊ x̋ x̌ x̍ x̎ x̏ x̐ x̑ x̒ x̓ x̔ x̕ x̖ x̗ x̘ x̙ x̚ x̛ x̜ x̝ x̞ x̟
x̀ x́ x̂ x̃ x̄ x̆ ẋ ẍ x̊ x̋ x̌ x̍ x̎ x̏ x̐ x̑ x̒ x̓ x̔ x̕ x̖ x̗ x̘ x̙ x̚ x̛ x̜ x̝ x̞ x̟

Below marks – GPOS anchor 2

x̠ x̡ x̢ x̣ x̤ x̥ x̦ x̧ x̨ x̩ x̪ x̫ x̬ x̭ x̮ x̯ x̰ x̱ x̲ x̳ x̴ x̵ x̶ x̷ x̸ x̹ x̺ x̻ x̼ x̽
x̠ x̡ x̢ x̣ x̤ x̥ x̦ x̧ x̨ x̩ x̪ x̫ x̬ x̭ x̮ x̯ x̰ x̱ x̲ x̳ x̴ x̵ x̶ x̷ x̸ x̹ x̺ x̻ x̼ x̽
x̠ x̡ x̢ x̣ x̤ x̥ x̦ x̧ x̨ x̩ x̪ x̫ x̬ x̭ x̮ x̯ x̰ x̱ x̲ x̳ x̴ x̵ x̶ x̷ x̸ x̹ x̺ x̻ x̼ x̽
x̠ x̡ x̢ x̣ x̤ x̥ x̦ x̧ x̨ x̩ x̪ x̫ x̬ x̭ x̮ x̯ x̰ x̱ x̲ x̳ x̴ x̵ x̶ x̷ x̸ x̹ x̺ x̻ x̼ x̽

Left angle – GPOS anchor 3

x᪽ *x᪽* x⃐ *x⃐*

Below-right marks – GPOS anchor 4

x᪾ xᪿ xᫀ
x᪾ xᪿ xᫀ
x᪾ xᪿ xᫀ
x᪾ xᪿ xᫀ

Cedilla – GPOS anchor 5

x̧ *x̧* x̧ *x̧*

Overlay marks – GPOS anchor 6

o̶ o̷ o̸ o̵ o̴
o̶ o̷ o̸ o̵ o̴
o̶ o̷ o̸ o̵ o̴
o̶ o̷ o̸ o̵ o̴

Double marks

x͡x x͡x x͞x x͟x x̃x x̂x x̡x
x͡x x͡x x͞x x͟x x̃x x̂x x̡x
x̆x x̋x x̏x x̂x x̆x x̊x ẍx
x̸x x̸x x̸x x̸x x̸x x̸x x̸x x̸x

Greek marks

t̀t́t̃t̄t̆ṫẗṭ
t̀t́t̃ṭ
t̂t̂ṫṫt̋t̊ẗṭ
t̀t́t̃t̄t̆ṫẗṭ
t̀t́t̃ṭ
t̂t̂ṫṫt̋t̊ẗṭ
t̀t́t̃t̄t̆ṫẗṭ
t̀t́t̃t
t̂t̂ṫṫt̋t̊ẗṭ
t̀t́t̃t̄t̆ṫẗṭ
t̀t́t̃t
t̂t̂ṫṫt̋t̊ẗṭ

Adding missing marks

Semibold and semibold italic lack many of the combining marks of regular and italic. The patch adds them, with mark anchors, in the range U+0332–U+036F. No mark is drawn from scratch: each is an outline that the designer drew in another style. A semibold mark is the regular outline (an above-mark is moved right by 12 units, with its anchor, as the designer's semibold above-marks are). A semibold italic mark is the italic outline moved right by 12 units, with the anchor of regular, as in the semibold italic anchor model. In semibold, U+0358–U+0363 were copies of the capital accents, and they are replaced. The patch also adds `dotaccent.cap` to italic and semibold italic: the dot of the precomposed İ, at the height it has in the precomposed dotted capitals.

The designer does not thicken below-marks between weights, but thickens above-marks by about 12 percent (mean stroke thickness, measured as twice the area of the outline divided by its perimeter: tilde 40 to 45, caron 36 to 42). So the added above-marks are made bolder. Where a mark is, or contains, another mark, the bold style's own mark is used. The rest of the outline is pushed outward by 7.5 percent of its stroke thickness on each side, about 2 to 3 units. These algorithmically emboldened marks are shown in the table on the next page, on a few bases that have above-anchors in all four patched fonts, with the designer's tilde and caron for comparison of weight. The regular and italic columns show the outlines that were copied.

Also added, but not shown in the table: the low line and double low line (U+0332, U+0333), the tilde, stroke and solidus overlays (U+0334–U+0337), the below-marks U+0339–U+033C, U+0347–U+0349, U+034D, U+034E, U+0353–U+0356, U+0359, U+035A, the double overline (U+033F), the dot above right (U+0358), the double marks (U+035C–U+0362) and the combining letters (U+0363–U+036F). These are copied without emboldening, since the designer's lines, below-marks and modifier letters are not thicker in the bold styles; the dot above right is the bold style's own dot. The double marks have no anchors, as in regular. In semibold, the overlays and the dot above right have no anchors, because the semibold lookup table has no anchor class for them. The half rings above (U+0351, U+0357) already exist in semibold, drawn to the right of the origin, so they sit beside the base; they are left as they are. Not added: the custom-width low lines and overlines of the next section, U+0344, U+0345 and U+0483.

		regular	semibold	italic	semibold italic	
U+0303	tilde (designer's, for comparison)	ãñũÕ	**ãñũÕ**	*ãñũÕ*	***ãñũÕ***	
U+030C	caron (designer's, for comparison)	ǎňǔǑ	**ǎňǔǑ**	*ǎňǔǑ*	***ǎňǔǑ***	
U+033D	x above	x̌x̌ǔx̌	**x̌x̌ǔx̌**	*x̌x̌ǔx̌*	***x̌x̌ǔx̌***	emboldened
U+033E	vertical tilde	a̾n̾ṵ̾o̾	**a̾n̾ṵ̾o̾**	*a̾n̾ṵ̾o̾*	***a̾n̾ṵ̾o̾***	emboldened
U+034A	not tilde above	ãn̋ũ̋Õ	**ãn̋ũ̋Õ**	*ãn̋ũ̋Õ*	***ãn̋ũ̋Õ***	emboldened
U+034B	homothetic above	ãn̑ũ̑Õ	**ãn̑ũ̑Õ**	*ãn̑ũ̑Õ*	***ãn̑ũ̑Õ***	own tilde; dots emboldened
U+034C	almost equal to above	ãn̆ũ̆Õ	**ãn̆ũ̆Õ**	*ãn̆ũ̆Õ*	***ãn̆ũ̆Õ***	emboldened
U+0350	right arrowhead above	ãn̛ữÕ	**ãn̛ữÕ**	*ãn̛ữÕ*	***ãn̛ữÕ***	emboldened
U+0351	left half ring above	á◌̜n̜ṵ̜Ó	**a◌̜n̜ṵ̜Oʿ**	*á◌̜n̜ṵ̜Ó*	***á◌̜n̜ṵ̜Ó***	emboldened (semibold italic only)
U+0352	fermata	ân̪ûÔ	**ân̪ûÔ**	*ân̪ûÔ*	***ân̪ûÔ***	own inverted breve; dot emboldened
U+0357	right half ring above	a̾n̾ṵ̾o̾	**a̾n̾ṵ̾Oʾ**	*a̾n̾ṵ̾o̾*	***a̾n̾ṵ̾o̾***	emboldened (semibold italic only)
U+035B	zigzag above	á◌̴n̴ṵ̴Ó	**á◌̴n̴ṵ̴Ó**	*á◌̴n̴ṵ̴Ó*	***á◌̴n̴ṵ̴Ó***	emboldened
U+0340	grave tone mark	àǹùÒ	**àǹùÒ**	*àǹùÒ*	***àǹùÒ***	own grave
U+0341	acute tone mark	áńúÓ	**áńúÓ**	*áńúÓ*	***áńúÓ***	own acute
U+0346	bridge above	ãn̪ũ̪Õ	**ãn̪ũ̪Õ**	*ãn̪ũ̪Õ*	***ãn̪ũ̪Õ***	own bridge below, raised

Overline and low line

U+0305 combining overline and U+0332 combining low line are applied by GSUB lookups that substitute a custom-width line mark according to width classes. GSUB lookup 51 effectively classifies the glyphs by width, and GSUB lookups 52–82 provide the appropriate custom-width mark for the base width class. Semibold and semibold italic do not have any combining low lines; the patch adds U+0332 and U+0333 with below-anchors, but not the custom-width variants. Custom-width line marks do not have anchors. They are positioned by HarfBuzz fallback shaping. The overlines are above the ascender line, so they are above all bases. The low lines are below the baseline, but above the descender line, and so they cut though any descenders. If a new glyph is added, and you want to support custom-width line marks for it, then remember to classify the new glyph in GSUB lookup 51.



Above-right marks – GPOS anchor 1

Libertinus has three combining marks with GPOS anchor 1 – U+031B horn, U+0315 comma above right, and U+0358 dot above right.

A few bases in Libertinus have this anchor (anchor 1), but none that possibly need it.

Horn

The Vietnamese alphabet has two horn vowels – Ơơ Ưư– and these vowels may take tonal marks. All these combinations are Unicode characters (with code points).

Ơơ Ưư Ớớ Ờờ Ởở Ỡỡ Ợợ Ứứ Ừừ Ửử Ữữ Ựự

Ơơ Ưư Ớớ Ờờ Ởở Ỡỡ Ợợ Ứứ Ừừ Ửử Ữữ Ựự

Ơơ Ưư Ớớ Ờờ Ởở Ỡỡ Ợợ Ứứ Ừừ Ửử Ữữ Ựự

Ơơ Ưư Ớớ Ờờ Ởở Ỡỡ Ợợ Ứứ Ừừ Ửử Ữữ Ựự

There is a Unicode character for the combining horn mark: ̛ `\char"031B` U+031B, but this is not used in practice. The Vietnamese vowels are usually entered in a LaTeX document as Unicode characters, also with `\usepackage[vietnamese]{babel}` for hyphenation and spacing.

Comma above right

There is a Unicode character for the combining comma above right mark: ̕ `\char"0315` U+0315.

U+0315 (comma above right) is used in rare cases when another comma-like mark does not fit.

In IPA, the glottalization diacritic mark uses U+0313 (comma above) as a combining mark. But when a glottalization mark is needed for a superscript consonant (1) or superscript consonant cluster (2), then the combining mark U+0315 (comma above right) is used instead, because its positioning is hopefully better than that of U+0313.

(1) *k̕wkt* [k̕^{wkx}t] ‘to scrape’ (Nuxalk)

(2) *p’ak* [p’^{kx}ak] ‘he/she cut it’ (Upper Chehalis)

In (1), the superscript consonant cluster is ^{wkx} and it is glottalized. In (2), the superscript consonant ^{x} is glottalized.

Because U+0315 is being used in superscript position on a smaller sized glyph, U+0315 should probably be smaller than U+0313, but it is not so in Libertinus.

Dot above right

There is a Unicode character for the combining dot above right mark: ̙ `\char"0358` U+0358 dot above right

U+0358 should be a small, above-right dot, used to mark palatalization or similar secondary articulations on superscript consonants, created for the same typographic reason U+0315 exists – the standard palatalization diacritic (U+0307, dot above) is too big and too high.

In Libertinus, U+0358 is a 104-diameter Bezier circle, and U+0307 is a smaller Bezier oval, inverting the desired size difference.

There are only a few IPA superscript letters that may serve as the possible host of U+0315 or U+0358 on superscript consonants or on superscript consonant clusters. In Libertinus, regrettably none of these characters has the anchor (anchor 1), so the mark (U+0315 and U+0358) is positioned by HarfBuzz fallback shaping, which seems to place the mark at the right boundary of the glyph at a constant vertical aspect (effectively at the superscript x-height).

These IPA marks U+0315 and U+0358 should only be used in regular (not italic and not semibold). Regrettably, in Libertinus serif semibold, U+0358 looks like a grave, not a dot, and there is no U+0358 glyph in Libertinus serif semibold italic.

There are three comma/apostrophe glyphs that could possibly mark glottalization: U+0313 combining comma above, U+0315 combining comma above right, and U+02BC modifier letter apostrophe. In Libertinus, U+0315 has the worst appearance, even though U+0315 is intended for superscript letters. In Gentium Plus (SIL Global), U+0315 is tightly above-right the superscript letter, as expected.

Libertinus

`[k\char"0313\char"02B7\char"1D4F\char"02E3\char"0313 t]` [kʾʷᵏˣʾt]

`[k\char"0313\char"02B7\char"1D4F\char"02E3\char"0315 t]` [kʾʷᵏˣʿt]

`[k\char"0313\char"02B7\char"1D4F\char"02E3\char"02BC t]` [kʾʷᵏˣʼt]

Gentium Plus

`[k\char"0313\char"02B7\char"1D4F\char"02E3\char"0313 t]` [kʾʷᵏˣʾt]

`[k\char"0313\char"02B7\char"1D4F\char"02E3\char"0315 t]` [kʾʷᵏˣʿt]

`[k\char"0313\char"02B7\char"1D4F\char"02E3\char"02BC t]` [kʾʷᵏˣʼt]

I patched Libertinus to add anchors to those IPA superscript letters to support U+0315.

Patched Libertinus, selected IPA superscript letters with U+0315 and U+0358

ʰʾɦʾʲʾɾʾɹʾɻʾʁʾwʾyʾʏʾlʾsʾxʾʕʾbʾkʾmʾpʾtʾɯʾvʾ◌͡◌ʾβʾɣʾðʾ

ʰʿɦʿʲʿɾʿɹʿɻʿʁʿwʿyʿʏʿlʿsʿxʿʕʿbʿkʿmʿpʿtʿɯʿvʿ◌͡◌ʿβʿɣʿðʿ

`[k\char"0313\char"02B7\char"1D4F\char"02E3\char"0315 t]` [kʾʷᵏˣʿt]

Gentium Plus, selected IPA superscript letters with U+0315 and U+0358

ʰʾɦʾʲʾɾʾɹʾɻʾʁʾwʾyʾʏʾlʾsʾxʾʕʾbʾkʾmʾpʾtʾɯʾvʾ◌͡◌ʾβʾɣʾðʾ

ʰʿɦʿʲʿɾʿɹʿɻʿʁʿwʿyʿʏʿlʿsʿxʿʕʿbʿkʿmʿpʿtʿɯʿvʿ◌͡◌ʿβʿɣʿðʿ

`[k\char"0313\char"02B7\char"1D4F\char"02E3\char"0315 t]` [kʾʷᵏˣʿt]

Here are the IPA superscript letters considered and their new anchor (anchor 1).

U+02B0: (347, 648), U+02B1: (339, 648), U+02B2: (193, 648), U+02B3: (284, 648), U+02B4: (274, 648), U+02B5: (319, 648), U+02B6: (308, 648), U+02B7: (497, 648), U+02B8: (368, 648), U+02E0: (341, 648), U+02E1: (193, 648), U+02E2: (270, 648), U+02E3: (344, 648), U+02E4: (282, 648), U+1D47: (319, 648), U+1D4F: (353, 648), U+1D50: (533, 648), U+1D56: (334, 648), U+1D57: (241, 648), U+1D5A: (525, 648), U+1D5B: (354, 648), U+1D5C: (508, 648), U+1D5D: (335, 648), U+1D5E: (372, 648), U+1D5F: (353, 648)

I did not change the anchor-1 of U+315 and U+0358. U+0358 has its anchor at the xMin and yMin of its bounding box. But U+0315 already has some clearance built in to its anchor: its anchor is 62 units below its yMin, but 25 units right of it xMin. So, I decided to provide x-clearance of 12 units and y-clearance of 80 units from the bases. All these IPA superscript letters already have a superscript meanline at $Y = 630$. So their anchor will be at $Y = 630 - 62 + 80 = 648$. The X coordinate depends on their bounding box and its xMax. To provide the desired clearance of 12 units, the anchor is at $X = \text{xMax} + 26 + 12$. These anchor coordinate values were optimized for U+0315 but they work fine for U+0358.

Bases

Here are some subgroups of bases.

Latin

ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyzÆæŒœØøÐðÞþßſĐđĦħıȷŁłƁƀ
ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyzÆæŒœØøÐðÞþßſĐđĦħıȷŁłƁƀ
ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyzÆæŒœØøÐðÞþßſĐđĦħıȷŁłƁƀ
ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyzÆæŒœØøÐðÞþßſĐđĦħıȷŁłƁƀ

IPA

ɒɔɕɗɖɑɋɜəɛɟfɠɣɦɬɪʞɯɲɵɭʒƨ|‖ǂ!ɐɑɒɓɔɕɖɗɘəɚɛɜɝɞɟɠɡɣɤɥɦɧɨɪɪɺɬɮɭɯɰɱɲɳɴɵœɷɸɹɻɽɾʀʁɕʃʅʆʇʈʉʊʌʍʎʏʐʑʒʓʔʕʖʗǀǁǂǃ
ɒɔɕɗɖɑɋɜəɛɟfɠɣɦɬɪʞɯɲɵɭʒƨ|‖ǂ!ɐɑɒɓɔɕɖɗɘəɚɛɜɝɞɟɠɡɣɤɥɦɧɨɪɪɺɬɮɭɯɰɱɲɳɴɵœɷɸɹɻɽɾʀʁɕʃʅʆʇʈʉʊʌʍʎʏʐʑʒʓʔʕʖʗǀǁǂǃ
ɒɔɕɗɖɑɋɜəɛɟfɠɣɦɬɪʞɯɲɵɭʒƨ|‖ǂ!ɐɑɒɓɔɕɖɗɘəɚɛɜɝɞɟɠɡɣɤɥɦɧɨɪɪɺɬɮɭɯɰɱɲɳɴɵœɷɸɹɻɽɾʀʁɕʃʅʆʇʈʉʊʌʍʎʏʐʑʒʓʔʕʖʗǀǁǂǃ
ɒɔɕɗɖɑɋɜəɛɟfɠɣɦɬɪʞɯɲɵɭʒƨ|‖ǂ!ɐɑɒɓɔɕɖɗɘəɚɛɜɝɞɟɠɡɣɤɥɦɧɨɪɪɺɬɮɭɯɰɱɲɳɴɵœɷɸɹɻɽɾʀʁɕʃʅʆʇʈʉʊʌʍʎʏʐʑʒʓʔʕʖʗǀǁǂǃ

Superscript consonants

ʰʱʲʳʴʵʷʸˡˢˣˤᵇᵏᵐᵖᵗᵠᵛ◌ᵝᵞᵟ
ʰʱʲʳʴʵʷʸˡˢˣˤᵇᵏᵐᵖᵗᵠᵛ◌ᵝᵞᵟ
ʰʱ ʲʳʴʵʷʸˡˢˣˤᵇᵏᵐᵖᵗᵠᵛ◌ᵝᵞᵟ
ʰʱʲʳʴʵʷʸˡˢˣˤᵇᵏᵐᵖᵗᵠᵛ◌ᵝᵞᵟ

Greek

ΑΒΓΔΕΖΗΘΙΚΛΜΝΞΟΠΡΣΤΥΦΧΨΩαβγδεζηθικλμνξοπρστυφχψω
ΑΒΓΔΕΖΗΘΙΚΛΜΝΞΟΠΡΣΤΥΦΧΨΩαβγδεζηθικλμνξοπρστυφχψω
ΑΒΓΔΕΖΗΘΙΚΛΜΝΞΟΠΡΣΤΥΦΧΨΩαβγδεζηθικλμνξοπρστυφχψω
ΑΒΓΔΕΖΗΘΙΚΛΜΝΞΟΠΡΣΤΥΦΧΨΩαβγδεζηθικλμνξοπρστυφχψω

Cyrillic

АБВГДЕЖЗИЙКЛМНОПРСТУФХЦЧШЩЪЫЬЭЮЯабвгдежзийклмнопрстуфхцчшщъыьэюяЁёЄєЇїЎўӘәӨөҺҥI
АБВГДЕЖЗИЙКЛМНОПРСТУФХЦЧШЩЪЫЬЭЮЯабвгдежзийклмнопрстуфхцчшщъыьэюяЁёЄєЇїЎўӘәӨөҺҥI
АБВГДЕЖЗИЙКЛМНОПРСТУФХЦЧШЩЪЫЬЭЮЯабвгдежзийклмнопрстуфхцчшщъыьэюяЁёЄєЇїЎўӘәӨөҺҥI
АБВГДЕЖЗИЙКЛМНОПРСТУФХЦЧШЩЪЫЬЭЮЯабвгдежзийклмнопрстуфхцчшщъыьэюяЁёЄєЇїЎўӘәӨөҺҥI

Precomposed capital vowels

ÀÁÂÃÄÅĀĂĄǍǞǠǺȀȂẠẢẤẦẨẪẬẮẰẲẴẶẸẺẼẾỀỂỄỆỈỊỌỎỐỒỔỖỘỚỜỞỠỢỤỦỨỪỬỮỰ
ŨŪŬŮŰŲǓǕǗǙǛỤỦỨỪỬỮỰ
ÀÁÂÃÄÅĀĂĄǍǞǠǺȀȂẠẢẤẦẨẪẬẮẰẲẴẶẸẺẼẾỀỂỄỆỈỊỌỎỐỒỔỖỘỚỜỞỠỢỤỦỨỪỬỮỰ
ŨŪŬŮŰŲǓǕǗǙǛỤỦỨỪỬỮỰ
ÀÁÂÃÄÅĀĂĄǍǞǠǺȀȂẠẢẤẦẨẪẬẮẰẲẴẶẸẺẼẾỀỂỄỆỈỊỌỎỐỒỔỖỘỚỜỞỠỢỤỦỨ
ỪỬỮŨŪŬŮŰŲǓǕǗǙǛỤỦỨỪỬỮỰ
ÀÁÂÃÄÅĀĂĄǍǞǠǺȀȂẠẢẤẦẨẪẬẮẰẲẴẶẸẺẼẾỀỂỄỆỈỊỌỎỐỒỔỖỘỚỜỞỠỢỤỦỨ
ỪỬỮŨŪŬŮŰŲǓǕǗǙǛỤỦỨỪỬỮỰ

Precomposed small vowels

àáâãäåāăąǎǟǡǻȁȃạảấầẩẫậắằẳẵặẹẻẽếềểễệỉịọỏốồổỗộớờởỡợụủứừửữự
àáâãäåāăąǎǟǡǻȁȃạảấầẩẫậắằẳẵặẹẻẽếềểễệỉịọỏốồổỗộớờởỡợụủứừửữự
àáâãäåāăąǎǟǡǻȁȃạảấầẩẫậắằẳẵặẹẻẽếềểễệỉịọỏốồổỗộớờởỡợụủứừửữự
àáâãäåāăąǎǟǡǻȁȃạảấầẩẫậắằẳẵặẹẻẽếềểễệỉịọỏốồổỗộớờởỡợụủứừửữự

Precomposed capital consonants

ḂḆÇĆĈČĊÇḈĎḐḒḌḎḐḞĜĞĠĢḠǦĤḢḤḦḪḨĴḰǨḴƘĹĻĽḶḼĿŁḺḸḻḽḾṀṂŃṆŇṄṊṈƝǸȎṔṖṕŔŖŘȐȒṘṞŚŜŞŠȘṢṤṦṠŠŞṨṩȘṧṨṪŢŤṰṮṬȚƬƮŨṼỤỦẂẀẄẂẊẌÝỲŶỸỴỾỶỹŸỵŶỴŹŻŽẐẔẒ
ḂḆÇĆĈČĊÇḈĎḐḒḌḎḐḞĜĞĠĢḠǦĤḢḤḦḪḨĴḰǨḴƘĹĻĽḶḼĿŁḺḸḻḽḾṀṂŃṆŇṄṊṈƝǸȎṔṖṕŔŖŘȐȒṘṞŚŜŞŠȘṢṤṦṠŠŞṨṩȘṧṨṪŢŤṰṮṬȚƬƮŨṼỤỦẂẀẄẂẊẌÝỲŶỸỴỾỶỹŸỵŶỴŹŻŽẐẔẒ
ḂḆÇĆĈČĊÇḈĎḐḒḌḎḐḞĜĞĠĢḠǦĤḢḤḦḪḨĴḰǨḴƘĹĻĽḶḼĿŁḺḸḻḽḾṀṂŃṆŇṄṊṈƝǸȎṔṖṕŔŖŘȐȒṘṞŚŜŞŠȘṢṤṦṠŠŞṨṩȘṧṨṪŢŤṰṮṬȚƬƮŨṼỤỦẂẀẄẂẊẌÝỲŶỸỴỾỶỹŸỵŶỴŹŻŽẐẔẒ
ḂḆÇĆĈČĊÇḈĎḐḒḌḎḐḞĜĞĠĢḠǦĤḢḤḦḪḨĴḰǨḴƘĹĻĽḶḼĿŁḺḸḻḽḾṀṂŃṆŇṄṊṈƝǸȎṔṖṕŔŖŘȐȒṘṞŚŜŞŠȘṢṤṦṠŠŞṨṩȘṧṨṪŢŤṰṮṬȚƬƮŨṼỤỦẂẀẄẂẊẌÝỲŶỸỴỾỶỹŸỵŶỴŹŻŽẐẔẒ

Precomposed small consonants

ƀḃḅƃɓƅćĉċčçḉƈďđḋḍḑḓḏđƌḟĝğġģḡǧǥķĥħḣḥḧḫẖĵǰḳḵķḳƙĺļľŀłḽḷḻḹŀɫḿṁṃńņňŉṅṇṋŋɲȯɑɋṕṗƥɼŗřȑȓṛṟɍśŝşšșṡṣßšșṩẛʂţťŧṱṭṯṫƭʃưṽỽẏŵẁẅẘẋẍýÿŷỳẏỷẙźżžẑẓẕȝƨƽpǯ
ƀḃḅƃɓƅćĉċčçḉƈďđḋḍḑḓḏđƌḟĝğġģḡǧǥķĥħḣḥḧḫẖĵǰḳḵķḳƙĺļľŀłḽḷḻḹŀɫḿṁṃńņňŉṅṇṋŋɲȯɑɋṕṗƥɼŗřȑȓṛṟɍśŝşšșṡṣßšșṩẛʂţťŧṱṭṯṫƭʃưṽỽẏŵẁẅẘẋẍýÿŷỳẏỷẙźżžẑẓẕȝƨƽpǯ
ƀḃḅƃɓƅćĉċčçḉƈďđḋḍḑḓḏđƌḟĝğġģḡǧǥķĥħḣḥḧḫẖĵǰḳḵķḳƙĺļľŀłḽḷḻḹŀɫḿṁṃńņňŉṅṇṋŋɲȯɑɋṕṗƥɼŗřȑȓṛṟɍśŝşšșṡṣßšșṩẛʂţťŧṱṭṯṫƭʃưṽỽẏŵẁẅẘẋẍýÿŷỳẏỷẙźżžẑẓẕȝƨƽpǯ

ƀḃḅƃɓƅćĉċčçḉƈďđḋḍḑḓḏđƌḟĝğġģḡǧǥķĥħḣḥḧḫẖĵǰḳḵķḳƙĺļľŀłḽḷḻḹŀɫḿṁṃńņňŉṅṇṋŋɲȯɑɋṕṗƥɼŗřȑȓṛṟɍśŝşšșṡṣßšșṩẛʂţťŧṱṭṯṫƭʃưṽỽẏŵẁẅẘẋẍýÿŷỳẏỷẙźżžẑẓẕȝƨƽpǯ

No marks: rare, digraph etc.

ȸ ȹ DZ Dz dz DŽ Dž dž LJ Lj lj NJ Nj nj
ȸ ȹ DZ Dz dz DŽ Dž dž LJ Lj lj NJ Nj nj
ȸ ȹ DZ Dz dz DŽ Dž dž LJ Lj lj NJ Nj nj
ȸ ȹ DZ Dz dz DŽ Dž dž LJ Lj lj NJ Nj nj

Small capitals

A B C D E F G H I J K L M N O P Q R S T U V W X Y Z Æ Ð Œ Þ IJ ß Ŋ
A B C D E F G H I J K L M N O P Q R S T U V W X Y Z Æ Ð Œ Þ IJ ß Ŋ
A B C D E F G H I J K L M N O P Q R S T U V W X Y Z Æ Ð Œ Þ IJ ß Ŋ
A B C D E F G H I J K L M N O P Q R S T U V W X Y Z Æ Ð Œ Þ IJ ß Ŋ
AÀÁÂÃÄÅĂĄ CÇĆČ DĎ EÈÉÊËĘĚ GĞ IÌÍÎÏI LĹŁĽ NÑŃŇ OÒÓÔÖŐÕØ RŔŘ SŚŞŠŞ TŢŦŤŢ UÙÚÛÜŰŮ YÝŸ ZŹŻŽ ÆÁÆ
AÀÁÂÃÄÅĂĄ CÇĆČ DĎ EÈÉÊËĘĚ GĞ IÌÍÎÏI LĹŁĽ NÑŃŇ OÒÓÔÖŐÕØ RŔŘ SŚŞŠŞ TŢŦŤŢ UÙÚÛÜŰŮ YÝŸ ZŹŻŽ ÆÁÆ
AÀÁÂÃÄÅĂĄ CÇĆČ DĎ EÈÉÊËĘĚ GĞ IÌÍÎÏI LĹŁĽ NÑŃŇ OÒÓÔÖŐÕØ RŔŘ SŚŞŠŞ TŢŦŤŢ UÙÚÛÜŰŮ YÝŸ ZŹŻŽ ÆÁÆ
AÀÁÂÃÄÅĂĄ CÇĆČ DĎ EÈÉÊËĘĚ GĞ IÌÍÎÏI LĹŁĽ NÑŃŇ OÒÓÔÖŐÕØ RŔŘ SŚŞŠŞ TŢŦŤŢ UÙÚÛÜŰŮ YÝŸ ZŹŻŽ ÆÁÆ

Precomposed characters of anchor-relevant bases and marks

àáâãāåạặầắẳẵấẫèéêẽēęệếềễẽìíîĩīịòóôõōọộốồỗùúûũūụỳýŷỹȳẏḍḷṇṛṣṭẓ
àáâãāåạặầắẳẵấẫèéêẽēęệếềễẽìíîĩīịòóôõōọộốồỗùúûũūụỳýŷỹȳẏḍḷṇṛṣṭẓ
àáâãāåạặầắẳẵấẫèéêẽēęệếềễẽìíîĩīịòóôõōọộốồỗùúûũūụỳýŷỹȳẏḍḷṇṛṣṭẓ
àáâãāåạặầắẳẵấẫèéêẽēęệếềễẽìíîĩīịòóôõōọộốồỗùúûũūụỳýŷỹȳẏḍḷṇṛṣṭẓ

Base and mark shaping – Linguistics notation

These grids display base and mark rendering in XeLaTeX documents in Libertinus serif regular, italic, semibold and semibold italic.

The grids use Latin in IPA bases and combining marks that are used in linguistics notation.

A black color means GPOS anchor positioning.

A blue color means a precomposed glyph.

A red color means HarfBuzz fallback shaping (no anchors, no precomposed glyph, no GSUB substitution).

A faded color – gray, light blue, or pink – means that the base and mark combination is not reasonably expected in notation in linguistics.

A gray color can also mean that mark glyph is missing in the font. In this case, just the gray base is printed. This only happens in Libertinus Serif Semibold and Semibold Italic for three below marks: o͇ (U+0333 double macron below), o͚ (U+033A inverted bridge below), and o͟ (U+033B square below). Otherwise, all considered base and mark combinations can be drawn in XeLaTeX documents in Libertinus without tofu for missing glyphs.

Regular, Bases requiring anchors for common marks, Common combining marks



Regular patched, Bases requiring anchors for common marks, Common combining marks



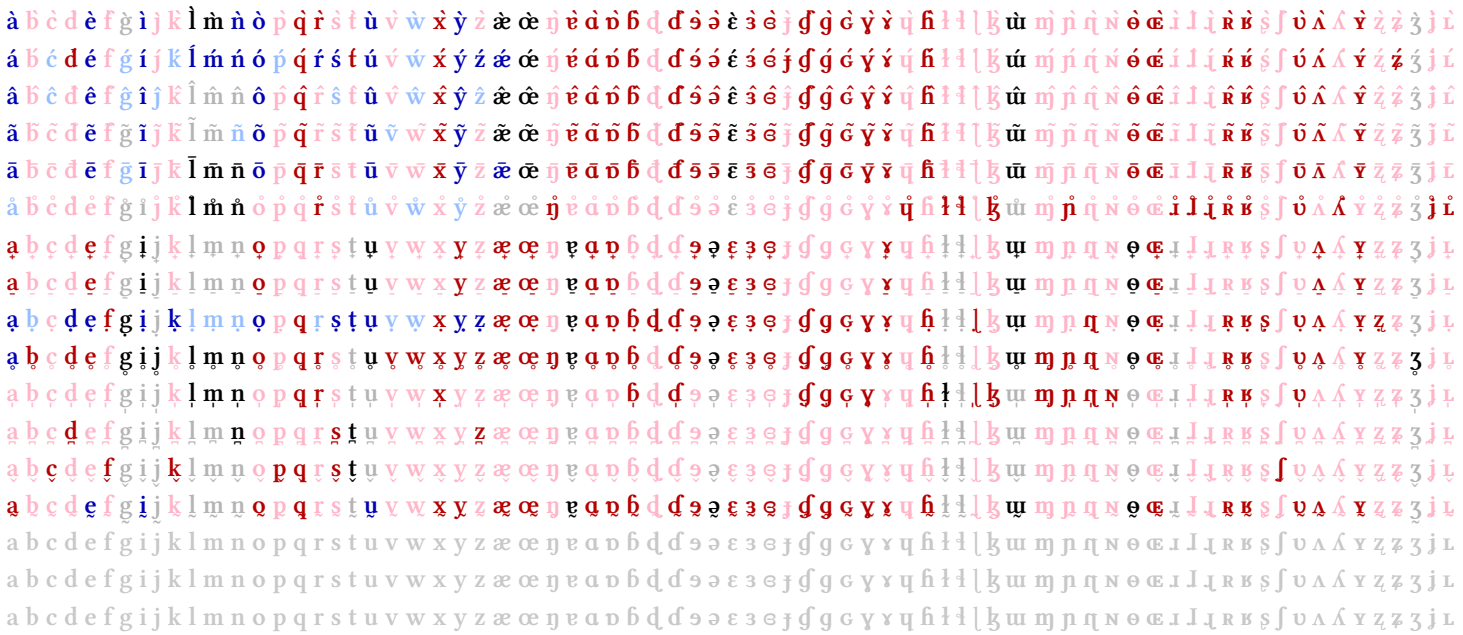
Italic, Bases requiring anchors for common marks, Common combining marks



Italic patched, Bases requiring anchors for common marks, Common combining marks



Semibold, Bases requiring anchors for common marks, Common combining marks



Semibold patched, Bases requiring anchors for common marks, Common combining marks



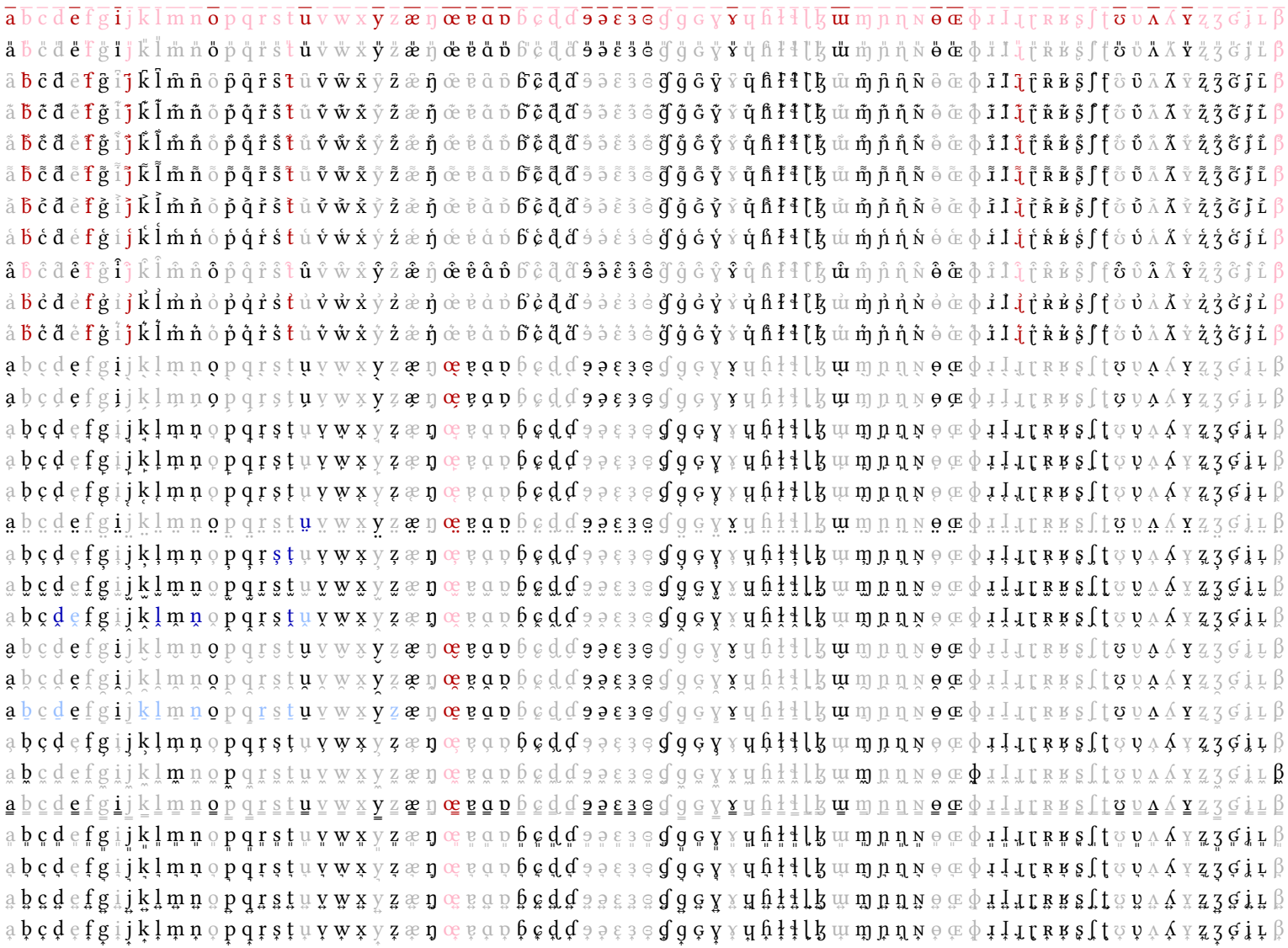
Semibold italic, Bases requiring anchors for common marks, Common combining marks



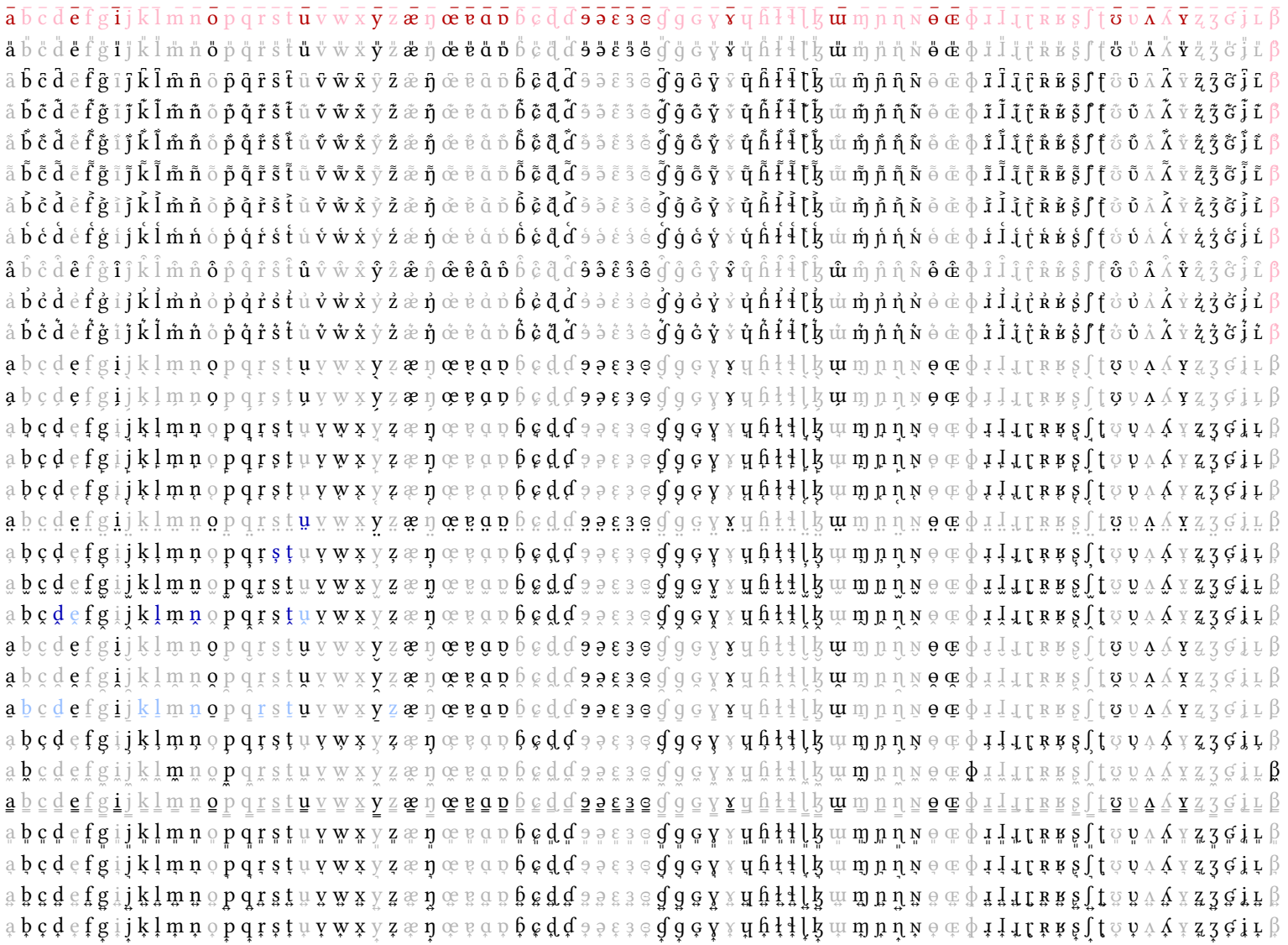
Semibold italic patched, Bases requiring anchors for common marks, Common combining marks



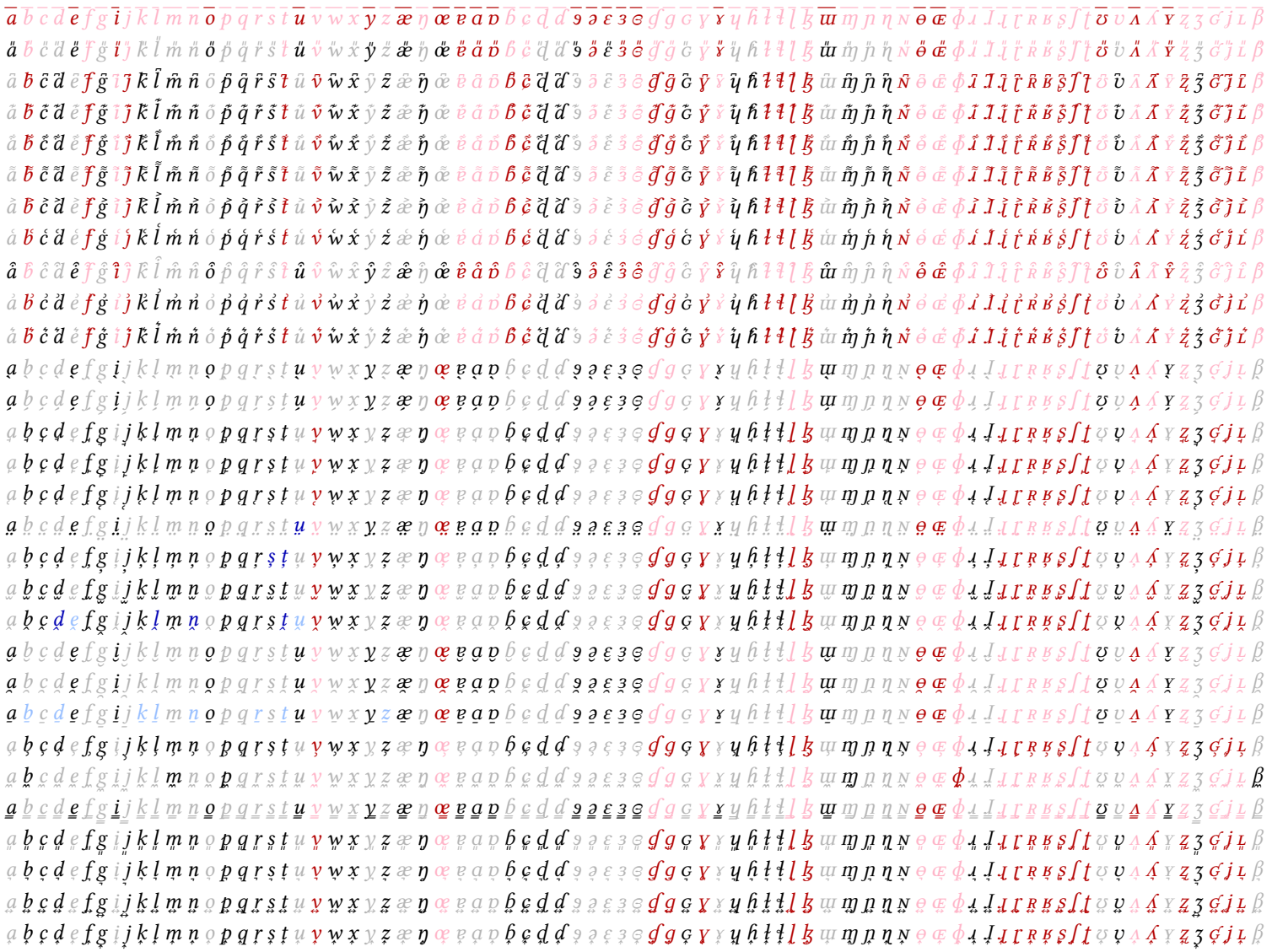
Regular, Bases requiring anchors for rare marks, Rare combining marks



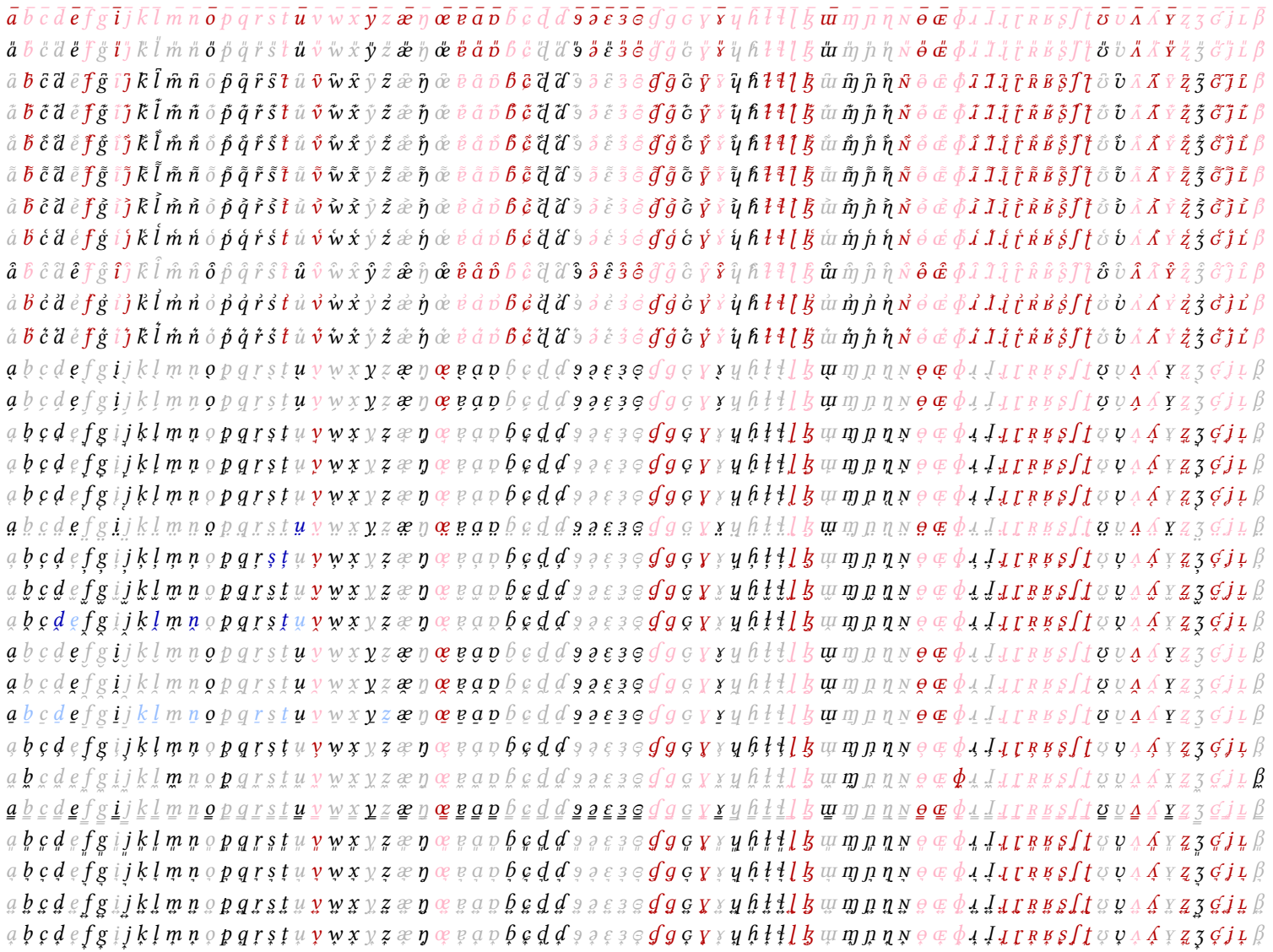
Regular patched, Bases requiring anchors for rare marks, Rare combining marks



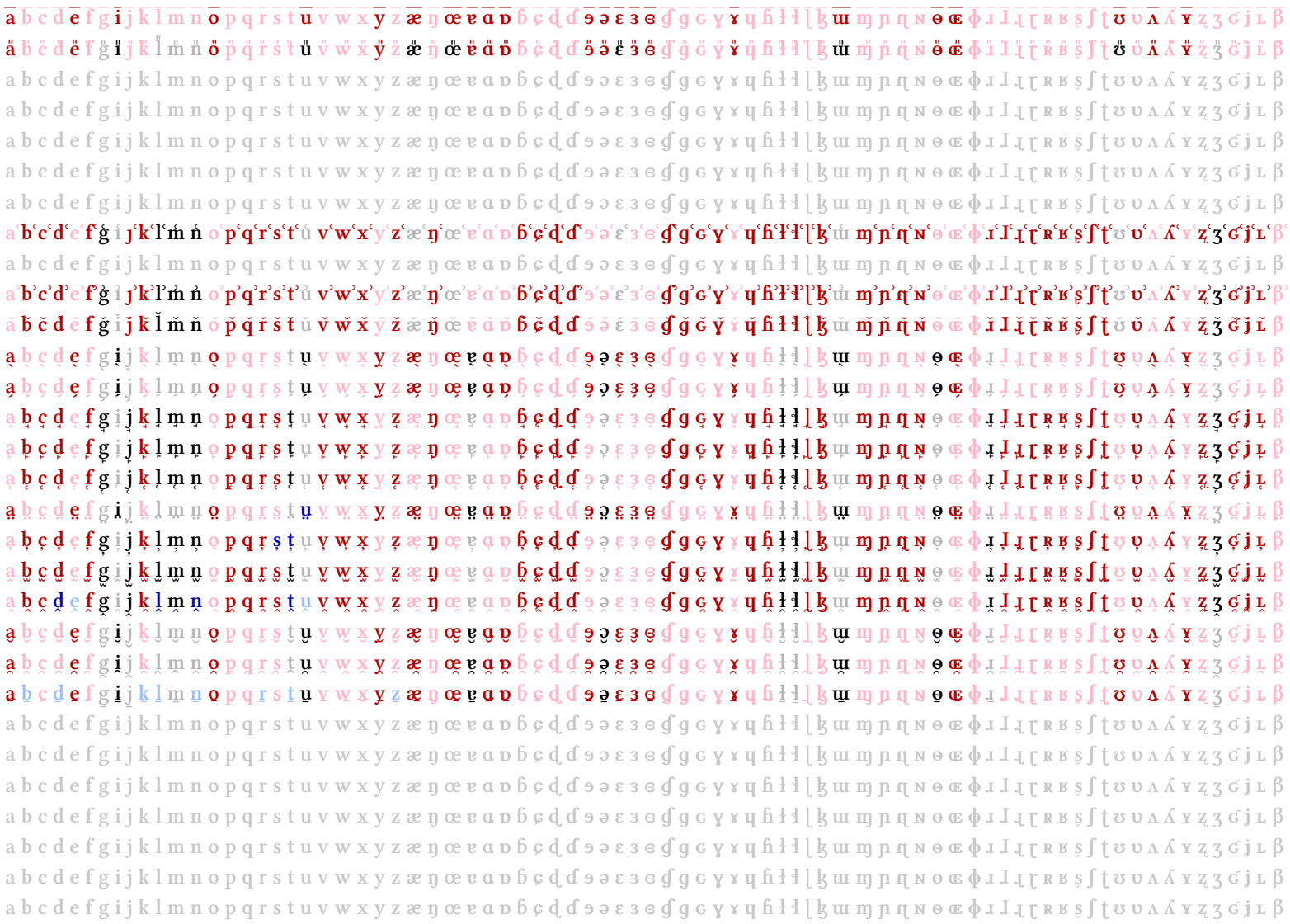
Italic, Bases requiring anchors for rare marks, Rare combining marks



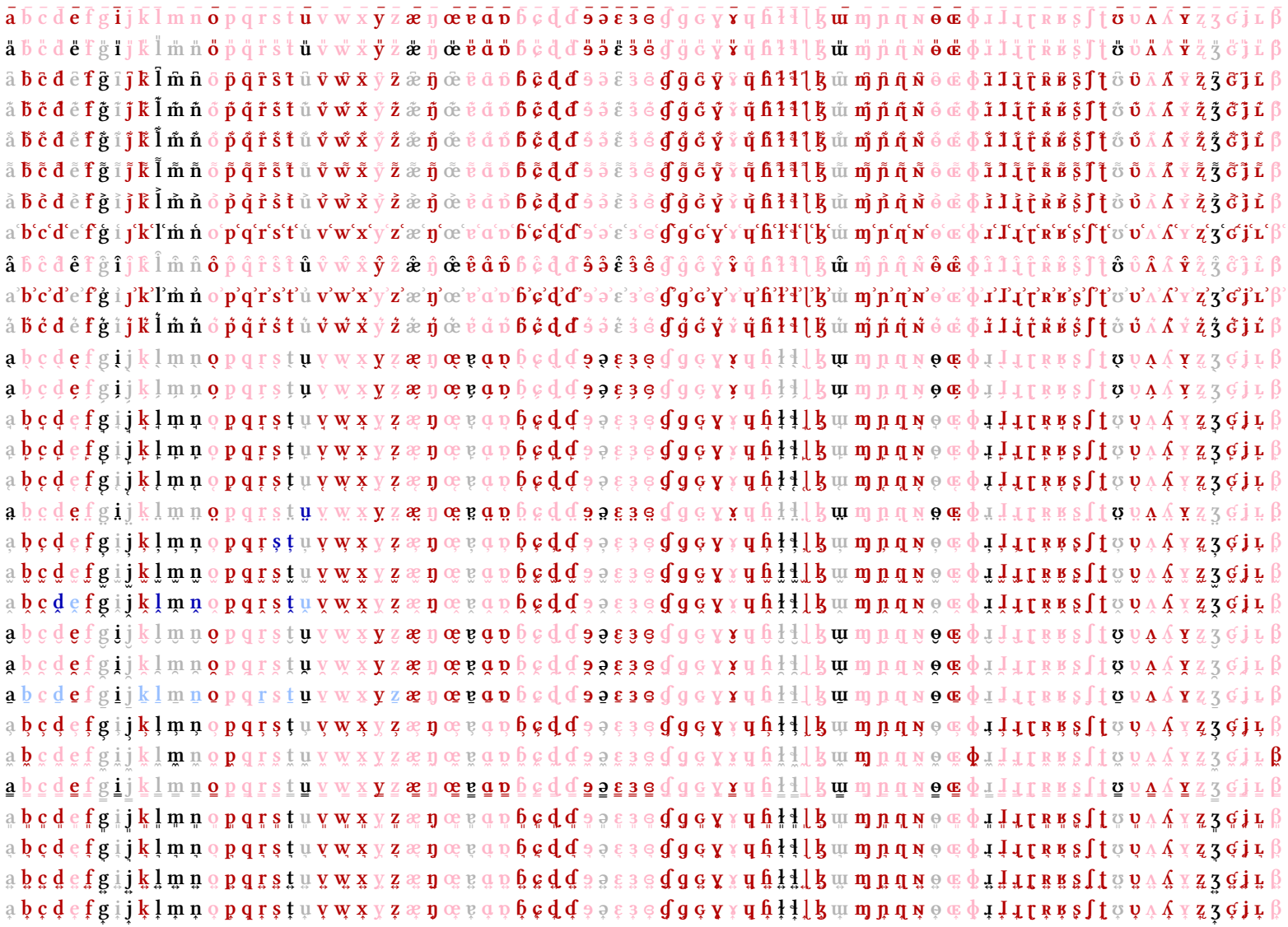
Italic patched, Bases requiring anchors for rare marks, Rare combining marks



Semibold, Bases requiring anchors for rare marks, Rare combining marks



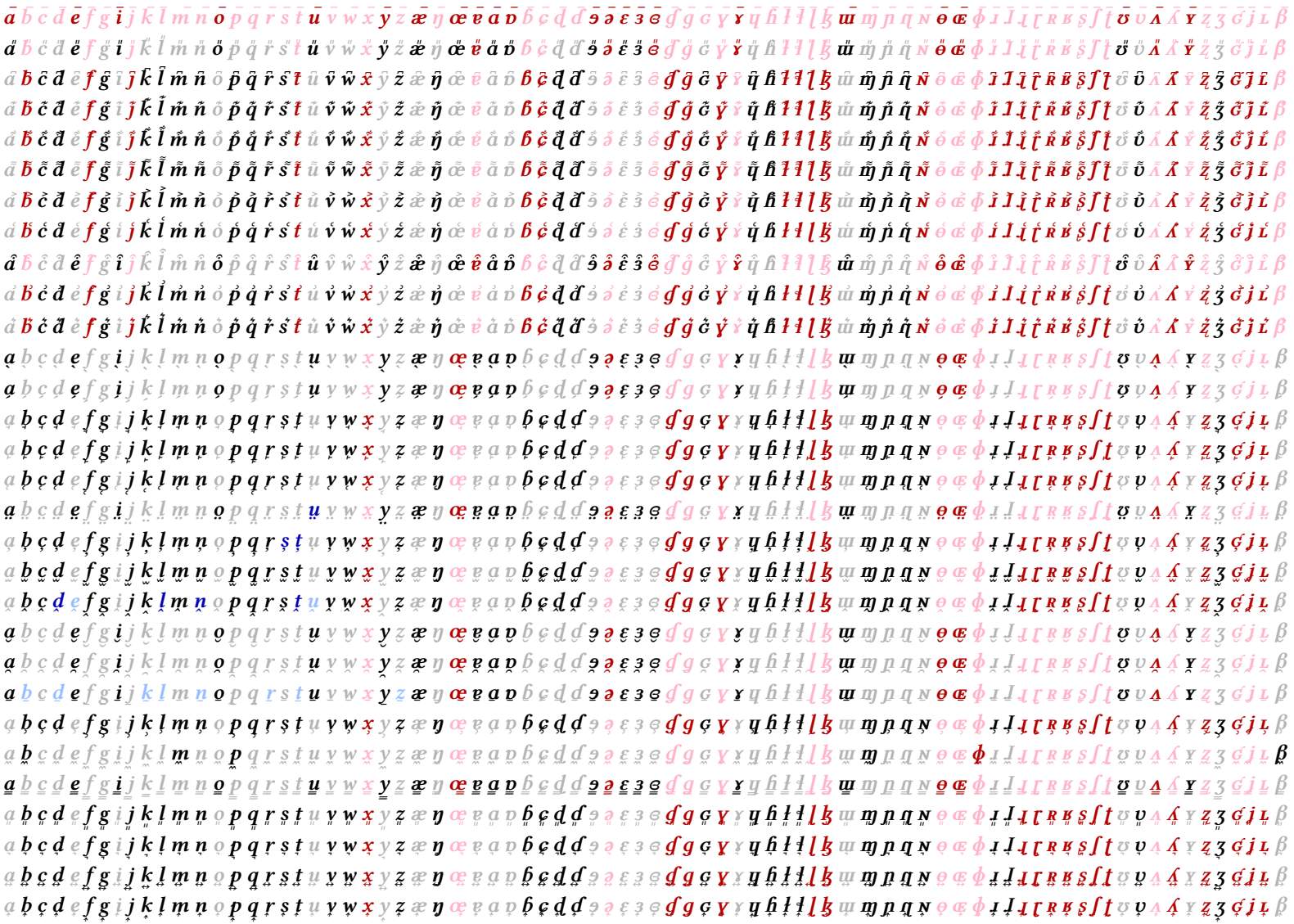
Semibold patched, Bases requiring anchors for rare marks, Rare combining marks



Semibold italic, Bases requiring anchors for rare marks, Rare combining marks



Semibold italic patched, Bases requiring anchors for rare marks, Rare combining marks



Base and mark shaping – Complete

These grids display base and mark rendering in XeLaTeX documents in Libertinus serif regular, italic, semibold and semibold italic. The grids use more complete sets of Latin and IPA bases and above and below marks.

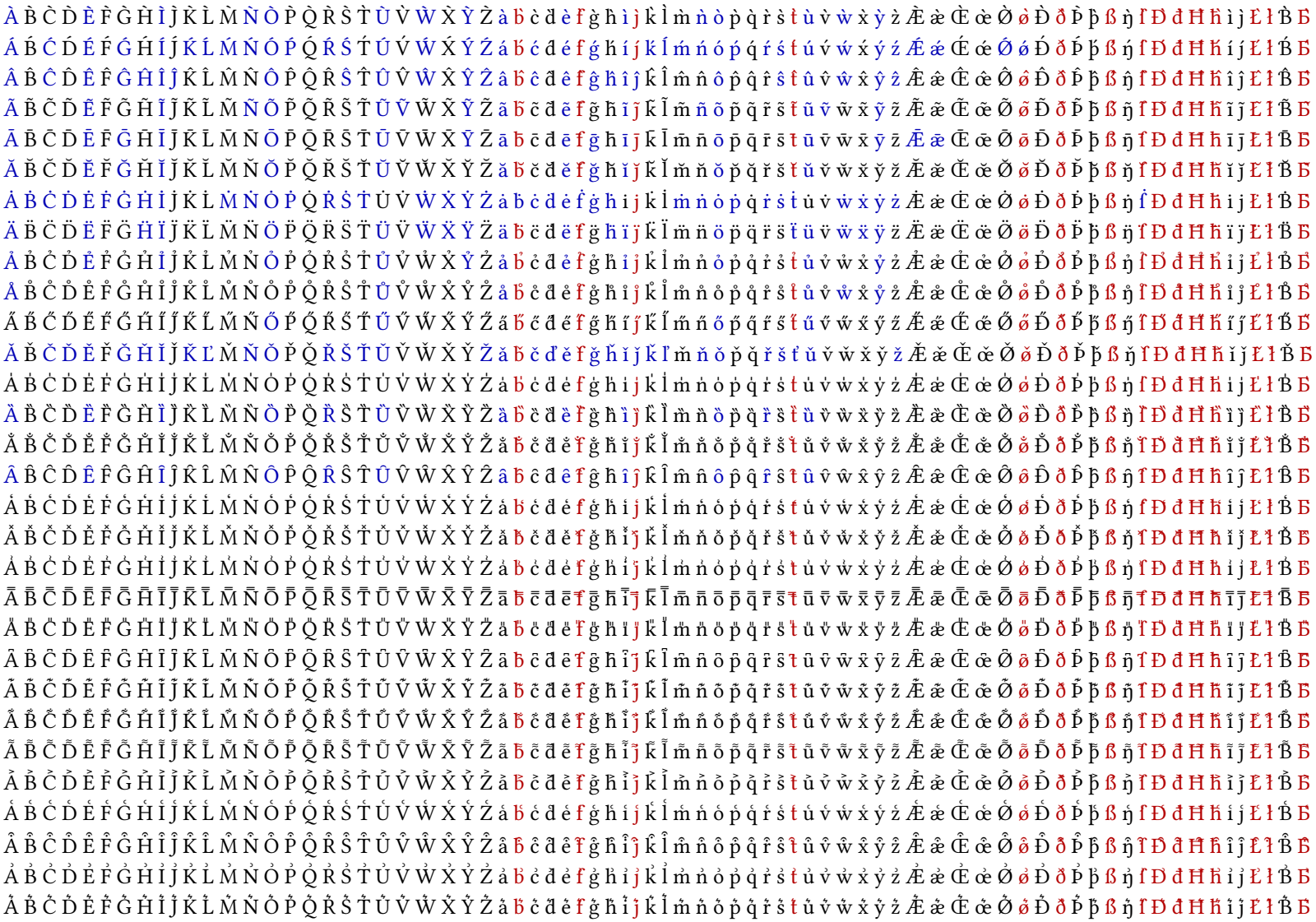
Blue indicates that XeLaTeX uses a precomposed Unicode character glyph.

Black indicates that the mark is positioned using GPOS anchors.

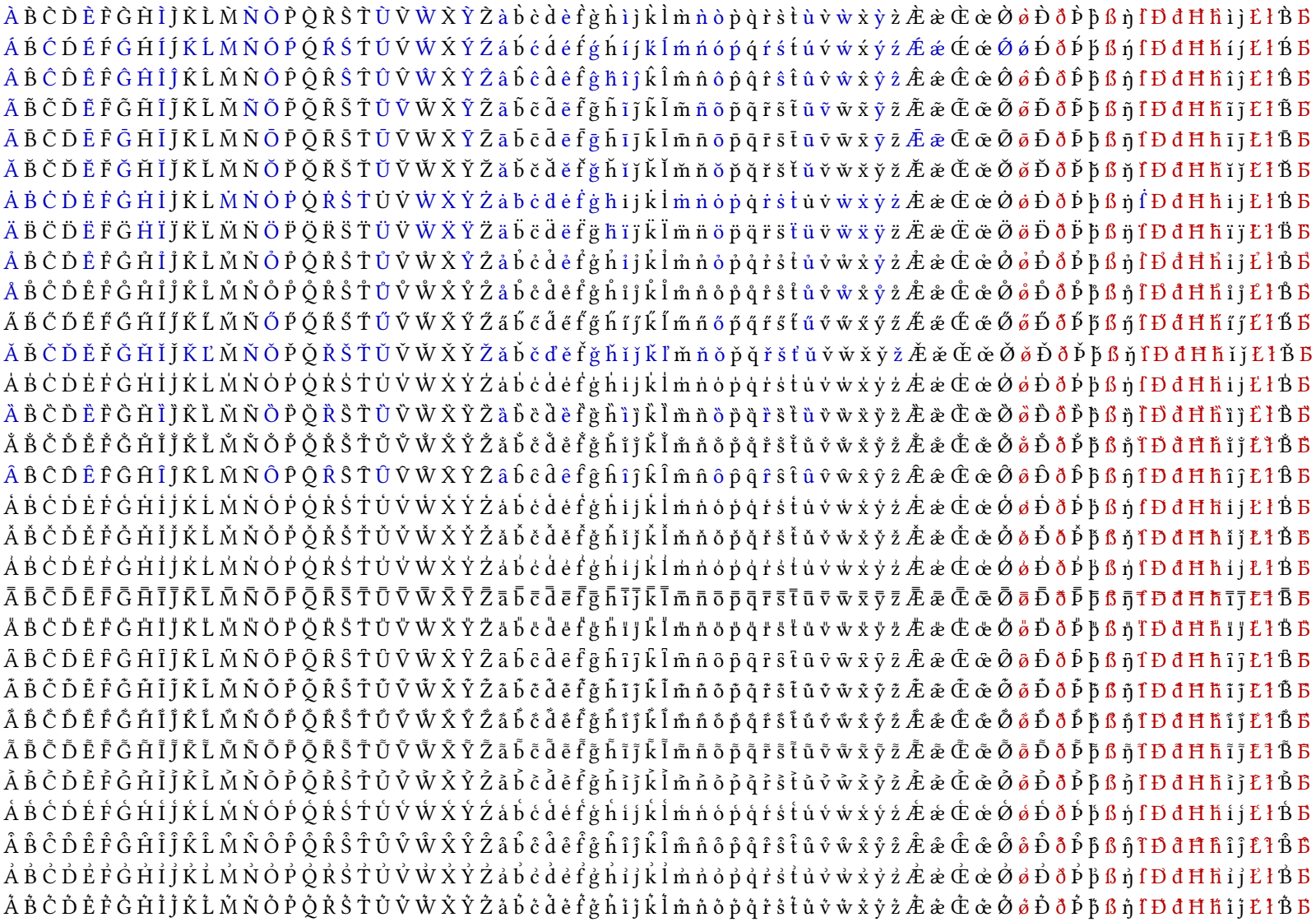
Red indicates HarfBuzz fallback shaping: no precomposed glyph and no GPOS anchors are used.

Light gray indicates that either the base or the mark glyph is missing from the font. In this case only the base glyph is shown.

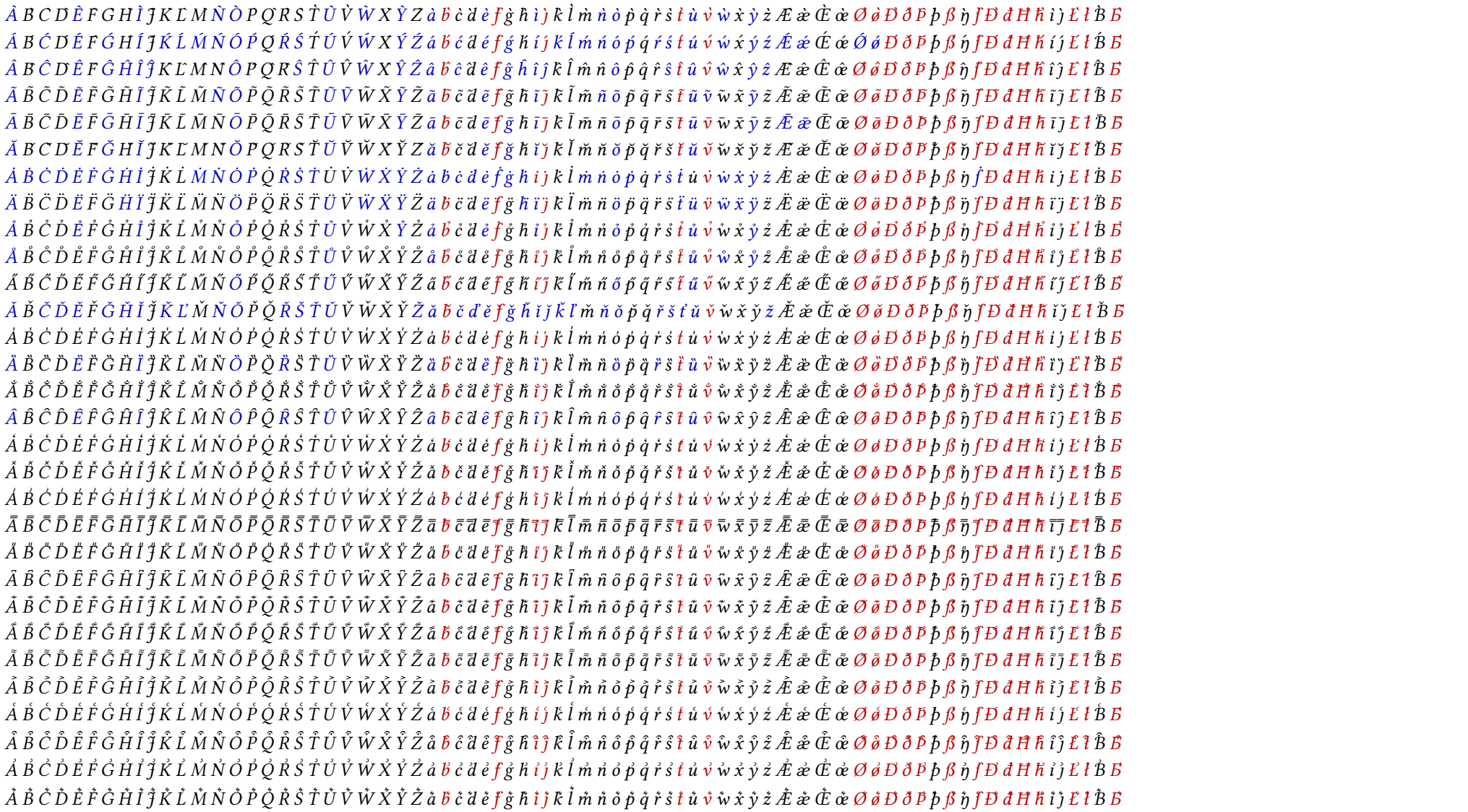
Regular, Latin, Above



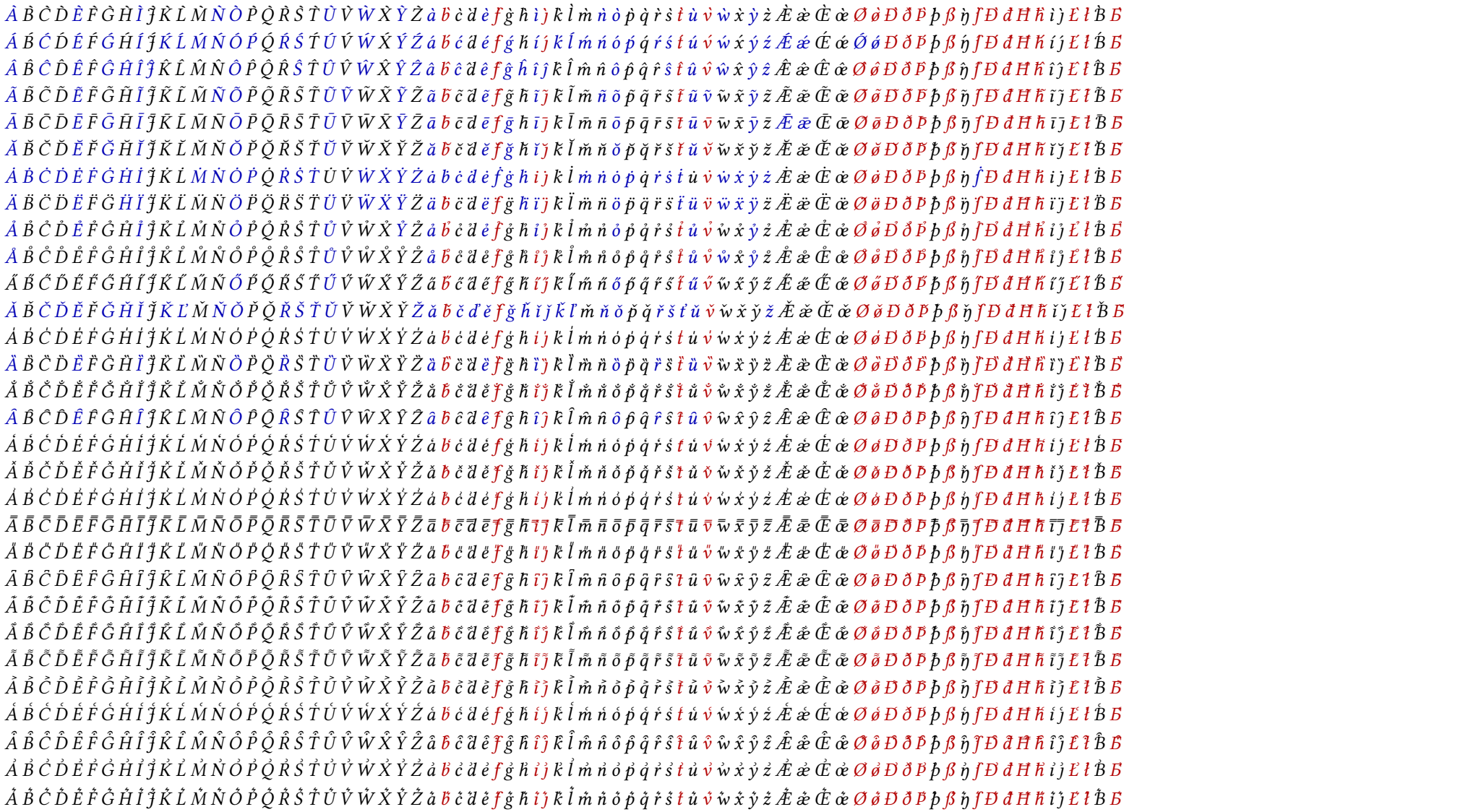
Regular patched, Latin, Above



Italic, Latin, Above



Italic patched, Latin, Above



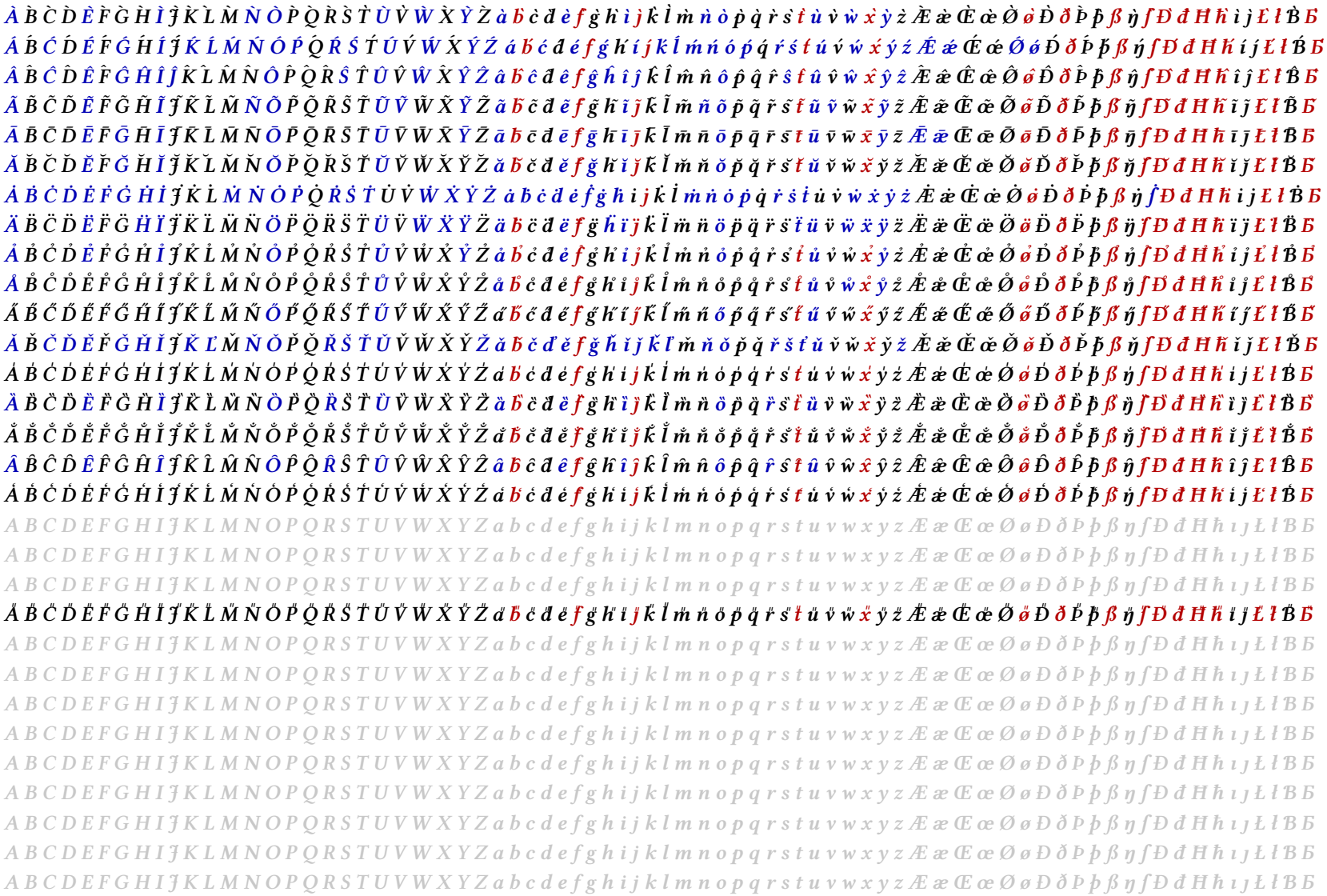
Semibold, Latin, Above



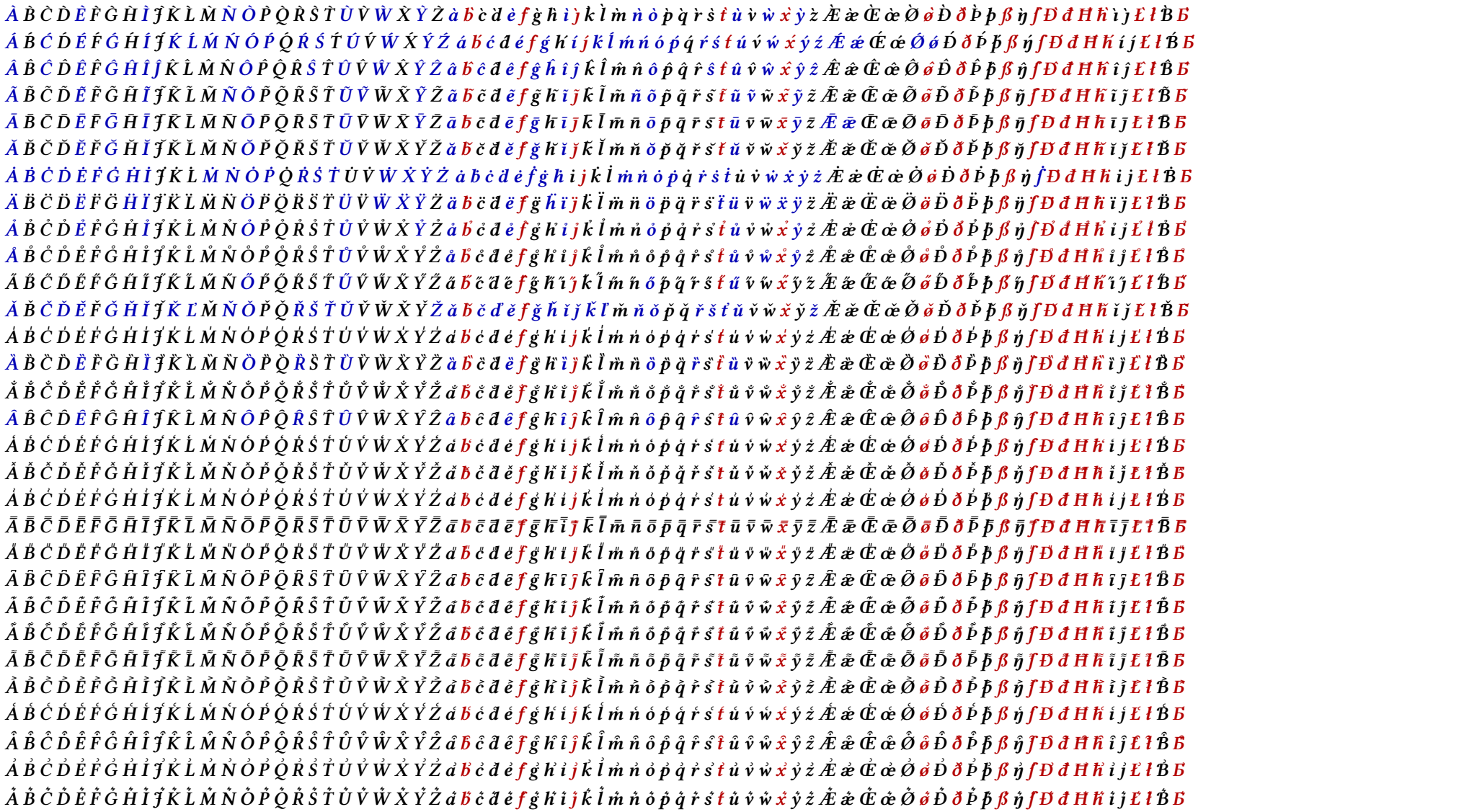
Semibold patched, Latin, Above



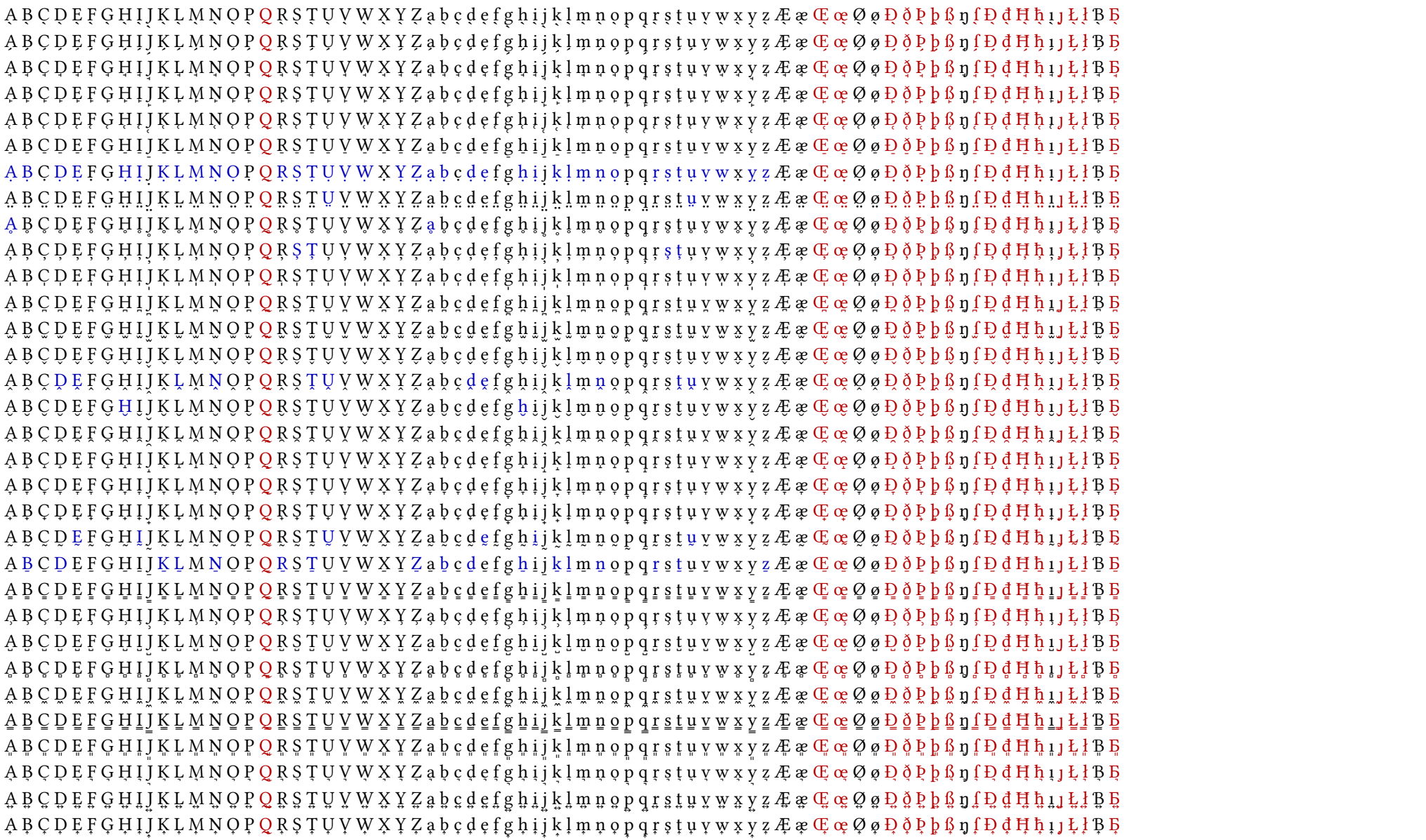
Semibold italic, Latin, Above



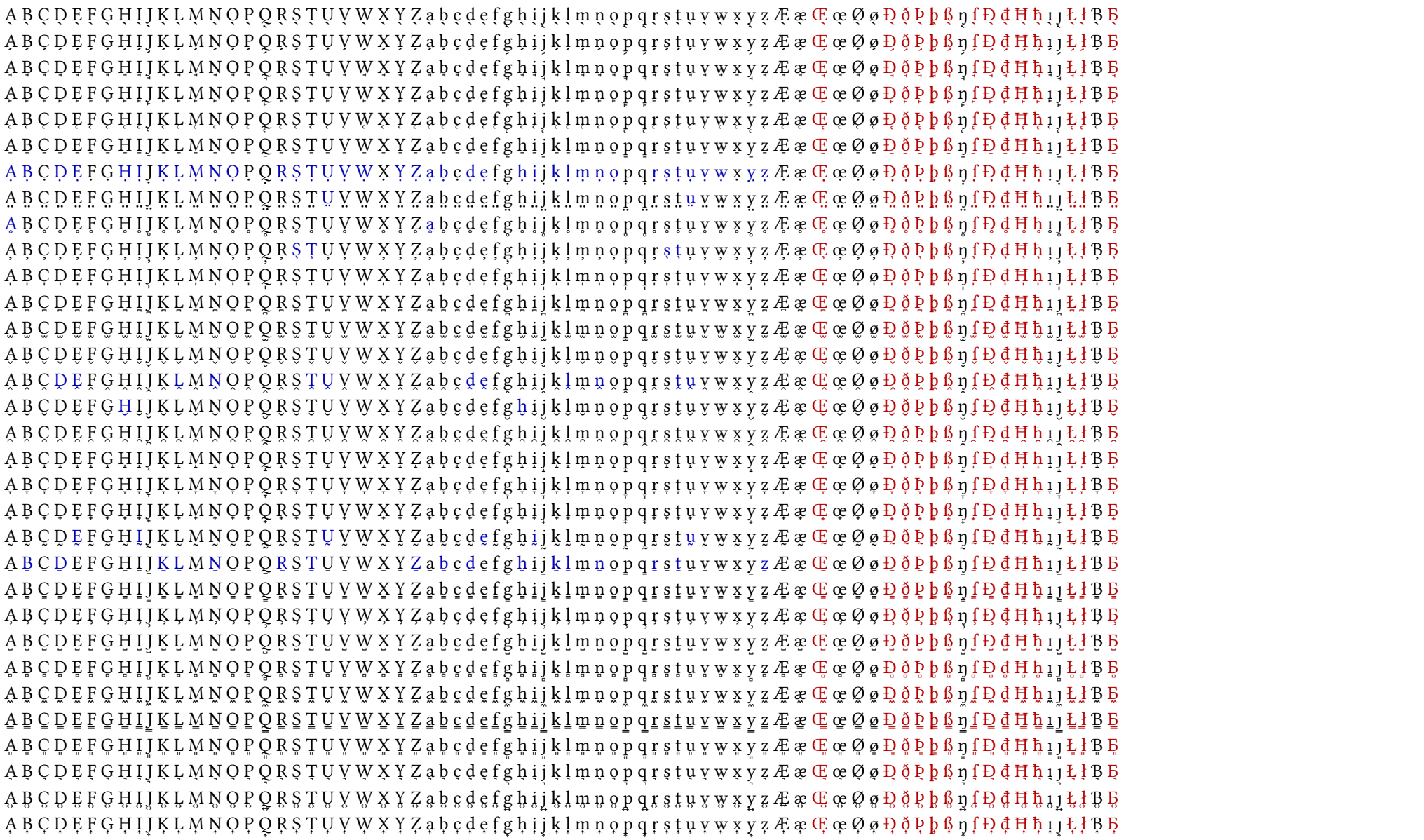
Semibold italic patched, Latin, Above



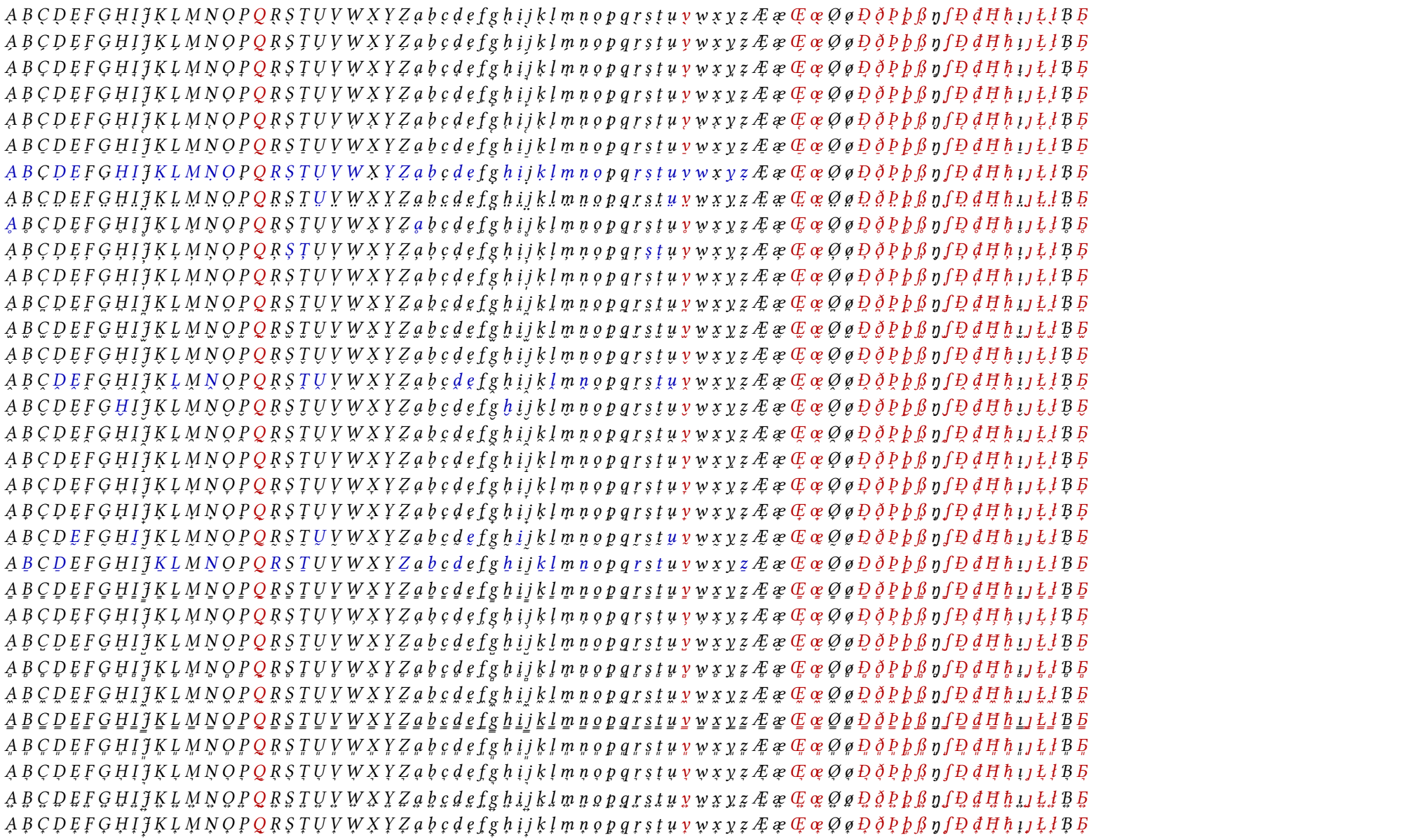
Regular, Latin, Below



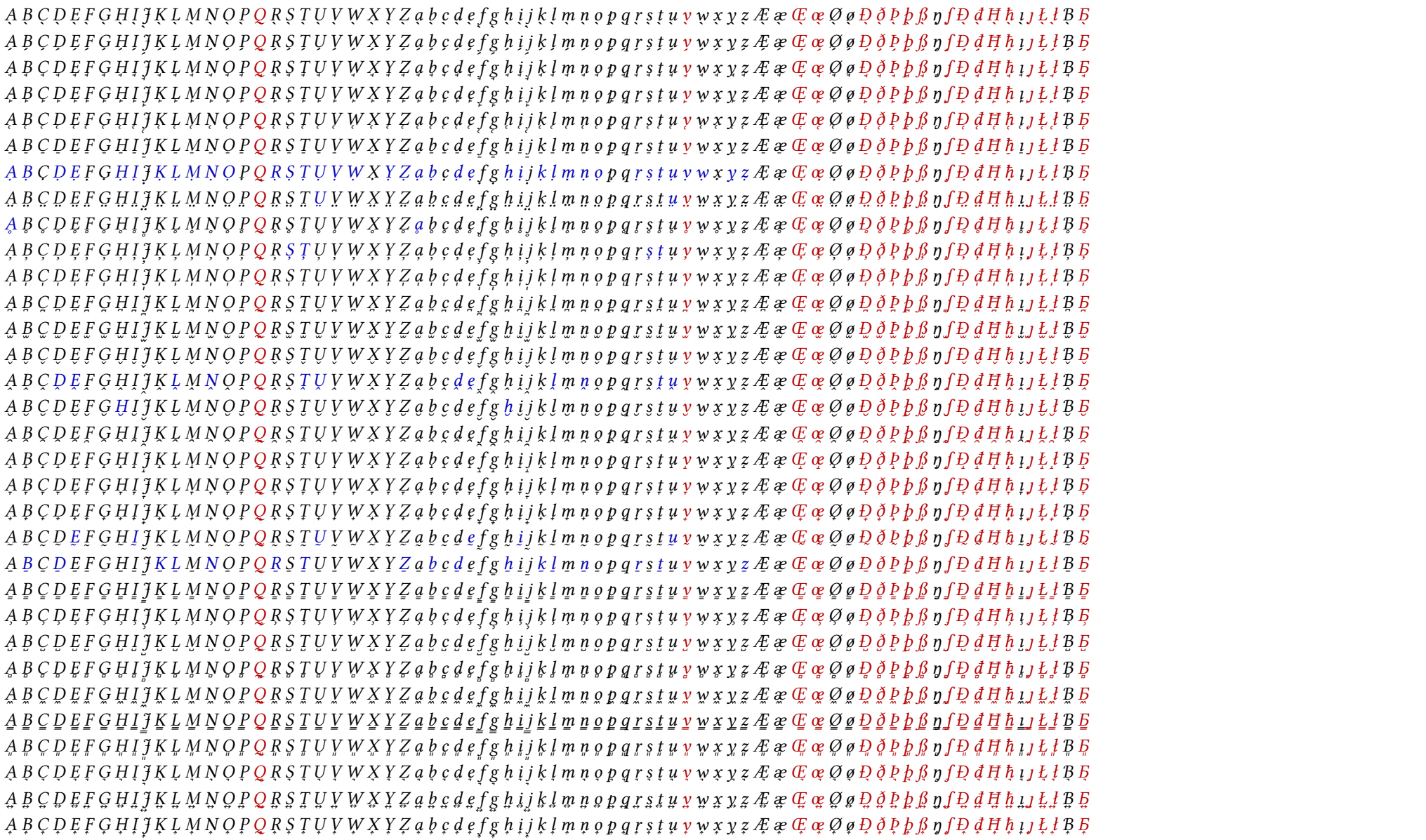
Regular patched, Latin, Below



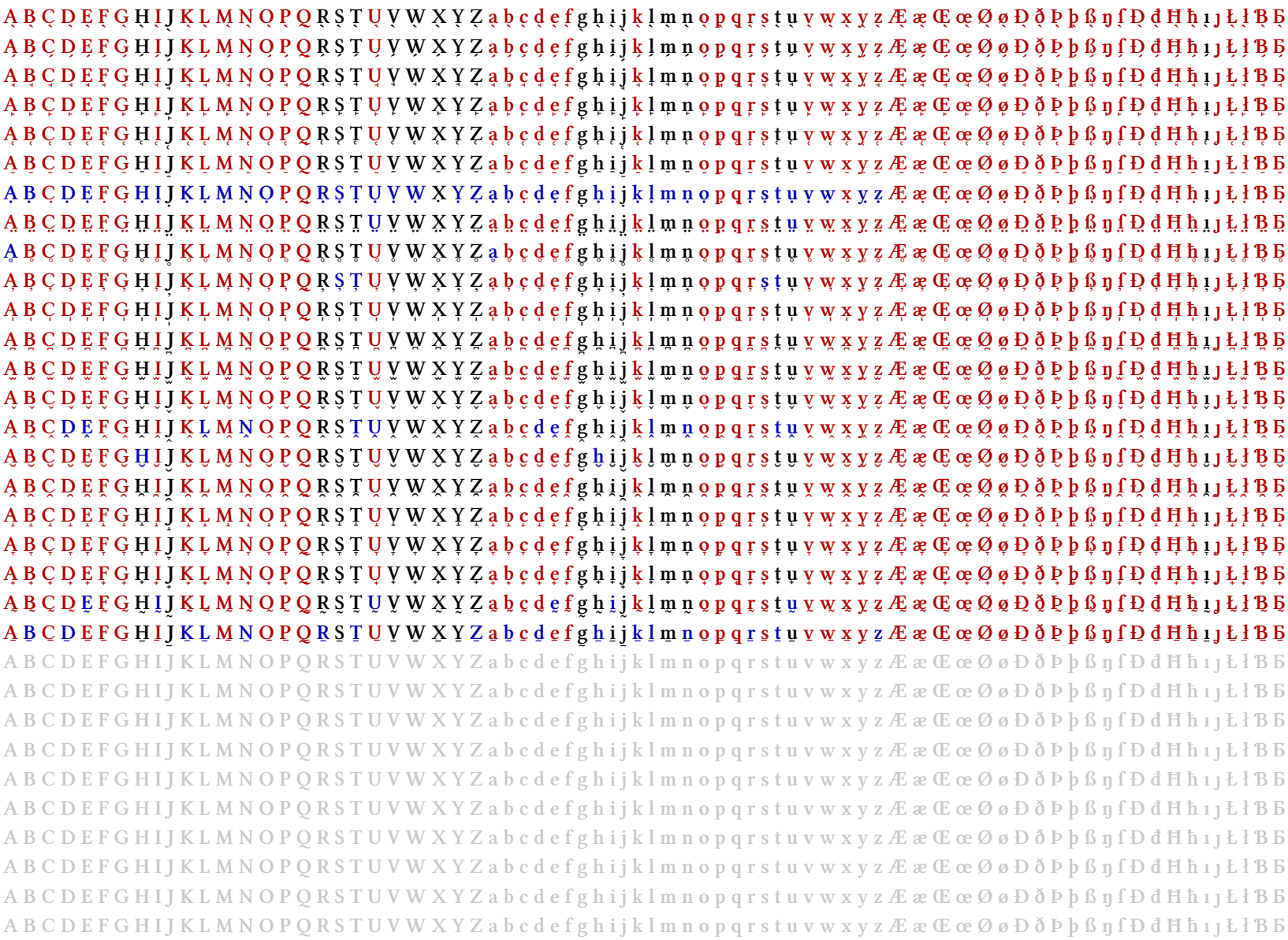
Italic, Latin, Below



Italic patched, Latin, Below



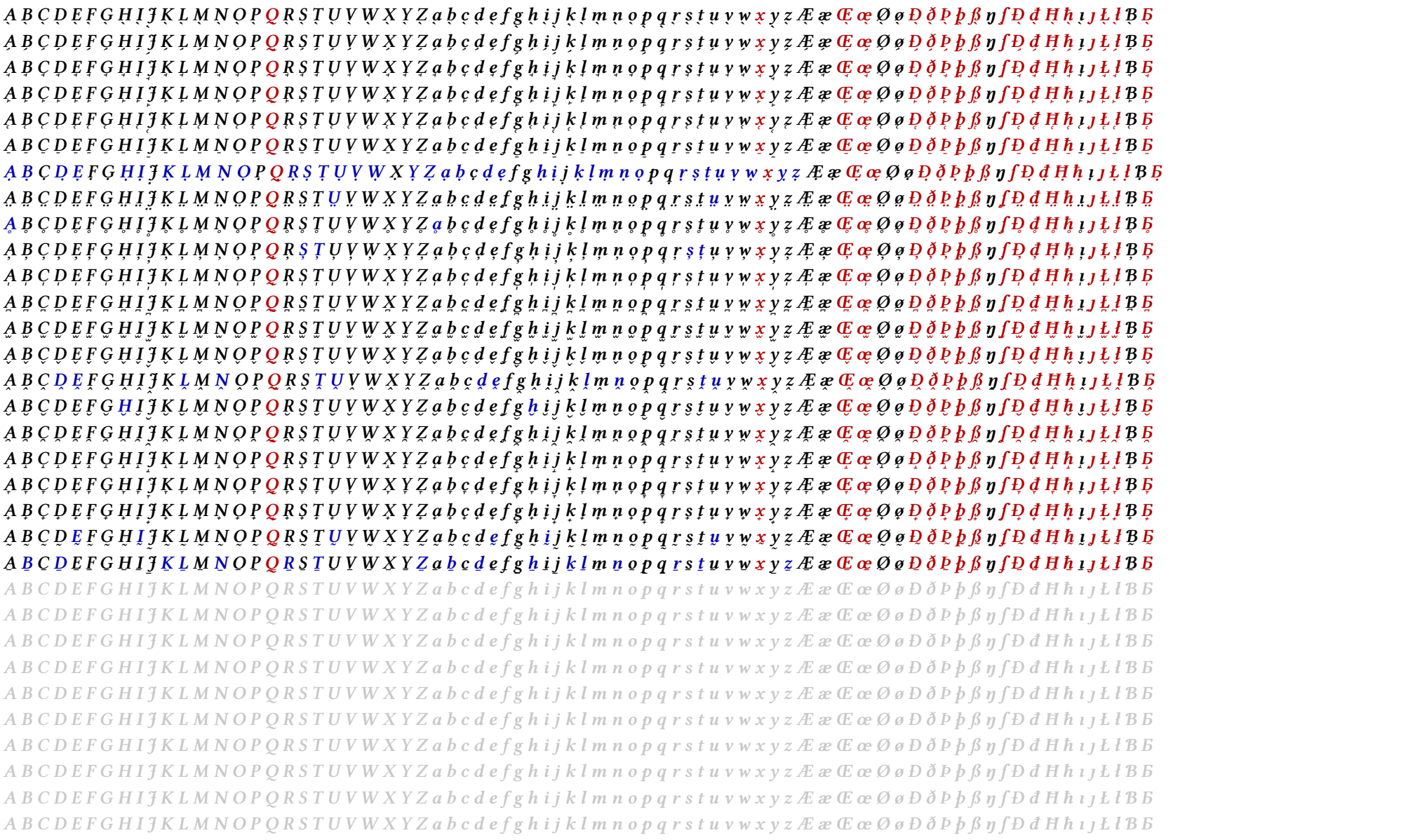
Semibold, Latin, Below



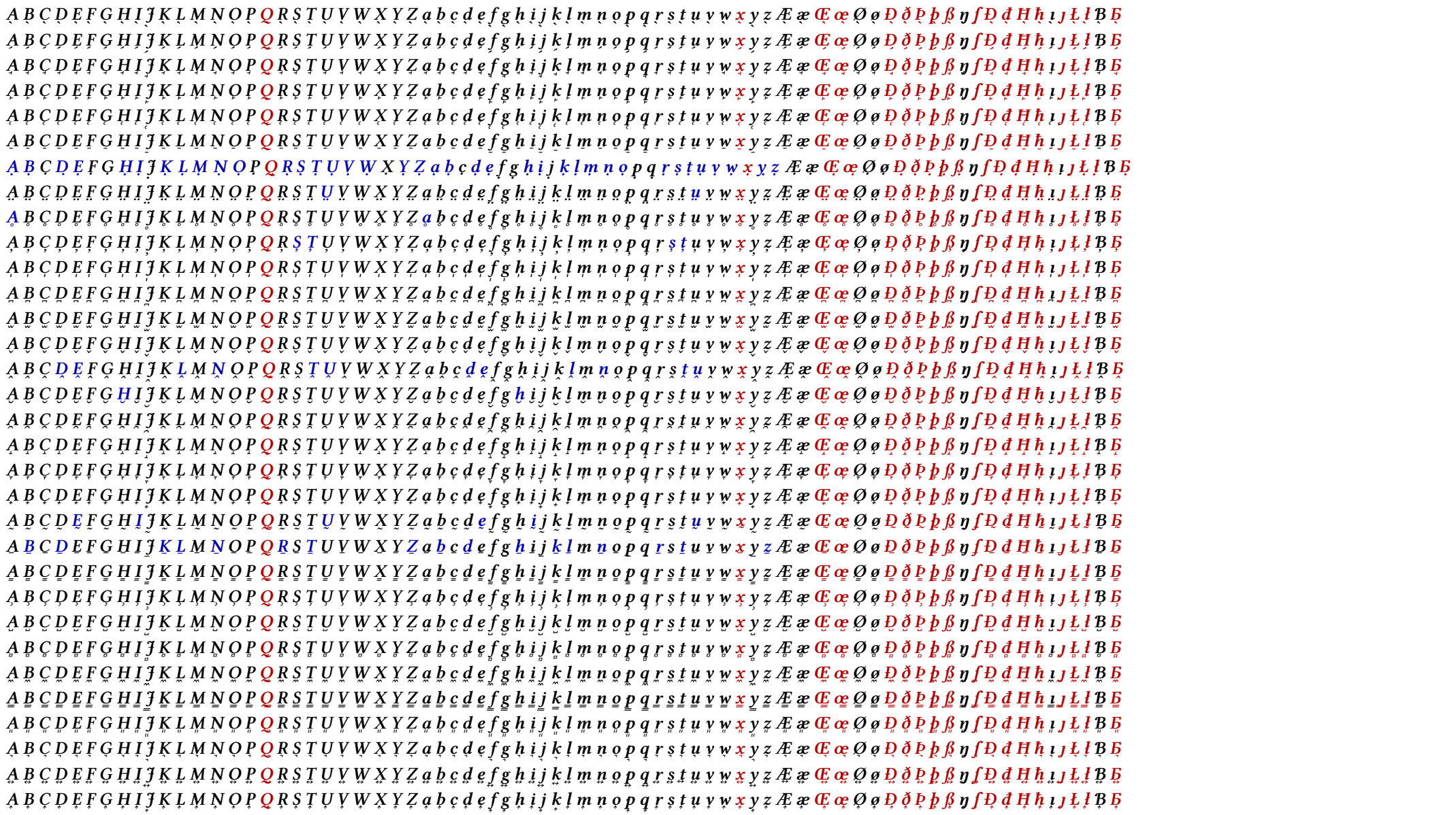
Semibold patched, Latin, Below



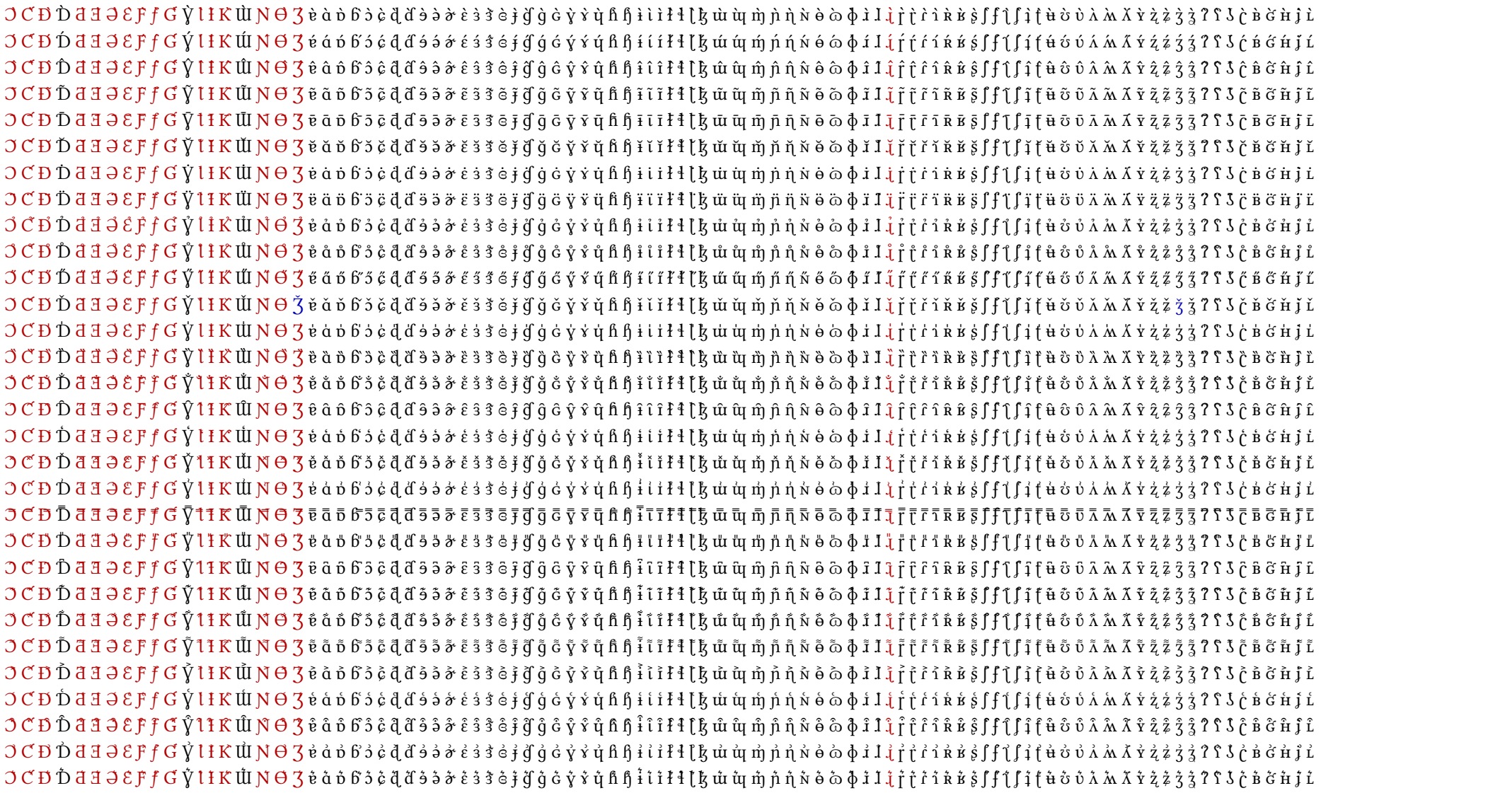
Semibold italic, Latin, Below



Semibold italic patched, Latin, Below



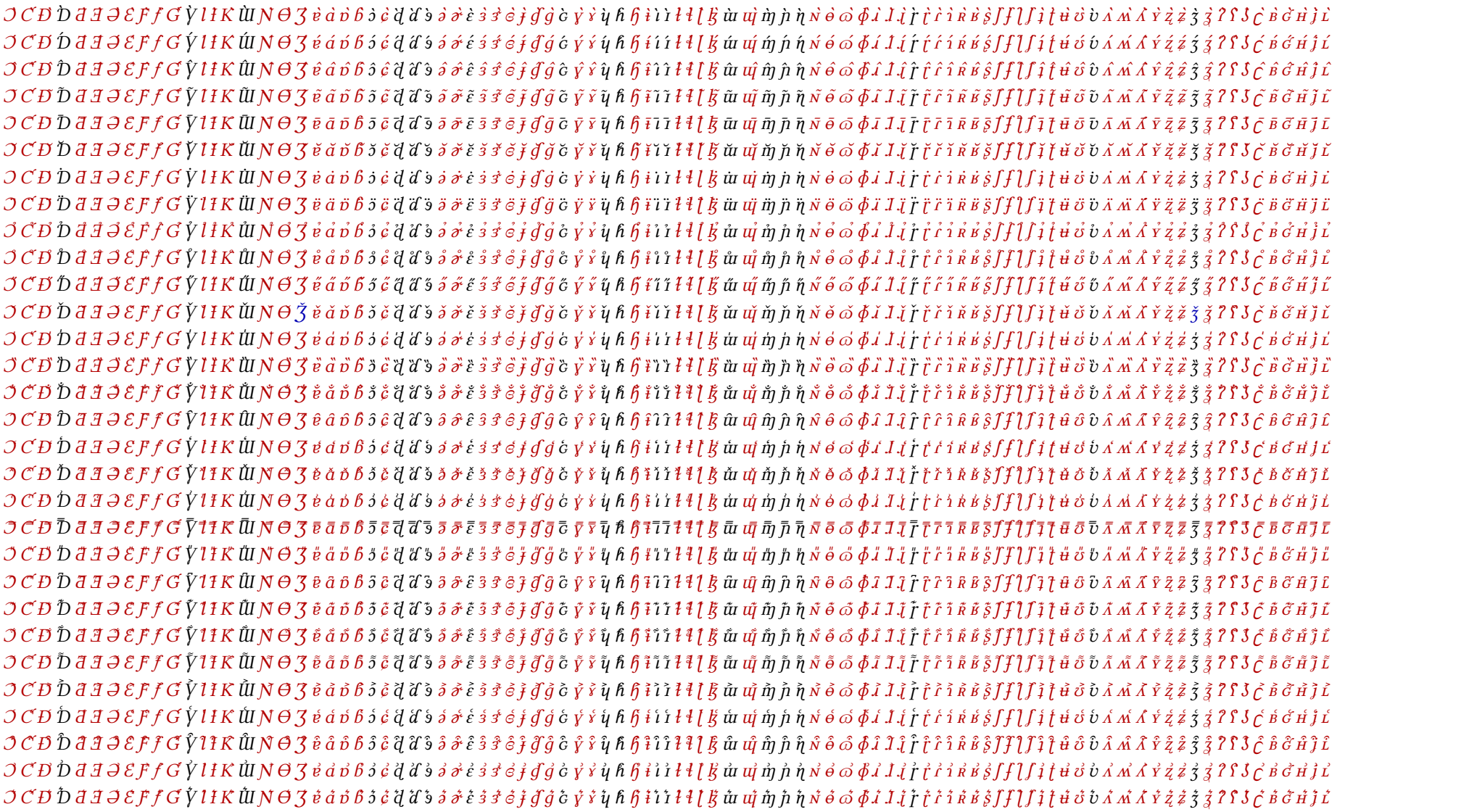
Regular, IPA, Above



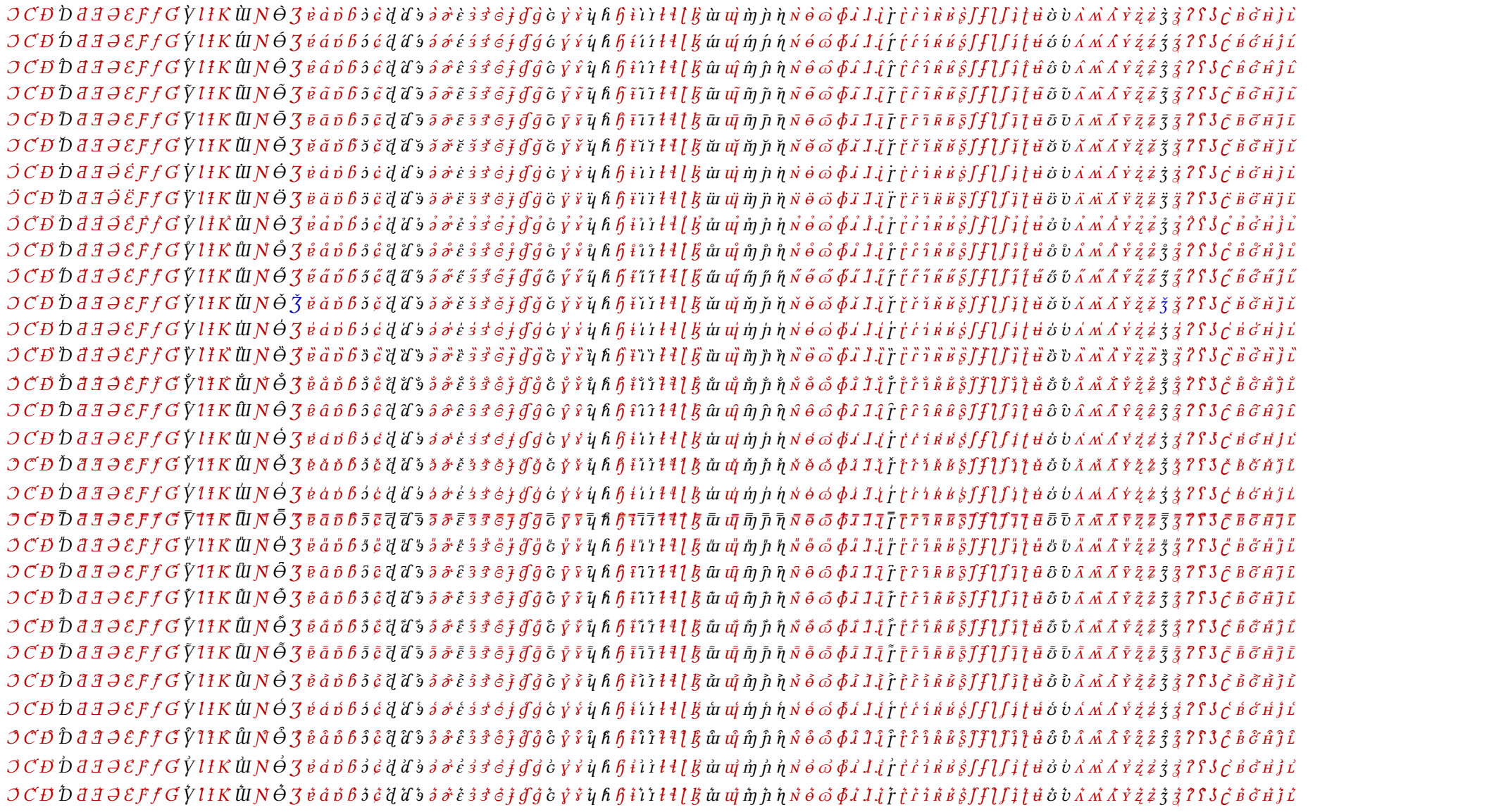
Regular patched, IPA, Above



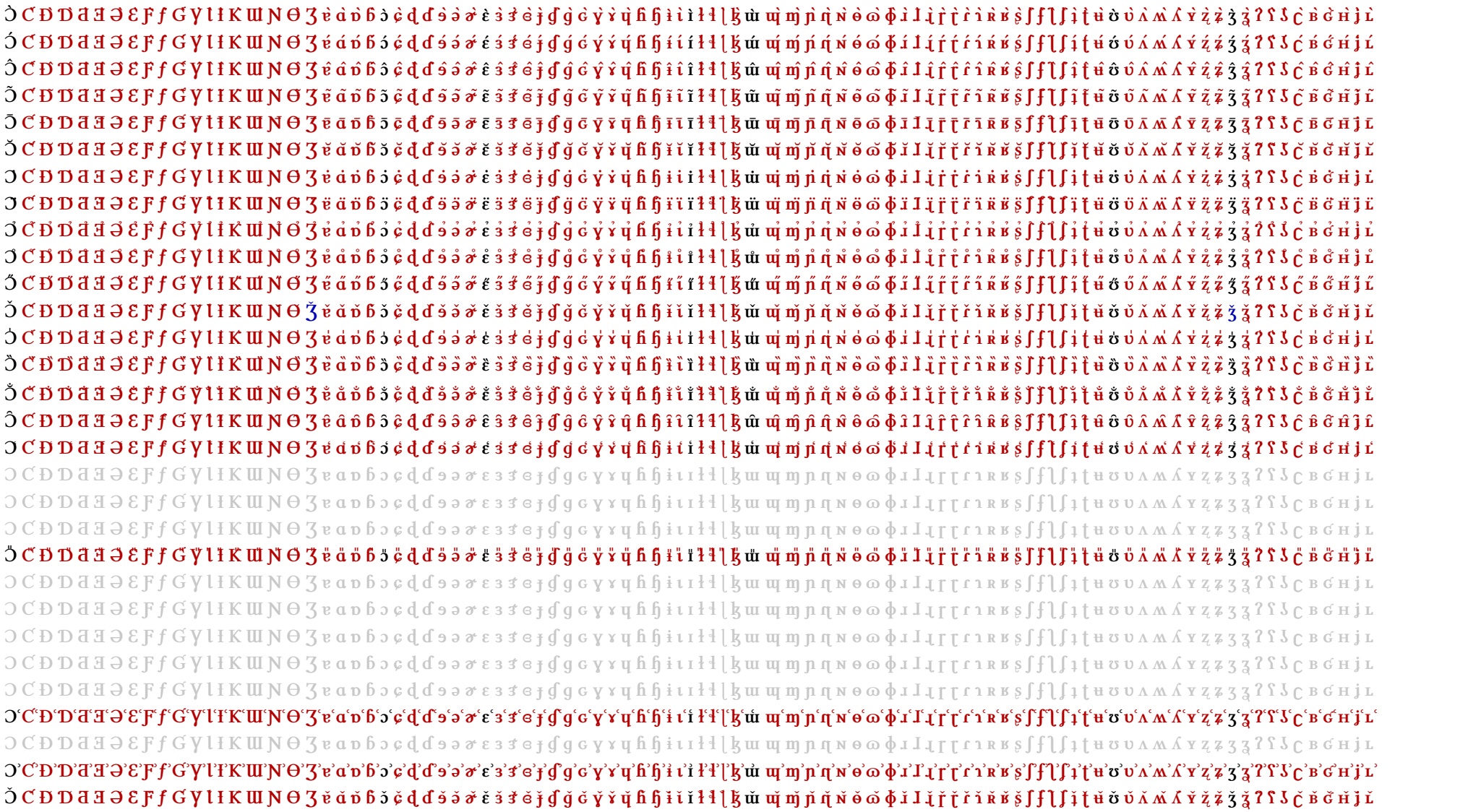
Italic, IPA, Above



Italic patched, IPA, Above



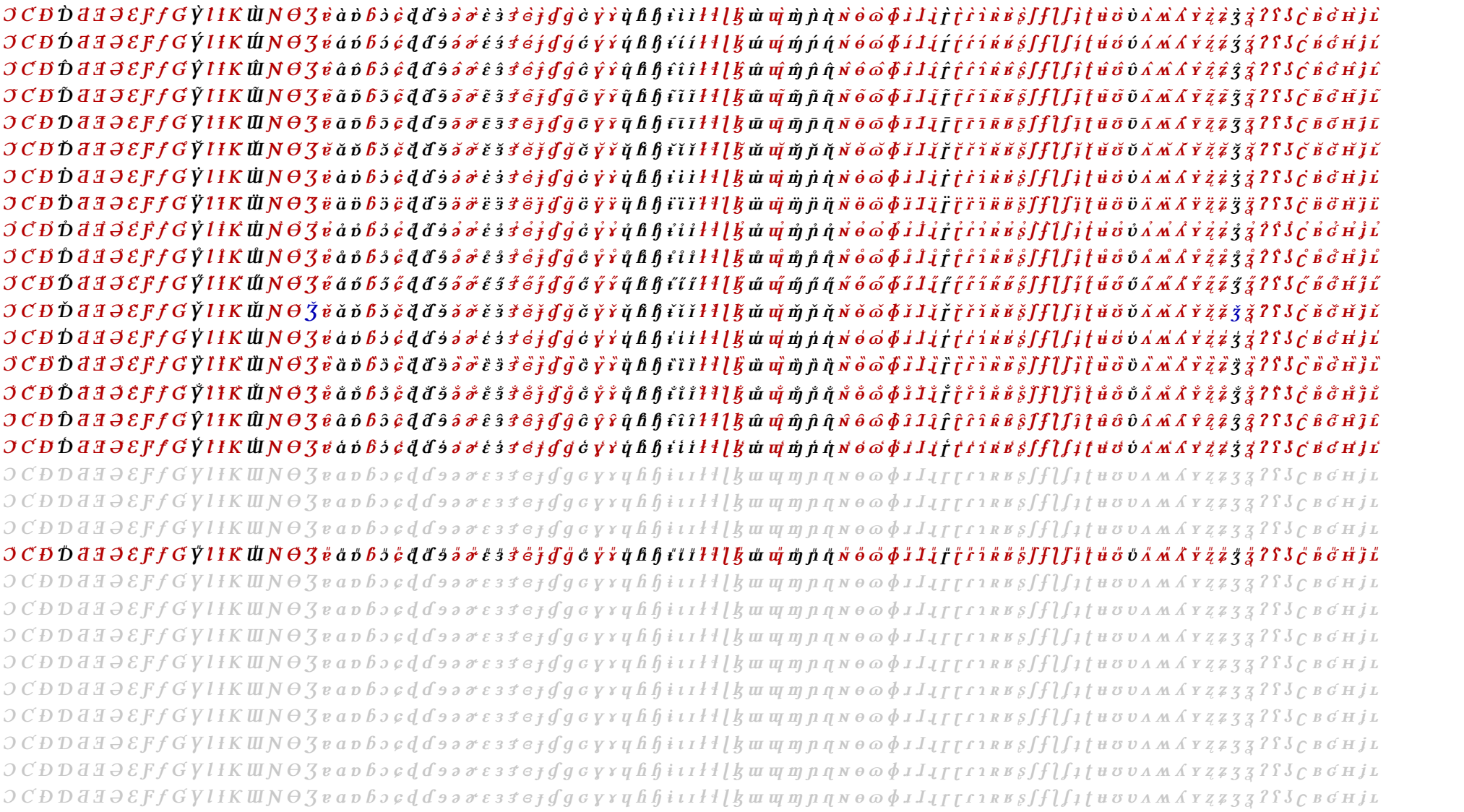
Semibold, IPA, Above



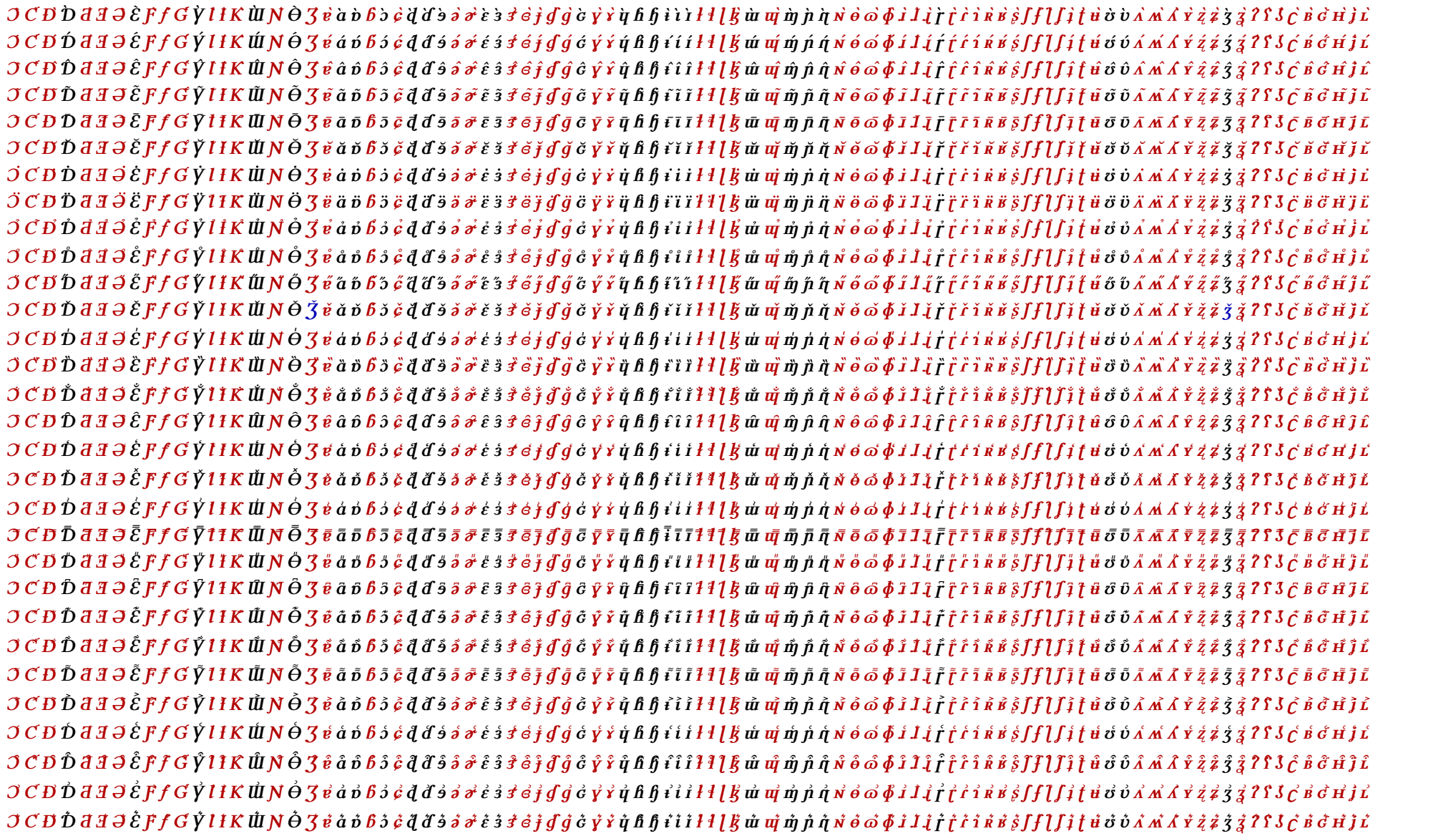
Semibold patched, IPA, Above



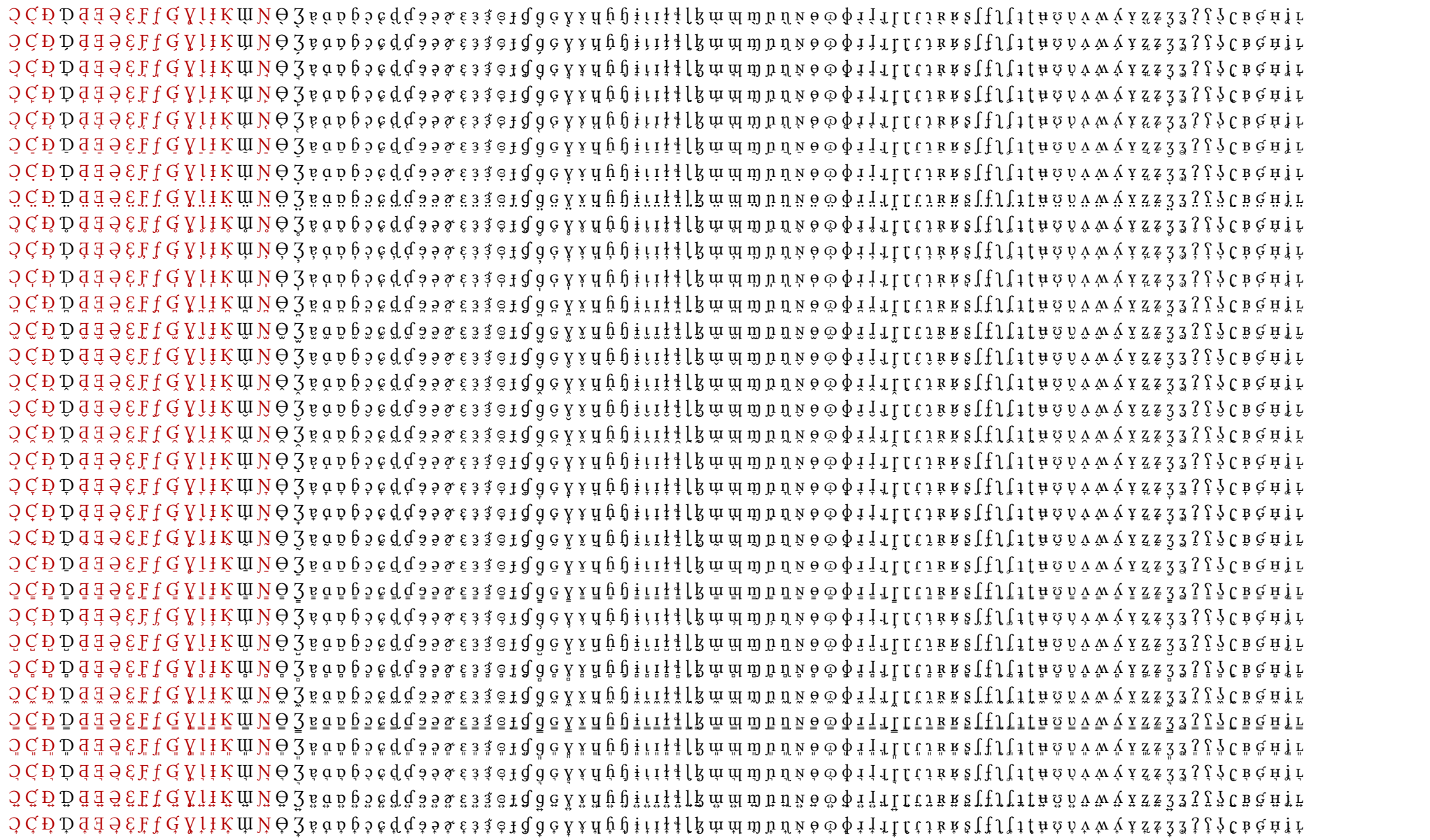
Semibold italic, IPA, Above



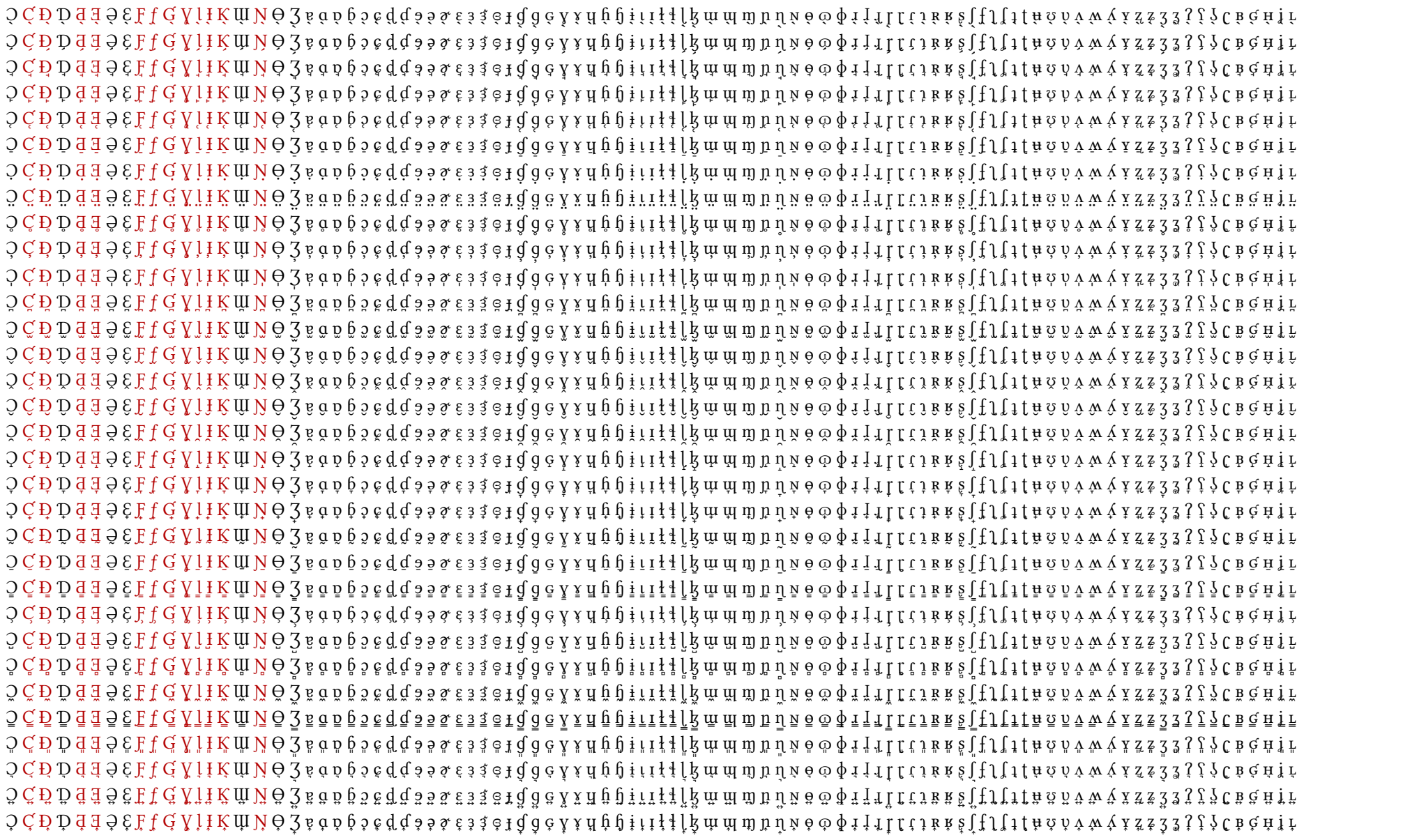
Semibold italic patched, IPA, Above



Regular, IPA, Below

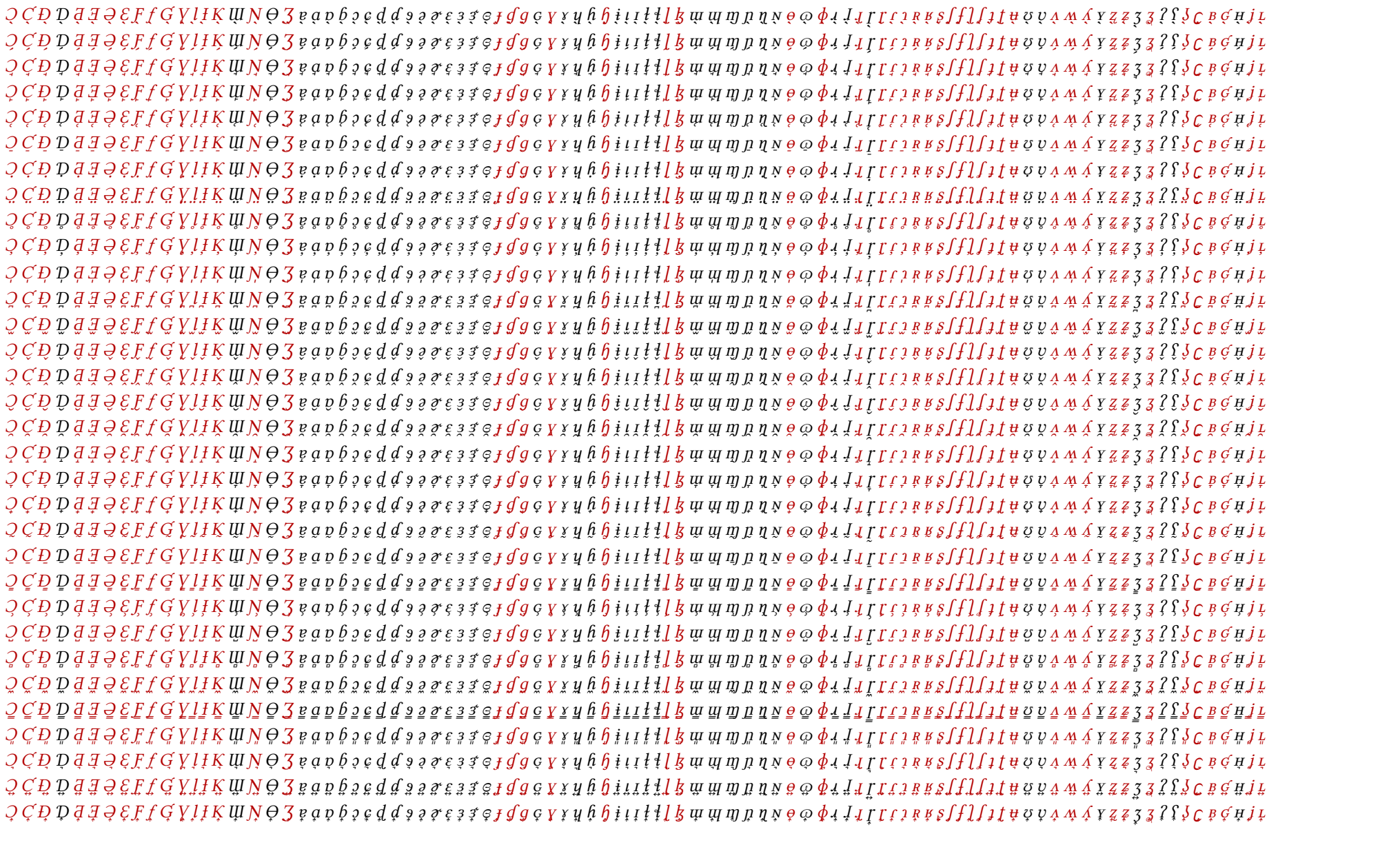


Regular patched, IPA, Below

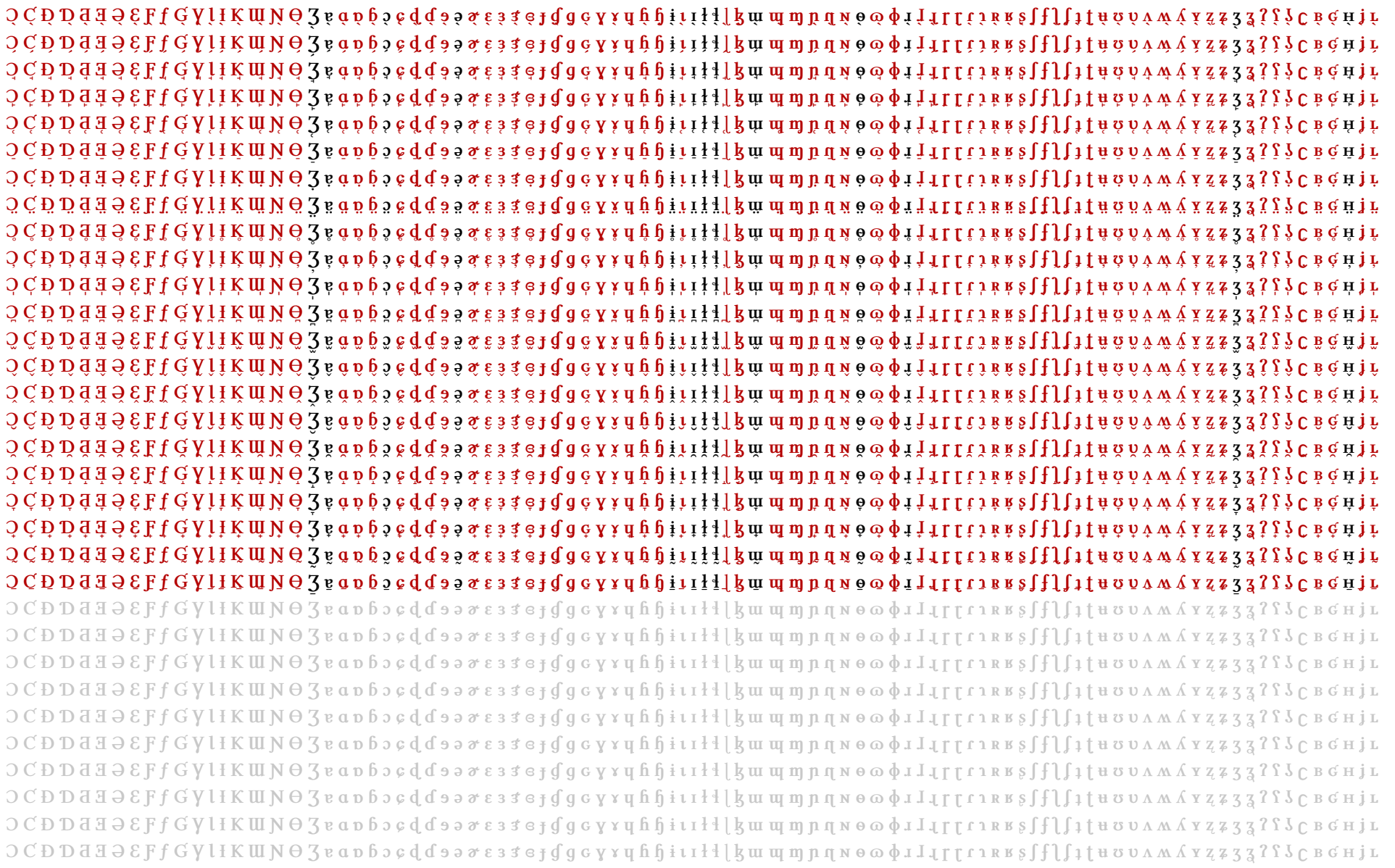


Italic, IPA, Below

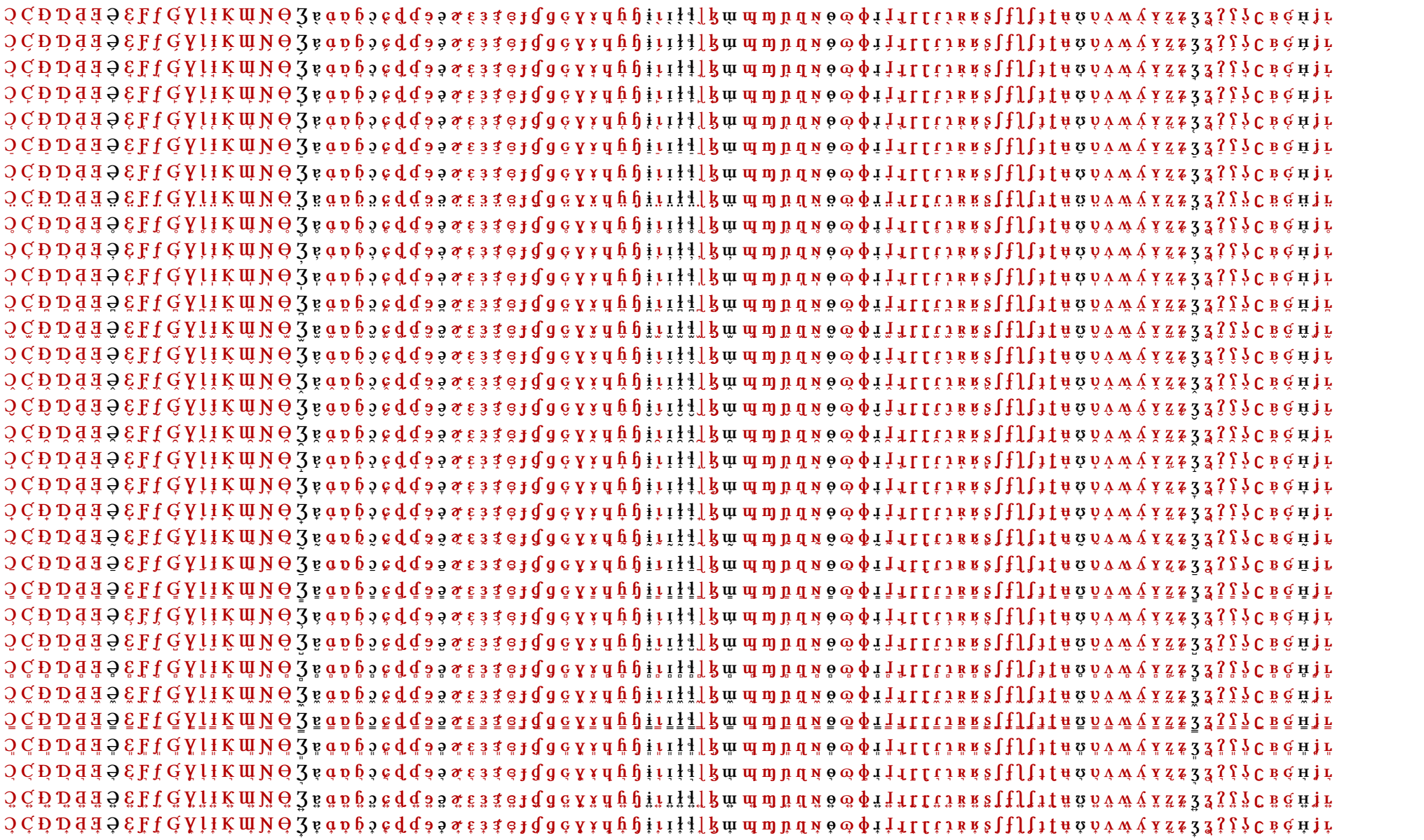
Italic patched, IPA, Below



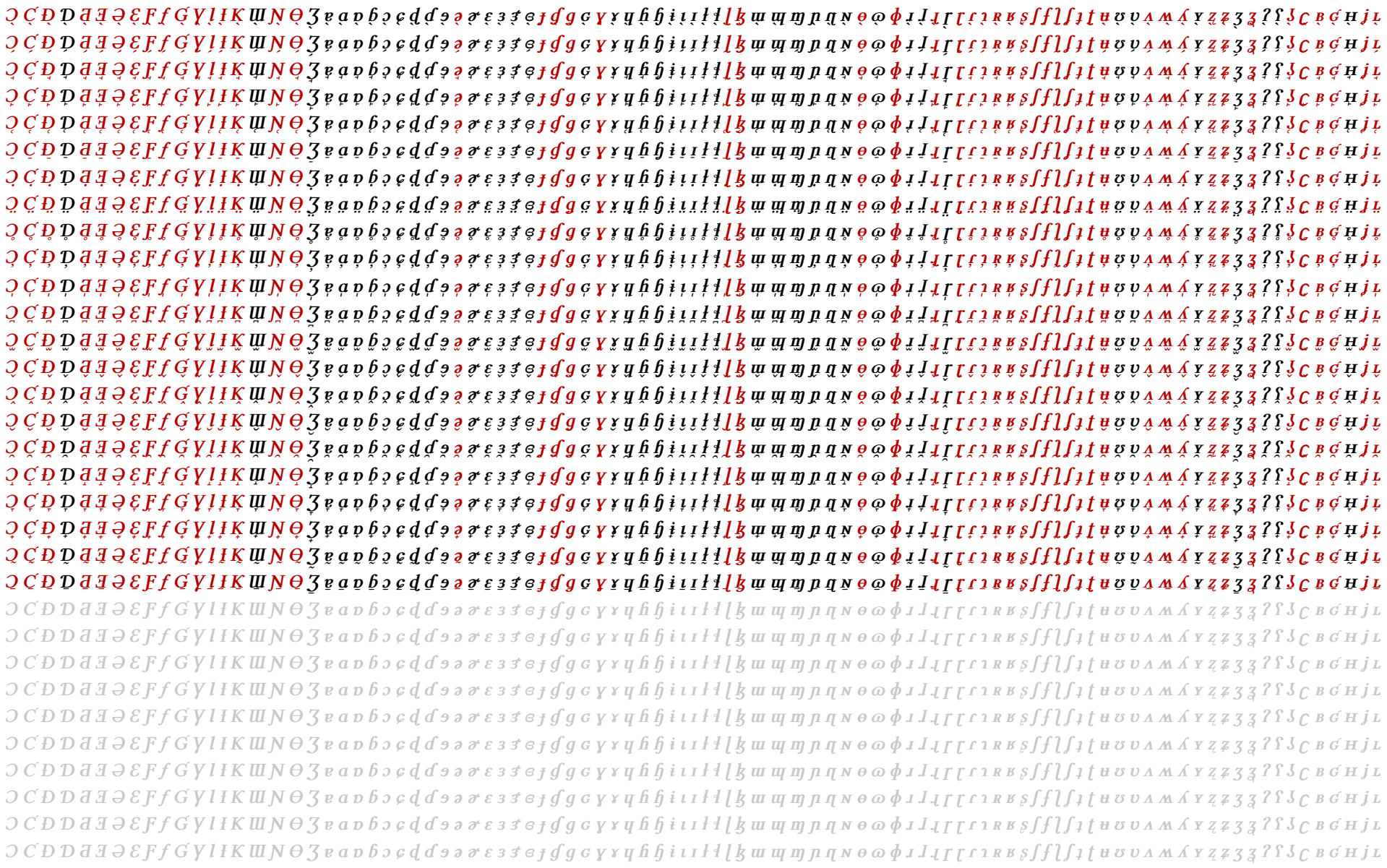
Semibold, IPA, Below



Semibold patched, IPA, Below



Semibold italic, IPA, Below



Semibold italic patched, IPA, Below



Capital alternates

These grids display capitals with alternate marks.

In the patched fonts, the mark anchor of each capital alternate is set by script (`font_patching_capmarks.py`) from the precomposed capitals A–Z. The Y coordinate is the capital anchor reference (850; 805 in semibold) less the height by which the mark is raised in the precomposed glyphs. The X coordinate of a symmetric mark is the center of its outline on its own axis; for the grave, acute, double acute and double grave it is the median position of the mark in the precomposed glyphs, relative to the midline of the base. In semibold italic both are shifted left by 150 units, as in the semibold italic anchor model. A human-curated anchor overrules the scripted one. Measured over the capitals that have an above-anchor, the median distance of an alternate mark from the midline falls from 142 to 17 units in italic and from 171 to 42 units in semibold italic; regular (16) and semibold (11) are unchanged.

Check these by eye:

- Italic. All ten curated anchors are kept and overrule the script. Their X coordinates differ from the scripted ones by 6 to 62 units: grave -45 (script 17), acute -65 (-37), caron -50 (-29), inverted breve -23 (-2), double grave -47 (-34), breve -10 (-1), circumflex 16 (24), hook -27 (-19), dieresis -75 (-69), double acute -109 (-103). The curated values fit the designer's capital anchors, which are left of the midline; the scripted values assume an anchor on the midline. If the capital anchors are moved to the midline, the curated values should be dropped.
- Regular. The curated grave $(-175, 723)$ and acute $(-210, 723)$ are kept; the script gives -162 and -218. The other eight curated anchors are within 4 units of the script and are commented out in `fontanchors_human`.
- Semibold italic. All anchors are scripted. The marks are still about 35 units left of the midline, because the designer's capital anchors are left of the candidates of the semibold italic anchor model; J and L are far off (their base anchors do not conform).
- Semibold. Only J, O, Q, R, S, T, V, W, X, Y, Z have an above-anchor; on the other capitals the alternates are placed by fallback. The hook above is raised by 89 units, and the double acute and double grave are lowered by 30 and 23 units, relative to the designer's anchors.
- The acute, grave and their doubles. The precomposed glyphs place these by eye, letter by letter, so the implied X varies by up to 190 units over the bases (most for the acute, because of L), and one anchor is a compromise.
- `dotaccent.cap` is new in italic and semibold italic, and in regular its anchor is raised by 50 units to match the precomposed dotted capitals.
- Not used as evidence, because the mark outline differs in the precomposed glyph: Ÿ and Ẅ (dieresis), Ľ (comma-style caron), and Ĵ in semibold italic.

Regular, Capitals, Marks with capital alternates

À B̀ C̀ D̀ È F̀ G̀ H̀ Ì J̀ K̀ L̀ M̀ Ǹ Ò P̀ Q̀ R̀ S̀ T̀ Ù V̀ Ẁ X̀ Ỳ Z̀ Ↄ Ə Ɛ Ɵ Σ Ʊ Ʊ Ƴ Ƶ Ę̀ Ḥ Ị̀ Ṛ Ṣ Ṭ Ṿ Ẓ Ạ̀ Ẹ̀ Ị̀ Ọ̀ Ụ̀
Á B́ Ć D́ É F́ Ǵ H́ Í J́ Ḱ Ĺ Ḿ Ń Ó Ṕ Q́ Ŕ Ś T́ Ú V́ Ẃ X́ Ý Ź Ↄ Ə Ɛ Ɵ Σ Ʊ Ʊ Ƴ Ƶ Ę́ Ḥ Í Ṛ Ṣ Ṭ Ṿ Ẓ Ạ́ Ẹ́ Í Ọ́ Ụ́
Â B̂ Ĉ D̂ Ê F̂ Ĝ Ĥ Î Ĵ Ǩ L̂ M̂ Ñ Ô P̂ Q̂ R̂ Ŝ T̂ Û V̂ Ŵ X̂ Ŷ Ẑ Ↄ Ə Ɛ Ɵ Σ Ʊ Ʊ Ƴ Ƶ Ę̂ Ḥ Î Ṛ Ṣ Ṭ Ṿ Ẓ Ậ Ệ Î Ộ Ụ̂
Ă B̆ Č D̆ Ĕ F̆ Ğ H̆ Ĭ J̆ Ǩ L̆ M̆ Ň Ŏ P̆ Q̆ Ř Š T̆ Ŭ V̆ W̆ X̆ Y̆ Ž Ↄ Ə Ɛ Ɵ Σ Ʊ Ʊ Ƴ Ƶ Ę̆ Ḥ Ĭ Ṛ Ṣ Ṭ Ṿ Ẓ Ặ Ẹ̆ Ĭ Ọ̆ Ụ̆
Ȧ Ḃ Ċ Ḋ Ė Ḟ Ġ Ḣ İ J̇ K̇ L̇ Ṁ Ṅ Ȯ Ṗ Q̇ Ṙ Ṡ Ṫ U̇ V̇ Ẇ Ẋ Ẏ Ż Ↄ Ə Ɛ Ɵ Σ Ʊ Ʊ Ƴ Ƶ Ę̇ Ḥ İ Ṛ Ṣ Ṭ Ṿ Ẓ Ạ̇ Ẹ̇ İ Ọ̇ Ụ̇
Ä B̈ C̈ D̈ Ë F̈ G̈ Ḧ Ï J̈ K̈ L̈ M̈ N̈ Ö P̈ Q̈ R̈ S̈ T̈ Ü V̈ Ẅ Ẍ Ÿ Z̈ Ↄ Ə Ɛ Ɵ Σ Ʊ Ʊ Ƴ Ƶ Ę̈ Ḥ Ï Ṛ Ṣ Ṭ Ṿ Ẓ Ạ̈ Ẹ̈ Ï Ọ̈ Ụ̈
Å B̊ C̊ D̊ E̊ F̊ G̊ H̊ Ů J̊ K̊ L̊ M̊ N̊ Ů P̊ Q̊ R̊ S̊ T̊ Ů V̊ W̊ Ẍ Y̊ Ż Ↄ Ə Ɛ Ɵ Σ Ʊ Ʊ Ƴ Ƶ Ę̊ Ḥ Ů Ṛ Ṣ Ṭ Ṿ Ẓ Ạ̊ Ẹ̊ Ů Ọ̊ Ụ̊
Ő B̋ C̋ D̋ Ű F̋ G̋ H̋ Ő J̋ K̋ L̋ M̋ N̋ Ő P̋ Q̋ R̋ S̋ T̋ Ű V̋ W̋ X̋ Ÿ Ż Ↄ Ə Ɛ Ɵ Σ Ʊ Ʊ Ƴ Ƶ Ę̋ Ḥ Ő Ṛ Ṣ Ṭ Ṿ Ẓ Ạ̋ Ẹ̋ Ő Ọ̋ Ụ̋
Ǎ B̌ Č Ď Ě F̌ Ǧ Ȟ Ǐ J̌ Ǩ Ľ M̌ Ň Ǒ P̌ Q̌ Ř Š Ť Ǔ V̌ W̌ X̌ Y̌ Ž Ↄ Ə Ɛ Ɵ Σ Ʊ Ʊ Ƴ Ƶ Ę̌ Ḥ Ǐ Ṛ Ṣ Ṭ Ṿ Ẓ Ạ̌ Ẹ̌ Ǐ Ọ̌ Ụ̌
Ȁ B̏ C̏ D̏ Ȅ F̏ G̏ H̏ Ȉ J̏ K̏ L̏ M̏ N̏ Ȍ P̏ Q̏ Ȑ S̏ T̏ Ȕ V̏ W̏ X̏ Ÿ Ż Ↄ Ə Ɛ Ɵ Σ Ʊ Ʊ Ƴ Ƶ Ę̏ Ḥ Ȉ Ṛ Ṣ Ṭ Ṿ Ẓ Ạ̏ Ẹ̏ Ȉ Ọ̏ Ụ̏
Â B̂ Ĉ D̂ Ê F̂ Ĝ Ĥ Î Ĵ Ǩ L̂ M̂ Ñ Ô P̂ Q̂ R̂ Ŝ T̂ Û V̂ Ŵ X̂ Ŷ Ẑ Ↄ Ə Ɛ Ɵ Σ Ʊ Ʊ Ƴ Ƶ Ę̂ Ḥ Î Ṛ Ṣ Ṭ Ṿ Ẓ Ậ Ệ Î Ộ Ụ̂

Regular patched, Capitals, Marks with capital alternates

À B̀ C̀ D̀ È F̀ G̀ H̀ Ì J̀ K̀ L̀ M̀ Ǹ Ò P̀ Q̀ R̀ S̀ T̀ Ù V̀ Ẁ X̀ Ỳ Z̀ Ɔ̀ ə̀ ɛ̀ ɵ̀ Ɛ̀ ʊ̀ Ù Ỳ Ƶ̀ Ḛ̀ Ḥ̀ Į̀ Ḛ̀ S̀ T̀ V̀ Ƶ̀ Ạ̀ Ḛ̀ Į̀ Ọ̀ Ụ̀

Á B́ Ć D́ É F́ Ǵ H́ Í J́ Ḱ Ĺ Ḿ Ń Ó Ṕ Q́ Ŕ Ś T́ Ú V́ Ẃ X́ Ý Ź Ɔ́ ə́ ɛ́ ɵ́ Ɛ́ ʊ́ Ú Ý Ƶ́ Ḛ́ Ḥ́ Į́ Ḛ́ Ś T́ V́ Ƶ́ Ạ́ Ḛ́ Į́ Ọ́ Ụ́

Â B̂ Ĉ D̂ Ê F̂ Ĝ Ĥ Î Ĵ K̂ L̂ M̂ N̂ Ô P̂ Q̂ R̂ Ŝ T̂ Û V̂ Ŵ X̂ Ŷ Ẑ Ɔ̂ ə̂ ɛ̂ ɵ̂ Ɛ̂ ʊ̂ Û Ŷ Ƶ̂ Ḛ̂ Ḥ̂ Į̂ Ḛ̂ Ŝ T̂ V̂ Ƶ̂ Ậ Ḛ̂ Į̂ Ộ Ụ̂

Ă B̆ Č D̆ Ĕ F̆ Ğ H̆ Ĭ J̆ Ǩ L̆ M̆ N̆ Ŏ P̆ Q̆ R̆ Š T̆ Ŭ V̆ W̆ X̆ Y̆ Ž Ɔ̆ ə̆ ɛ̆ ɵ̆ Ɛ̆ ʊ̆ Ŭ Y̆ Ž̆ Ḛ̆ Ḥ̆ Į̆ Ḛ̆ Š T̆ V̆ Ž̆ Ặ Ḛ̆ Į̆ Ọ̆ Ụ̆

Ȧ Ḃ Ċ Ḋ Ė Ḟ Ġ Ḣ İ J̇ K̇ L̇ Ṁ Ṅ Ȯ Ṗ Q̇ Ṙ Ṡ Ṫ U̇ V̇ Ẇ Ẋ Ẏ Ż Ɔ̇ ə̇ ɛ̇ ɵ̇ Ɛ̇ ʊ̇ U̇ Ẏ Ż̧ Ȩ̰ Ḩ̣ Į̧ Ȩ̰ Ṣ Ṭ V̇ Ż̧ Ạ̇ Ȩ̰ Į̧ Ọ̧ Ụ̧

Ä B̈ C̈ D̈ Ë F̈ G̈ Ḧ Ï J̈ K̈ L̈ M̈ N̈ Ö P̈ Q̈ R̈ S̈ T̈ Ü V̈ Ẅ Ẍ Ÿ Z̈ Ɔ̈ ə̈ ɛ̈ ɵ̈ Ɛ̈ ʊ̈ Ü Ÿ Z̈ Ḛ̈ Ḧ Ï Ḛ̈ S̈ T̈ V̈ Z̈ Ä Ḛ̈ Ï Ö Ü

Å B̊ C̊ D̊ E̊ F̊ G̊ H̊ Į̊ J̊ K̊ L̊ M̊ N̊ Ů P̊ Q̊ R̊ S̊ T̊ Ů V̊ W̊ Ẍ Y̊ Ż Ɔ̊ ə̊ ɛ̊ ɵ̊ Ɛ̊ ʊ̊ Ů Y̊ Ż̊ Ḛ̊ Ḥ̊ Į̊ Ḛ̊ S̊ T̊ V̊ Ż̊ Å Ḛ̊ Į̊ Ọ̊ Ụ̊

Ő B̋ C̋ D̋ Ḝ F̋ G̋ H̋ Ḯ J̋ K̋ L̋ M̋ N̋ Ő P̋ Q̋ R̋ S̋ T̋ Ű V̋ W̋ X̋ Ÿ Ż Ɔ̋ ə̋ ɛ̋ ɵ̋ Ɛ̋ ʊ̋ Ű Ÿ Ż̋ Ḝ̋ Ḥ̋ Į̋ Ḝ̋ S̋ T̋ V̋ Ż̋ Ą̋ Ḝ̋ Į̋ Ǫ̋ Ų̋

Ă B̌ Č Ď Ě F̌ Ǧ Ȟ Ǐ J̌ Ǩ Ľ M̌ Ň Ǒ P̌ Q̌ Ř Š Ť Ǔ V̌ W̌ X̌ Y̌ Ž Ɔ̌ ə̌ ɛ̌ ɵ̌ Ɛ̌ ʊ̌ Ǔ Y̌ Ž̌ Ḛ̌ Ḥ̌ Į̌ Ḛ̌ Š Ť V̌ Ž̌ Ą̌ Ḛ̌ Į̌ Ǫ̌ Ų̌

Ä B̈ C̈ D̈ Ë F̈ G̈ Ḧ Ï J̈ K̈ L̈ M̈ N̈ Ö P̈ Q̈ R̈ S̈ T̈ Ü V̈ Ẅ Ẍ Ÿ Z̈ Ɔ̈ ə̈ ɛ̈ ɵ̈ Ɛ̈ ʊ̈ Ü Ÿ Z̈ Ḛ̈ Ḧ Ï Ḛ̈ S̈ T̈ V̈ Z̈ Ä Ḛ̈ Ï Ö Ü

Â B̂ Ĉ D̂ Ê F̂ Ĝ Ĥ Î Ĵ K̂ L̂ M̂ N̂ Ô P̂ Q̂ R̂ Ŝ T̂ Û V̂ Ŵ X̂ Ŷ Ẑ Ɔ̂ ə̂ ɛ̂ ɵ̂ Ɛ̂ ʊ̂ Û Ŷ Ƶ̂ Ḛ̂ Ḥ̂ Į̂ Ḛ̂ Ŝ T̂ V̂ Ƶ̂ Ą̂ Ḛ̂ Į̂ Ǫ̂ Ų̂

Italic, Capitals, Marks with capital alternates

Â B C D È F G H Ì J K L M Ǹ Ò P Q R S T̀ Ù V̀ Ẁ X Ỳ Z Ɔ Ə Ɛ Θ Σ Ʊ U Y Ƶ Ȩ Ḥ Į Ɍ Ș Ț V̇ Ẓ Ą Ę Ị Ǫ Ų

Á B Ć D É F Ǵ H Í J Ḱ Ĺ Ḿ Ń Ó Ṕ Q Ŕ Ś T́ Ú V́ Ẃ X Ý Ź Ɔ Ə Ɛ Θ Σ Ʊ U Y Ƶ Ȩ Ḥ Į Ɍ Ș Ț V̇ Ẓ Ą Ę Ị Ǫ Ų

Â B Ĉ D Ê F Ĝ Ĥ Î J̇ K L M N Ô P Q R Ŝ T̂ Û V̂ Ŵ X Ŷ Ẑ Ɔ Ə Ɛ Θ Σ Ʊ U Y Ƶ Ȩ Ḥ Į Ɍ Ș Ț V̇ Ẓ Ą̂ Ę̂ Į̇ Ǫ̇ Ų

Ă B C D Ĕ F Ğ H Ǐ J K L M N Ŏ P Q R S Ť Ŭ V̆ W̆ X Y̆ Z Ɔ Ə Ɛ Θ Σ Ʊ U Y Ƶ Ȩ Ḥ Į Ɍ Ș Ț V̇ Ẓ Ą̆ Ę̆ Į̆ Ǫ̆ Ų̆

Ȧ Ḃ Ċ Ḋ Ė Ḟ Ġ Ḣ İ J̇ Ṙ Ḷ Ṁ Ṅ Ȯ Ṗ Q̇ Ṙ Ṡ Ṫ U̇ V̇ Ẇ Ẋ Ẏ Ż Ɔ Ə Ɛ Θ Σ Ʊ U Y Ƶ Ȩ Ḥ Į Ɍ Ṣ Ṭ V̇ Ẓ Ą Ę Į Ǫ Ų

Ä B̈ C̈ D̈ Ë F̈ G̈ Ḧ Ï J̈ K̈ L̈ M̈ N̈ Ö P̈ Q̈ R̈ S̈ T̈ Ü V̈ Ẅ Ẍ Ÿ Z̈ Ɔ Ə Ɛ Θ Σ Ʊ U Y Ƶ Ȩ Ḥ Į Ɍ Ș Ț V̇ Ẓ Ą Ę Į Ǫ Ų

Å Ḃ Ḉ Ḋ Ḛ Ḟ Ġ Ḣ İ J̊ Ḳ Ḻ Ṁ Ṅ Ȯ Ṗ Q̊ Ṙ Ṡ Ṫ Ů V̊ W̊ Ẋ Y̊ Ż Ɔ Ə Ɛ Θ Σ Ʊ U Y Ƶ Ȩ Ḥ Į Ɍ Ș Ț V̇ Ẓ Ą Ę Į Ǫ Ų

Ő B̋ C̋ D̋ Ő F̋ G̋ H̋ Í J̋ K̋ L̋ M̋ N̋ Ő P̋ Q̋ R̋ S̋ T̋ Ű V̋ W̋ X̋ Ÿ Ż Ɔ Ə Ɛ Θ Σ Ʊ U Y Ƶ Ȩ Ḥ Į Ɍ Ș Ț V̇ Ẓ Ą Ę Į Ǫ Ų

Ă Ǎ Č Ď Ě F̌ Ǧ Ȟ Ǐ J̌ Ǩ Ľ M̌ Ň Ǒ P̌ Q̌ Ř Š Ť Ǔ V̌ W̌ X̌ Y̌ Ž Ɔ Ə Ɛ Θ Σ Ʊ U Y Ƶ Ȩ Ḥ Į Ɍ Ș Ț V̇ Ẓ Ą Ę Į Ǫ Ų

Ä B̈ C̈ D̈ Ë F̈ G̈ Ḧ Ï J̈ K̈ L̈ M̈ N̈ Ö P̈ Q̈ R̈ S̈ T̈ Ü V̈ Ẅ Ẍ Ÿ Z̈ Ɔ Ə Ɛ Θ Σ Ʊ U Y Ƶ Ȩ Ḥ Į Ɍ Ș Ț V̇ Ẓ Ą Ę Į Ǫ Ų

Â B̂ Ĉ D̂ Ê F̂ Ĝ Ĥ Î J̊ K̂ L̂ M̂ N̂ Ô P̂ Q̂ R̂ Ŝ T̂ Û V̂ Ŵ X̂ Ŷ Ẑ Ɔ Ə Ɛ Θ Σ Ʊ U Y Ƶ Ȩ Ḥ Į Ɍ Ș Ț V̇ Ẓ Ą Ę Į Ǫ Ų

Italic patched, Capitals, Marks with capital alternates

Semibold, Capitals, Marks with capital alternates

À B C D È F G H Ì J K L M N Ò P Q R S T Ù V W X Y Z Ɔ Ə Ɛ Ɵ Σ Ʊ Ʊ ʏ Ƶ Ę Ḥ Į Ṛ Ş Ṭ Ṿ Ẓ Ạ Ẹ Ị Ọ Ụ
Á B Ć D É F G H Í J Ḱ Ĺ Ḿ Ń Ó Ṕ Q Ŕ Ś Ť Ú V Ẃ X Ý Ź Ɔ Ə Ɛ Ɵ Σ Ʊ Ʊ ʏ Ƶ Ę Ḥ Į Ṛ Ş Ṭ Ṿ Ẓ Ạ Ẹ Ị Ọ Ụ
Â B Ĉ D Ê F Ĝ Ĥ Î Ĵ K L M N Ô P Q R Ŝ T Û V Ŵ X Ŷ Ẑ Ɔ Ə Ɛ Ɵ Σ Ʊ Ʊ ʏ Ƶ Ę Ḥ Į Ṛ Ş Ṭ Ṿ Ẓ Ậ Ệ Ị Ộ Ụ
Ă B C D Ĕ F Ğ H Ĭ J K L M N Ŏ P Q Ř Š Ť Ŭ V W X Y Ž Ɔ Ə Ɛ Ɵ Σ Ʊ Ʊ ʏ Ƶ Ę Ḥ Į Ṛ Ş Ṭ Ṿ Ẓ Ặ Ẹ Ị Ŏ Ŭ
Ȧ Ḃ Ċ Ḋ Ė Ḟ Ġ Ḣ İ J K L Ṁ Ṅ Ȯ Ṗ Q Ṙ Ṡ Ṫ U V Ẇ Ẋ Ẏ Ż Ɔ Ə Ɛ Ɵ Σ Ʊ Ʊ ʏ Ƶ Ę Ḥ Į Ṛ Ṣ Ṭ Ṿ Ẓ Ạ Ẹ Ị Ọ Ụ
Ä B C D Ë F G Ḧ Ï J K L M N Ö P Q R S T Ü V Ẅ Ẍ Ÿ Ż Ɔ Ə Ɛ Ɵ Σ Ʊ Ʊ ʏ Ƶ Ę Ḥ Į Ṛ Ṣ Ṭ Ṿ Ẓ Ạ Ẹ Ị Ọ Ụ
Å Ḃ Ċ Ḋ Ḛ Ḟ Ġ Ḣ Ḭ J Ḳ Ḷ Ṁ Ṅ Ṏ Ṗ Q Ṛ Ṡ Ṭ Ů V W̊ Ẍ Y̊ Ż Ɔ Ə Ɛ Ɵ Σ Ʊ Ʊ ʏ Ƶ Ę Ḥ Į Ṛ Ṣ Ṭ Ṿ Ẓ Ạ Ẹ Ị Ọ Ụ
Ǻ Ḃ Ć Ḏ Ḛ Ḟ Ǵ Ḥ Ḯ J Ḱ Ḹ Ḿ Ń Ǿ Ṕ Q Ŕ Ś Ṭ Ű V Ẃ Ẍ Ý Ź Ɔ Ə Ɛ Ɵ Σ Ʊ Ʊ ʏ Ƶ Ę Ḥ Į Ṛ Ṣ Ṭ Ṿ Ẓ Ạ Ẹ Ị Ọ Ụ
Ǎ B Č Ď Ě F Ǧ Ȟ Ǐ J Ǩ Ľ M Ň Ǒ P Q Ř Š Ť Ǔ V W X Y Ž Ɔ Ə Ɛ Ɵ Σ Ʊ Ʊ ʏ Ƶ Ę Ḥ Į Ṛ Ṣ Ṭ Ṿ Ẓ Ạ Ẹ Ị Ǒ Ǔ
Ä B C D Ë F G Ḧ Ï J K L M N Ö P Q R S T Ü V Ẅ Ẍ Ÿ Ż Ɔ Ə Ɛ Ɵ Σ Ʊ Ʊ ʏ Ƶ Ę Ḥ Į Ṛ Ṣ Ṭ Ṿ Ẓ Ạ Ẹ Ị Ọ Ụ
Â B C D Ê F G H Î J K L M N Ô P Q R S T Û V Ŵ X Ŷ Ž Ɔ Ə Ɛ Ɵ Σ Ʊ Ʊ ʏ Ƶ Ę Ḥ Į Ṛ Ṣ Ṭ Ṿ Ẓ Ạ Ẹ Ị Ọ Ụ

Semibold patched, Capitals, Marks with capital alternates

À B C D È F G H Ì J K L M N Ò P Q R S T Ù V W X Y Z Ɔ Ə Ɛ Ɵ Ɔ̀ Ʊ̀ Ʊ̀ Y Ƶ Ę Ḥ Į Ṛ Ṣ Ṭ Ų̀ Ƶ̀ Ạ Ẹ Ị Ọ Ụ
Á B Ć D É F G H Í J K Ĺ M Ń Ó P Q Ŕ Ś T Ú V Ẃ X Ý Ź Ɔ́ Ə́ Ɛ́ Ɵ́ Ɔ́ Ʊ́ Ʊ́ Y Ƶ́ Ę Ḥ Į Ṛ Ṣ Ṭ Ų́ Ƶ́ Ạ Ẹ Ị Ọ Ụ
Â B Ĉ D Ê F Ĝ Ĥ Î Ĵ K L M N Ô P Q R Ŝ T Û V Ŵ X Ŷ Ẑ Ɔ̂ Ə̂ Ɛ̂ Ɵ̂ Ɔ̂ Ʊ̂ Ʊ̂ Ŷ Ƶ̂ Ę Ḥ Į Ṛ Ṣ Ṭ Ų̂ Ƶ̂ Ậ Ệ Ị Ộ Ụ
Ă B C D Ĕ F Ğ H Ĭ J K L M N Ŏ P Q Ř Š Ť Ŭ V W X Y Ž Ɔ̆ Ə̆ Ɛ̆ Ɵ̆ Ɔ̆ Ʊ̆ Ʊ̆ Y Ž̆ Ę Ḥ Į Ṛ Ṣ Ṭ Ų̆ Ž̆ Ặ Ệ Ị Ọ̆ Ụ
Ȧ Ḃ Ċ Ḋ Ė Ḟ Ġ Ḣ İ J K L̇ Ṁ Ṅ Ȯ Ṗ Q Ṙ Ṡ Ṫ U̇ V Ẇ Ẋ Ẏ Ż Ɔ̇ Ə̇ Ɛ̇ Ɵ̇ Ɔ̇ Ʊ̇ Ʊ̇ Ẏ Ż Ę̇ Ḥ̇ Į̇ Ṛ Ṣ Ṭ Ų̇ Ż Ạ̇ Ẹ̇ Į̇ Ọ̇ Ụ
Ä B C D Ë F G H Ï J K L M N Ö P Q R S T Ü V W X Y Ž Ɔ̈ Ə̈ Ɛ̈ Ɵ̈ Ɔ̈ Ʊ̈ Ʊ̈ Ÿ Ž̈ Ę̈ Ḥ̈ Į̈ Ṛ Ṣ Ṭ Ų̈ Ž̈ Ạ̈ Ẹ̈ Į̈ Ọ̈ Ụ
Å B C D Ḛ F G H Ḭ J K L M N Ǒ P Q R S T Ů V W X Y Ž Ɔ̊ Ə̊ Ɛ̊ Ɵ̊ Ɔ̊ Ʊ̊ Ʊ̊ Y Ž̊ Ę̊ Ḥ̊ Į̊ Ṛ Ṣ Ṭ Ų̊ Ž̊ Ạ̊ Ḛ̊ Į̊ Ọ̊ Ụ̊
Ấ B C D Ế F G H Ố J K L M N Ớ P Q R S T Ứ V W X Y Ž Ɔ̋ Ə̋ Ɛ̋ Ɵ̋ Ɔ̋ Ʊ̋ Ʊ̋ Y Ž̋ Ę̋ Ḥ̋ Į̋ Ṛ Ṣ Ṭ Ų̋ Ž̋ Ạ̋ Ḛ̋ Į̋ Ọ̋ Ụ̋
Ă B Č D Ě F Ğ H Ǐ J K Ľ M N Ǒ P Q Ř Š Ť Ŭ V W X Y Ž Ɔ̌ Ə̌ Ɛ̌ Ɵ̌ Ɔ̌ Ʊ̌ Ʊ̌ Y Ž̌ Ę̌ Ḥ̌ Į̌ Ṛ̌ Ṣ̌ Ṭ̌ Ų̌ Ž̌ Ạ̌ Ḛ̌ Į̌ Ǒ Ǔ
Ȁ B C D Ȅ F G H Ȉ J K L M N Ȍ P Q R S T Ȕ V W X Y Ž Ɔ̏ Ə̏ Ɛ̏ Ɵ̏ Ɔ̏ Ʊ̏ Ʊ̏ Y Ž̏ Ę̏ Ḥ̏ Į̏ Ṛ̏ Ṣ̏ Ṭ̏ Ų̏ Ž̏ Ạ̏ Ḛ̏ Į̏ Ọ̏ Ụ̏
Â B C D Ê F G H Î J K L M N Ô P Q R S T Û V W X Ŷ Ž Ɔ̂ Ə̂ Ɛ̂ Ɵ̂ Ɔ̂ Ʊ̂ Ʊ̂ Ŷ Ƶ̂ Ę Ḥ Į Ṛ Ṣ Ṭ Ų̂ Ƶ̂ Ạ Ẹ Ị Ọ Ụ

Semibold italic, Capitals, Marks with capital alternates



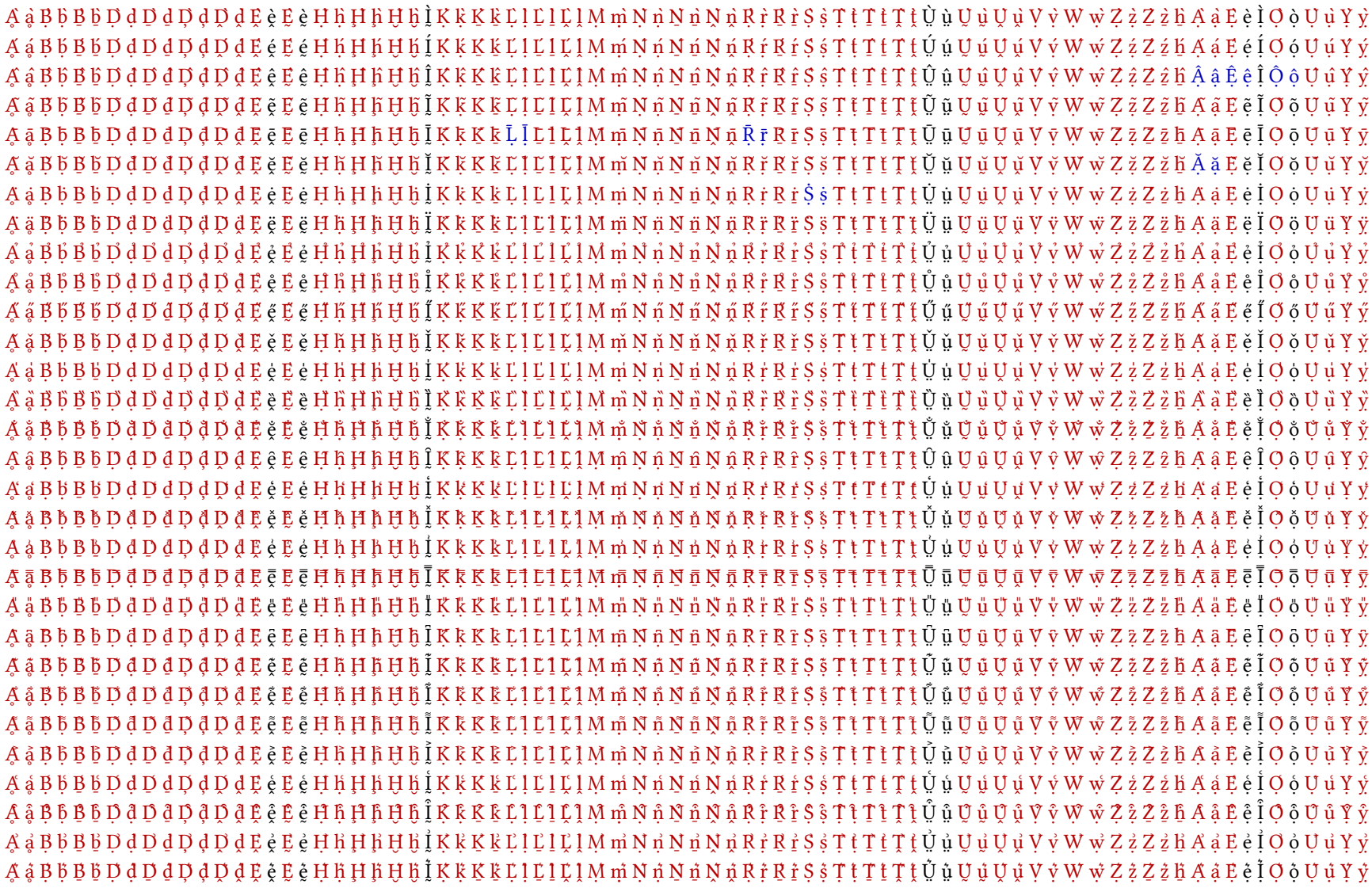
Semibold italic patched, Capitals, Marks with capital alternates



Precomposed characters

These grids display precomposed characters with referenced glyphs, where a precomposed character can inherit an anchor from the references glyph.

Regular, Precomposed bases that can inherit an above anchor, Above



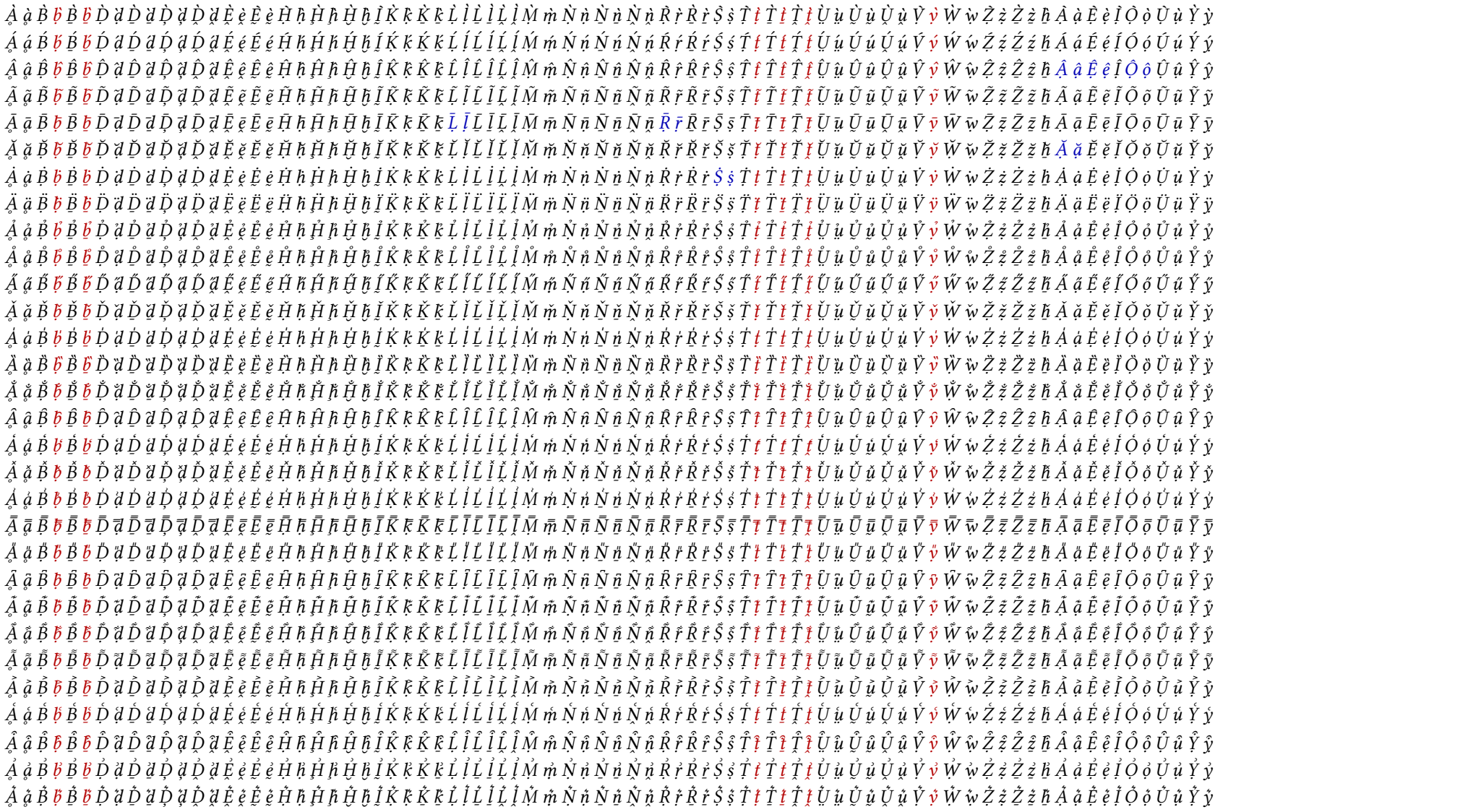
Regular patched, Precomposed bases that can inherit an above anchor, Above



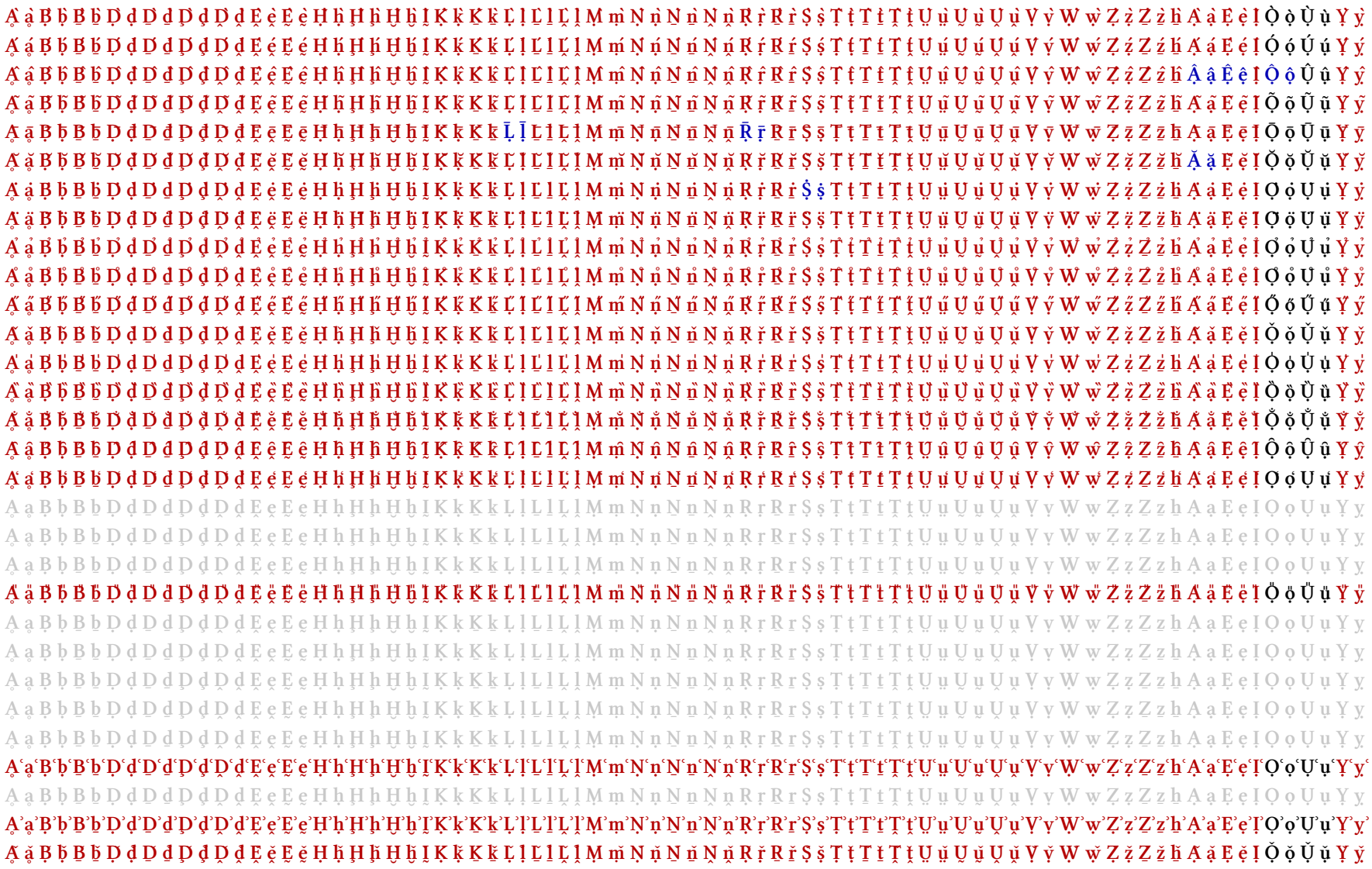
Italic, Precomposed bases that can inherit an above anchor, Above



Italic patched, Precomposed bases that can inherit an above anchor, Above



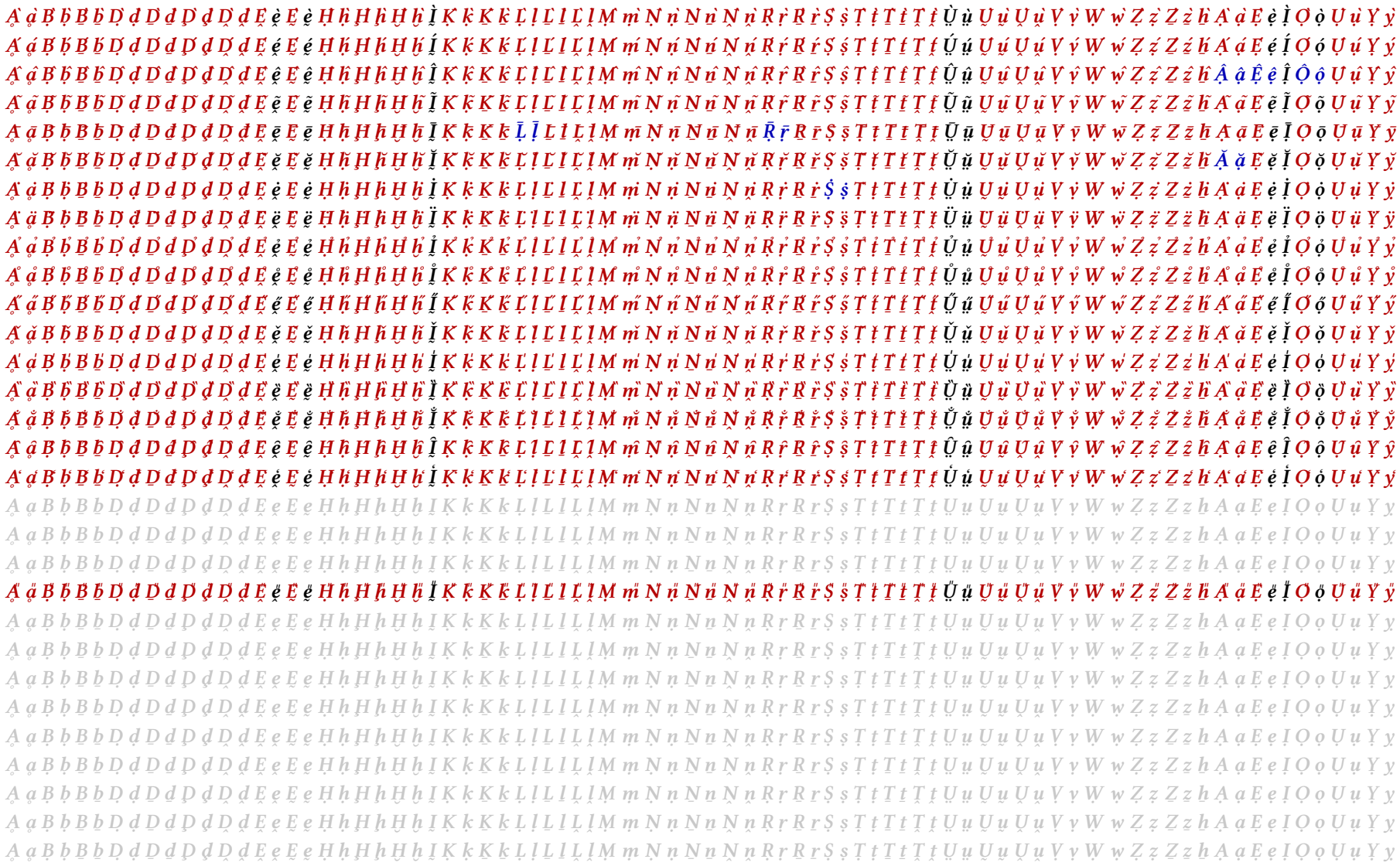
Semibold, Precomposed bases that can inherit an above anchor, Above



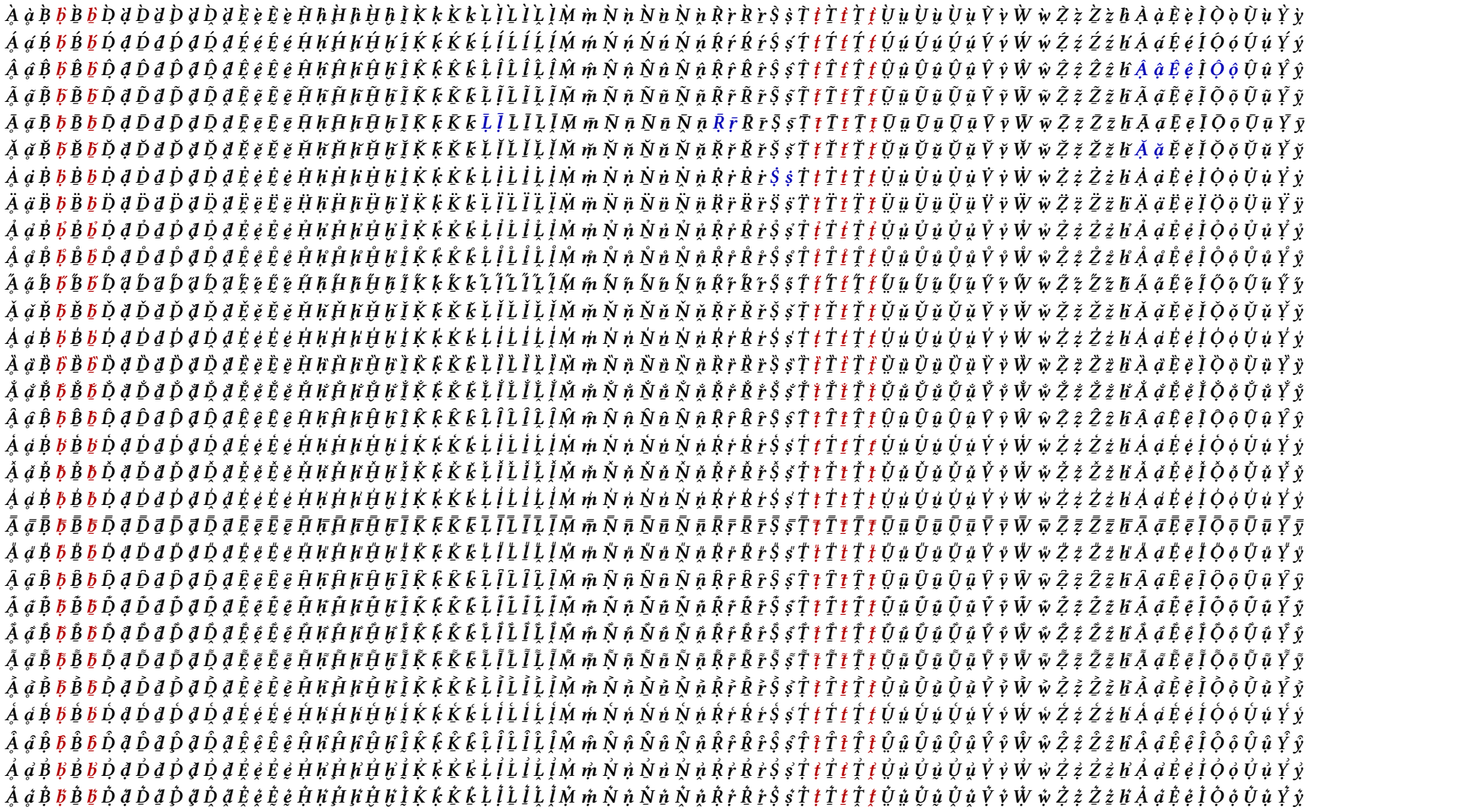
Semibold patched, Precomposed bases that can inherit an above anchor, Above



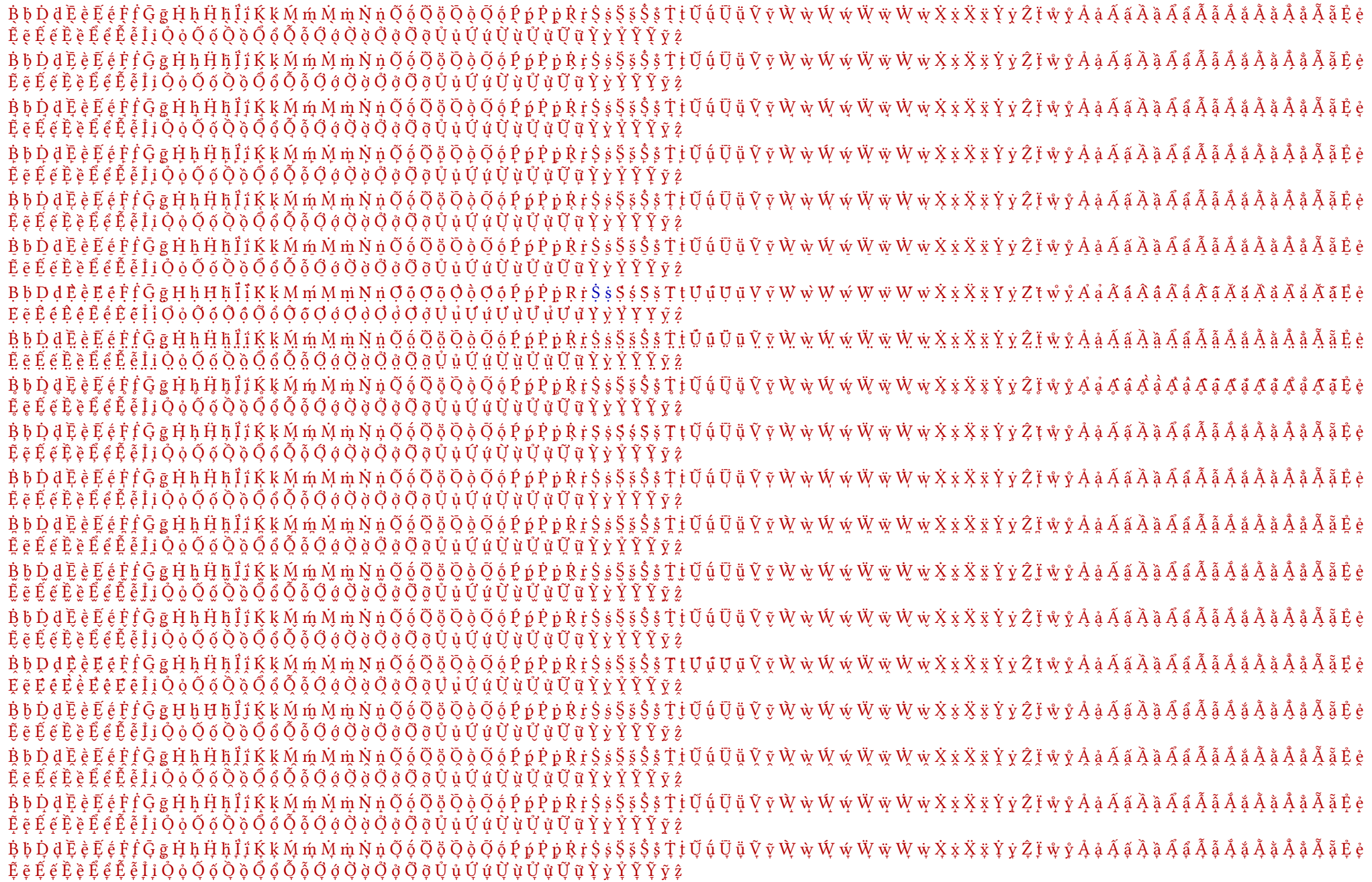
Semibold italic, Precomposed bases that can inherit an above anchor, Above



Semibold italic patched, Precomposed bases that can inherit an above anchor, Above



Regular, Precomposed bases that can inherit a below anchor, Below



Ḅ ḅ Ḍ ḍ Ḕ ḕ Ḗ ḗ Ḟ ḟ Ḡ ḡ Ḥ ḥ Ḧ ḧ Ḯ ḯ Ḱ ḱ Ḵ ḵ Ḿ ḿ Ṁ ṁ Ṉ ṉ Ṓ ṓ Ṑ ṑ Ṏ ṏ Ṍ ṍ Ṕ ṕ Ṗ ṗ Ṙ ṙ Ṣ ṣ Ṩ ṩ Ṧ ṧ Ṭ ṭ Ṵ ṵ Ṷ ṷ Ṹ ṹ Ṽ ṽ Ẁ ẁ Ẃ ẃ Ẅ ẅ Ẇ ẇ Ẋ ẋ Ẍ ẍ Ẏ ẏ Ẑ ẑ ẘ ẙ Ả ả Ấ ấ Ầ ầ Ẩ ẩ Ẫ ẫ Ắ ắ Ằ ằ Ẳ ẳ Ẵ ẵ Ặ ặ Ẹ ẹ
Ẽ ẽ Ế ế Ề ề Ể ể Ễ ễ Ệ ệ Ỉ ỉ Ị ị Ọ ọ Ố ố Ồ ồ Ổ ổ Ỗ ỗ Ộ ộ Ớ ớ Ờ ờ Ở ở Ỡ ỡ Ợ ợ Ụ ụ Ủ ủ Ứ ứ Ừ ừ Ử ử Ữ ữ Ự ự Ỳ ỳ Ỵ ỵ Ỷ ỷ Ỹ ỹ Ỿ ỿ Ⱬ ⱬ Ⱳ ⱳ Ⱶ ⱶ ⱷ ⱷ ⱸ ⱸ ⱹ ⱹ ⱺ ⱺ ⱻ ⱻ ⱼ ⱼ ⱽ ⱽ Ⱦ ⱦ Ⱨ ⱨ Ⱪ ⱪ

Regular patched, Precomposed bases that can inherit a below anchor, Below

Ḅ ḅ Ḍ ḍ Ḕ ḕ Ḗ ḗ Ḟ ḟ Ḡ ḡ Ḥ ḥ Ḧ ḧ Ḯ ḯ Ḵ ḵ Ḹ ḹ Ḻ ḻ Ṉ ṉ Ṓ ṓ Ṑ ṑ Ṝ ṝ Ṟ ṟ Ṩ ṩ Ṯ ṯ Ṳ ṳ Ṷ ṷ Ṽ ṽ Ẉ ẉ Ẍ ẍ Ẏ ẏ Ẓ ẓ ẘ ẙ Ạ ạ Ả ả Ấ ấ Ầ ầ Ẩ ẩ Ẫ ẫ Ắ ắ Ằ ằ Ẳ ẳ Ẵ ẵ Ặ ặ Ẹ ẹ
Ẽ ẽ Ế ế Ề ề Ể ể Ễ ễ Ệ ệ Ỉ ỉ Ị ị Ọ ọ Ố ố Ồ ồ Ổ ổ Ỗ ỗ Ộ ộ Ớ ớ Ờ ờ Ở ở Ỡ ỡ Ợ ợ Ụ ụ Ứ ứ Ừ ừ Ử ử Ữ ữ Ự ự Ỳ ỳ Ỵ ỵ Ỷ ỷ Ỹ ỹ

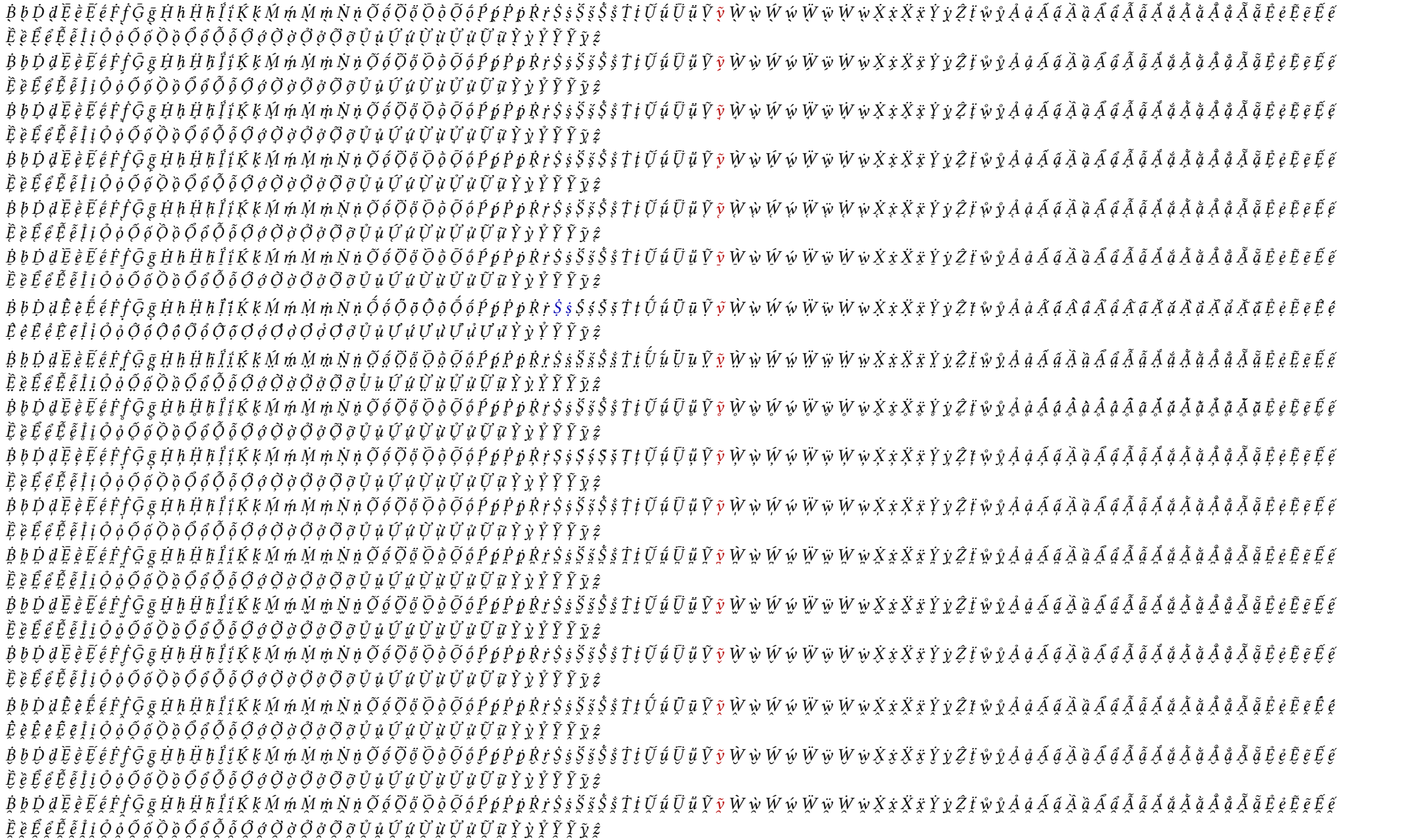
Ḅ ḅ Ḍ ḍ Ḕ ḕ Ḗ ḗ Ḟ ḟ Ḡ ḡ Ḥ ḥ Ḧ ḧ Ḯ ḯ Ḵ ḵ Ḿ ḿ Ṁ ṁ Ṅ ṅ Ṓ ṓ Ṑ ṑ Ṓ ṓ Ṍ ṍ Ṏ ṏ Ṕ ṕ Ṗ ṗ Ṙ ṙ Ṣ ṣ Ṩ ṩ Ṩ ṩ Ṫ ṫ Ṵ ṵ Ṷ ṷ Ṹ ṹ Ṻ ṻ Ṽ ṽ Ẁ ẁ Ẃ ẃ Ẅ ẅ Ẁ ẁ Ẅ ẅ Ẋ ẋ Ẍ ẍ Ẏ ẏ Ẓ ẓ ẘ ẙ Ạ ạ Ả ả Ấ ấ Ầ ầ Ẩ ẩ Ẫ ẫ Ắ ắ Ằ ằ Ẳ ẳ Ẵ ẵ Ặ ặ Ẹ ẹ
Ẽ ẽ Ế ế Ề ề Ể ể Ễ ễ Ệ ệ Ỉ ỉ Ị ị Ọ ọ Ố ố Ồ ồ Ổ ổ Ỗ ỗ Ộ ộ Ớ ớ Ờ ờ Ở ở Ỡ ỡ Ợ ợ Ụ ụ Ủ ủ Ứ ứ Ừ ừ Ử ử Ữ ữ Ự ự Ỳ ỳ Ỵ ỵ Ỷ ỷ Ỹ ỹ ỻ ỽ

Italic, Precomposed bases that can inherit a below anchor, Below



Ḃ ḃ Ḍ ḍ Ḕ ḕ Ḗ ḗ Ḟ ḟ Ḡ ḡ Ḥ ḥ Ḧ ḧ Ḯ ḯ Ḱ ḱ Ḿ ḿ Ṁ ṁ Ṅ ṅ Ṓ ṓ Ṑ ṑ Ṏ ṏ Ṕ ṕ Ṗ ṗ Ṙ ṙ Ṡ ṡ Ṣ ṣ Ṥ ṥ Ṫ ṫ Ṻ ṻ Ṹ ṹ Ṽ ṽ Ẁ ẁ Ẃ ẃ Ẅ ẅ Ẇ ẇ Ẋ ẋ Ẍ ẍ Ẏ ẏ Ẑ ẑ ẘ ẙ Ả ả Ấ ấ Ầ ầ Ẩ ẩ Ẫ ẫ Ắ ắ Ằ ằ Ẳ ẳ Ẵ ẵ Ặ ặ Ẹ ẹ Ẽ ẽ Ế ế
Ề ề Ể ể Ễ ễ Ệ ệ Ỉ ỉ Ị ị Ọ ọ Ố ố Ồ ồ Ổ ổ Ỗ ỗ Ộ ộ Ớ ớ Ờ ờ Ở ở Ỡ ỡ Ợ ợ Ụ ụ Ứ ứ Ừ ừ Ử ử Ữ ữ Ự ự Ỳ ỳ Ỷ ỷ Ỹ ỹ Ỵ ỵ

Italic patched, Precomposed bases that can inherit a below anchor, Below



Ḃ ḃ Ḍ ḍ Ḕ ḕ Ḗ ḗ Ḟ ḟ Ḡ ḡ Ḥ ḥ Ḧ ḧ Ḯ ḯ Ḱ ḱ Ḿ ḿ Ṁ ṁ Ṅ ṅ Ṓ ṓ Ṑ ṑ Ṓ ṓ Ṍ ṍ Ṕ ṕ Ṗ ṗ Ṙ ṙ Ṡ ṡ Ṣ ṣ Ṥ ṥ Ṫ ṫ Ṹ ṹ Ṻ ṻ Ṽ ṽ Ẁ ẁ Ẃ ẃ Ẅ ẅ Ẇ ẇ Ẋ ẋ Ẍ ẍ Ẏ ẏ Ẑ ẑ ẘ ẙ Ả ả Ấ ấ Ầ ầ Ẩ ẩ Ẫ ẫ Ắ ắ Ằ ằ Ẳ ẳ Ẵ ẵ Ặ ặ Ẹ ẹ Ẽ ẽ Ế ế
Ề ề Ể ể Ễ ễ Ệ ị Ỏ ỏ Ố ố Ồ ồ Ổ ổ Ỗ ỗ Ộ ộ Ờ ờ Ở ở Ỡ ỡ Ợ ợ Ụ ụ Ứ ứ Ừ ừ Ử ử Ữ ữ Ỳ ỳ Ỷ ỷ Ỹ ỹ Ỿ ỿ

Semibold, Precomposed bases that can inherit a below anchor, Below

Ḅ ḅ Ḍ ḍ Ḕ ḕ Ḗ ḗ Ḟ ḟ Ḡ ḡ Ḥ ḥ Ḧ ḧ Ḫ ḫ Ḭ ḭ Ḵ ḵ Ḷ ḷ Ḹ ḹ Ḻ ḻ Ṉ ṉ Ṓ ṓ Ṕ ṕ Ṙ ṙ Ṣ ṣ Ṥ ṥ Ṧ ṧ Ṩ ṩ Ṫ ṫ Ṱ ṱ Ṳ ṳ Ṵ ṵ Ṷ ṷ Ẉ ẉ Ẍ ẍ Ẏ ẏ Ẓ ẓ ẘ ẙ Ạ ạ Ả ả Ấ ấ Ầ ầ Ẩ ẩ Ẫ ẫ Ậ ậ Ắ ắ Ằ ằ Ẳ ẳ Ẵ ẵ Ặ ặ Ẹ ẹ Ẽ ẽ Ế ế Ề ề Ể ểỄ ễ Ệ ệ Ỉ ỉ Ị ị Ọ ọ Ố ố Ồ ồ Ổ ổ Ỗ ỗ Ộ ộ Ớ ớ Ờ ờ Ở ở Ỡ ỡ Ợ ợ Ụ ụ Ủ ủ Ứ ứ Ừ ừ Ử ử Ữ ữ Ự ự Ỳ ỳ Ỵ ỵ Ỷ ỷ Ỹ ỹ Ỿ ỿ Ⱬ ⱬ

Ḅ ḅ Ḍ ḍ Ḕ ḕ Ḗ ḗ Ḟ ḟ Ḡ ḡ Ḥ ḥ Ḧ ḧ Ḫ ḫ Ḭ ḭ Ḵ ḵ Ḷ ḷ Ḹ ḹ Ḻ ḻ Ṉ ṉ Ṓ ṓ Ṕ ṕ Ṙ ṙ Ṣ ṣ Ṥ ṥ Ṧ ṧ Ṩ ṩ Ṫ ṫ Ṱ ṱ Ṳ ṳ Ṵ ṵ Ṷ ṷ Ẉ ẉ Ẍ ẍ Ẏ ẏ Ẓ ẓ ẘ ẙ Ạ ạ Ả ả Ấ ấ Ầ ầ Ẩ ẩ Ẫ ẫ Ậ ậ Ắ ắ Ằ ằ Ẳ ẳ Ẵ ẵ Ặ ặ Ẹ ẹ Ẽ ẽ Ế ế Ề ề Ể ể

Ḃ ḃ Ḋ ḋ Ḕ ḕ Ḗ ḗ Ḟ ḟ Ḡ ḡ Ḥ ḥ Ḧ ḧ Ḭ ḭ Ḱ ḱ Ḿ ḿ Ṁ ṁ Ṅ ṅ Ṓ ṓ Ṑ ṑ Ṙ ṙ Ṡ ṡ Ṧ ṧ Ṩ ṩ Ṫ ṫ Ṻ ṻ Ṽ ṽ Ẁ ẁ Ẃ ẃ Ẅ ẅ Ẇ ẇ Ẋ ẋ Ẍ ẍ Ẏ ẏ Ẓ ẓ ẘ ẙ Ạ ạ Ả ả Ấ ấ Ầ ầ Ẩ ẩ Ẫ ẫ Ậ ậ Ắ ắ Ằ ằ Ẳ ẳ Ẵ ẵ
Ặ ặ Ẹ ẹ Ẽ ẽ Ế ế Ề ề Ể ể Ễ ễ Ệ ệ Ỉ ỉ Ị ị Ọ ọ Ỏ ỏ Ố ố Ồ ồ Ổ ổ Ỗ ỗ Ộ ộ Ớ ớ Ờ ờ Ở ở Ỡ ỡ Ợ ợ Ụ ụ Ủ ủ Ứ ứ Ừ ừ Ử ử Ữ ữ Ự ự Ỳ ỳ Ỷ ỷ Ỹ ỹ Ỵ ỵ

Semibold patched, Precomposed bases that can inherit a below anchor, Below

Ḅ ḅ Ḍ ḍ Ḕ ḕ Ḗ ḗ Ḟ ḟ Ḡ ḡ Ḥ ḥ Ḧ ḧ Ḯ ḯ Ḳ ḳ Ḵ ḵ Ḿ ḿ Ṁ ṁ Ṉ ṉ Ṓ ṓ Ṑ ṑ Ṓ ṓ Ṍ ṍ Ṕ ṕ Ṗ ṗ Ṛ ṛ Ṣ ṣ Ṩ ṩ Ṩ ṩ Ṭ ṭ Ṳ ṳ Ṷ ṷ Ṽ ṽ Ẁ ẁ Ẃ ẃ Ẅ ẅ Ẁ ẁ Ẋ ẋ Ẍ ẍ Ẏ ẏ Ẓ ẓ ẘ ẙ Ạ ạ Ấ ấ Ầ ầ Ẩ ẩ Ẫ ẫ Ậ ậ Ắ ắ Ằ ằ Ẳ ẳ Ẵ ẵ
Ẹ ẹ Ẽ ẽ Ế ế Ề ề Ể ể Ễ ễ Ệ ệ Ỉ ỉ Ị ị Ọ ọ Ố ố Ồ ồ Ổ ổ Ỗ ỗ Ộ ộ Ờ ờ Ở ở Ỡ ỡ Ợ ợ Ụ ụ Ứ ứ Ừ ừ Ử ử Ữ ữ Ự ự Ỳ ỳ Ỷ ỷ Ỹ ỹ ỵ ỿ

Ḅ ḅ Ḍ ḍ Ḕ ḕ Ḗ ḗ Ḟ ḟ Ḡ ḡ Ḥ ḥ Ḧ ḧ Ḫ ḫ Ḭ ḭ Ḳ ḳ Ḵ ḵ Ḷ ḷ Ḹ ḹ Ḻ ḻ Ṃ ṃ Ṁ ṁ Ṇ ṇ Ṓ ṓ Ṑ ṑ Ṓ ṓ Ṑ ṑ Ṕ ṕ Ṗ ṗ Ṙ ṙ Ṣ ṣ Ṩ ṩ Ṩ ṩ Ṭ ṭ Ṵ ṵ Ṷ ṷ Ṹ ṹ Ṻ ṻ Ẁ ẁ Ẃ ẃ Ẅ ẅ Ẇ ẇ Ẉ ẉ Ẋ ẋ Ẍ ẍ Ẏ ẏ Ẑ ẑ ẘ ẙ Ạ ạ Ả ả Ấ ấ Ầ ầ Ẩ ẩ Ẫ ẫ Ậ ậ Ắ ắ Ằ ằ Ẳ ẳ Ẵ ẵ
Ặ ặ Ẹ ẹ Ẽ ẽ Ế ế Ề ề Ể ể Ễ ễ Ệ ệ Ỉ ỉ Ị ị Ọ ọ Ỏ ỏ Ố ố Ồ ồ Ổ ổ Ỗ ỗ Ộ ộ Ớ ớ Ờ ờ Ở ở Ỡ ỡ Ợ ợ Ụ ụ Ủ ủ Ứ ứ Ừ ừ Ử ử Ữ ữ Ự ự Ỳ ỳ Ỷ ỷ Ỹ ỹ Ỵ ỵ

Semibold italic, Precomposed bases that can inherit a below anchor, Below

Ḅ ḅ Ḍ ḍ Ḕ ḕ Ḗ ḗ Ḟ ḟ Ḡ ḡ Ḥ ḥ Ḧ ḧ Ḯ ḯ Ḳ ḳ Ḵ ḵ Ḷ ḷ Ḹ ḹ Ḻ ḻ Ṉ ṉ Ṋ ṋ Ṍ ṍ Ṏ ṏ Ṑ ṑ Ṓ ṓ Ṕ ṕ Ṗ ṗ Ṛ ṛ Ṣ ṣ Ṥ ṥ Ṧ ṧ Ṩ ṩ Ṭ ṭ Ṷ ṷ Ṹ ṹ Ṻ ṻ Ṽ ṽ Ẁ ẁ Ẃ ẃ Ẅ ẅ Ẇ ẇ Ẋ ẋ Ẍ ẍ Ẏ ẏ Ẑ ẑ ẘ ẙ Ả ả Ấ ấ Ầ ầ Ẩ ẩ Ẫ ẫ Ắ ắ Ằ ằ Ẳ ẳ Ẵ ẵ
Ặ Ẹ ẹ Ệ ệ Ỉ ỉ Ọ ọ Ố ố Ồ ồ Ổ ổ Ỗ ỗ Ộ ộ Ớ ớ Ờ ờ Ở ở Ỡ ỡ Ợ ợ Ụ ụ Ứ ứ Ừ ừ Ử ử Ữ ữ Ỳ ỳ Ỵ Ỷ Ỹ ỹ ỿ

Ḃ ḃ Ḋ ḋ Ḕ ḕ Ḗ ḗ Ḟ ḟ Ḡ ḡ Ḣ ḣ Ḥ ḥ Ḧ ḧ Ḭ ḭ Ḱ ḱ Ḳ ḳ Ḵ ḵ Ḷ ḷ Ḹ ḹ Ḻ ḻ Ḽ ḽ Ḿ ḿ Ṁ ṁ Ṃ ṃ Ṅ ṅ Ṇ ṇ Ṉ ṉ Ṋ ṋ Ṍ ṍ Ṏ ṏ Ṑ ṑ Ṓ ṓ Ṕ ṕ Ṗ ṗ Ṙ ṙ Ṛ ṛ Ṝ ṝ Ṟ ṟ Ṡ ṡ Ṣ ṣ Ṥ ṥ Ṧ ṧ Ṩ ṩ Ṫ ṫ Ṭ ṭ Ṯ ṯ Ṱ ṱ Ṳ ṳ Ṵ ṵ Ṷ ṷ Ṹ ṹ Ṻ ṻ Ṽ ṽ Ṿ ṿ Ẁ ẁ Ẃ ẃ Ẅ ẅ Ẇ ẇ Ẉ ẉ Ẋ ẋ Ẍ ẍ Ẏ ẏ Ẑ ẑ Ẓ ẓ Ẕ ẕ Ả ả Ấ ấ Ầ ầ Ẩ ẩ Ẫ ẫ Ậ ậ Ắ ắ Ằ ằ Ẳ ẳ Ẵ ẵ Ặ ặ Ẹ ẹ Ẽ ẽ Ế ế Ề ề Ể ể Ễ ễ Ệ ệ Ỉ ỉ Ị ị Ọ ọ Ỏ ỏ Ố ố Ồ ồ Ổ ổ Ỗ ỗ Ộ ộ Ớ ớ Ờ ờ Ở ở Ỡ ỡ Ợ ợ Ụ ụ Ủ ủ Ứ ứ Ừ ừ Ử ử Ữ ữ Ự ự Ỳ ỳ Ỵ ỵ Ỷ ỷ Ỹ ỹ Ỿ ỿ

Ḃ ḃ Ḋ ḋ Ḕ ḕ Ḗ ḗ Ḟ ḟ Ḡ ḡ Ḣ ḣ Ḥ ḥ Ḧ ḧ Ḭ ḭ Ḱ ḱ Ḳ ḳ Ḵ ḵ Ḷ ḷ Ḹ ḹ Ḻ ḻ Ḽ ḽ Ḿ ḿ Ṁ ṁ Ṃ ṃ Ṅ ṅ Ṇ ṇ Ṉ ṉ Ṋ ṋ Ṍ ṍ Ṏ ṏ Ṑ ṑ Ṓ ṓ Ṕ ṕ Ṗ ṗ Ṙ ṙ Ṛ ṛ Ṝ ṝ Ṟ ṟ Ṡ ṡ Ṣ ṣ Ṥ ṥ Ṧ ṧ Ṩ ṩ Ṫ ṫ Ṭ ṭ Ṯ ṯ Ṱ ṱ Ṳ ṳ Ṵ ṵ Ṷ ṷ Ṹ ṹ Ṻ ṻ Ṽ ṽ Ṿ ṿ Ẁ ẁ Ẃ ẃ Ẅ ẅ Ẇ ẇ Ẉ ẉ Ẋ ẋ Ẍ ẍ Ẏ ẏ Ẑ ẑ Ẓ ẓ Ẕ ẕ Ả ả Ấ ấ Ầ ầ Ẩ ẩ Ẫ ẫ Ậ ậ Ắ ắ Ằ ằ Ẳ ẳ Ẵ ẵ Ặ ặ Ẹ ẹ Ẽ ẽ Ế ế Ề ề Ể ể Ễ ễ Ệ ệ Ỉ ỉ Ị ị Ọ ọ Ỏ ỏ Ố ố Ồ ồ Ổ ổ Ỗ ỗ Ộ ộ Ớ ớ Ờ ờ Ở ở Ỡ ỡ Ợ ợ Ụ ụ Ủ ủ Ứ ứ Ừ ừ Ử ử Ữ ữ Ự ự Ỳ ỳ Ỵ ỵ Ỷ ỷ Ỹ ỹ Ỿ ỿ

Semibold italic patched, Precomposed bases that can inherit a below anchor, Below



Ḃ ḃ Ḍ ḍ Ḕ ḕ Ḗ ḗ Ḟ ḟ Ḡ ḡ Ḥ ḥ Ḧ ḧ Ḯ ḯ Ḳ ḳ Ḿ ḿ Ṁ ṁ Ṇ ṇ Ṓ ṓ Ṑ ṑ Ṓ ṓ Ṗ ṗ Ṕ ṕ Ṙ ṙ Ṡ ṡ Ṣ ṣ Ṫ ṫ Ṹ ṹ Ṻ ṻ Ṽ ṽ Ẁ ẁ Ẃ ẃ Ẅ ẅ Ẇ ẇ Ẋ ẋ Ẍ ẍ Ẏ ẏ Ẑ ẑ ẘ ẙ Ả ả Ấ ấ Ầ ầ Ẩ ẩ Ẫ ẫ Ắ ắ Ằ ằ Ẳ ẳ Ẵ ẵ
Ặ Ẹ ẹ Ẽ ẽ Ế ế Ề ề Ể ể Ễ ễ Ệ ệ Ỉ ỉ Ọ ọ Ố ố Ồ ồ Ổ ổ Ỗ ỗ Ộ ộ Ờ ờ Ở ở Ỡ ỡ Ợ ụ Ủ ủ Ứ ứ Ừ ừ Ử ử Ữ ữ Ỳ ỳ Ỷ ỷ Ỹ ỹ Ỵ ỵ

1. ”i” not losing its dot when there is a tone diacritic AND a tilde below

   i̱ í̱

   í̱

   ì̱

2. troubles with capital I

   I̱ I̱ I̱

   Solution: Use combnining grapheme joiner (CGJ)

   I̱ I̱ I̱

3. double acute critic shifted right in bold face

   a̋ a̋ a̱ a̱̋ a̱̋ a̱̋

   Solution: Patch anchor of double acute

   a̋ a̋ a̱ a̱̋ a̱̋ a̱̋ **a̋ a̋ a̱ a̱̋ a̱̋ a̱̋** ***a̋ a̋ a̱ a̱̋ a̱̋ a̱̋***

4. wrong tilde placement together with caron (and grave? and acute? and double grave? OK, it seems it’s rather about vowels...) with some vowels

   ǎ̱ ǎ̱ ǎ̱ ǎ̱

   ù̱ ù̱ ù̱ ù̱

   Solution: Use combnining grapheme joiner (CGJ)

   ǎ̱ ǎ̱ ǎ̱ ǎ̱

   ù̱ ù̱ ù̱ ù̱

5. wrong acute placement in bold face

   é *é* **é** ***é***

   á *á* **á** ***á***

   ó *ó* **ó** ***ó***

   ú *ú* **ú** ***ú***

   è *è* **è** ***è***

   à *à* **à** ***à***

   ò *ò* **ò** ***ò***

   ù *ù* **ù** ***ù***

   Solution: Patch above-anchor of acute and grave in (semi)bold.

   é *é* **é** ***é*** á *á* **á** ***á***

   ó *ó* **ó** ***ó*** ú *ú* **ú** ***ú***

   è *è* **è** ***è*** à *à* **à** ***à***

   ò *ò* **ò** ***ò*** ù *ù* **ù** ***ù***

6. italic examples with tilde below

Flą̰

+ *Flą̰*

gṵ̀ ą̰

gṵ̀ą̰

+ *gṵ̀ą̰*

lɛ̰̃ ɔ̰

+ *lɛ̰̃ ɔ̰*

mṵ

kṵ̃,

kṵ̃

+ *kṵ̃*

nṵ̃

+ *nṵ̃*

nɔ̰̌

nṵ̀

nṵ̀

N-ą̰

vɔ̰ṵ̃ṵ̃

fɔ̰ṵ̃fɔ̰ṵ̃

lɛ̰̃ ɔ̰

Flą̰

gṵ̀ ą̰

gṵ̀ą̰

lɛ̰̃ ɔ̰

lɛ̰̃ ɔ̰

mṵ

kṵ̃,

kṵ̃

nṵ̃

nɔ̰̌

nɔ̰̌

nù̱

nù̱

N-ą̆

7. Ɛ̏ bold italics

   Ɛ̏

   Ɛ̀ Ɛ̏ **Ɛ̀ Ɛ̏**

   Solution: Set above-anchor for open-E, patch GSUB to use capital alternate mark glyph ”uni”